Occupation counts of rarely switching Markov chains: stratified local laws, support selection and the boundary layer

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Abstract

A Markov chain on d states runs for N steps with transition matrix $I+(T/N)Q$. On each face whose states communicate within it and carry start mass, we prove a local limit theorem for the occupation counts with an explicit constant and relative error $O(1/T)$, uniformly for $1 \le T \le \varepsilon N$ with $\varepsilon \max_i q_i < 1$ and on compact subsets of the open face. Up to $O(1)$, $-\log P_N(n)$ is $|F|-1$ times $\log(N/\sqrt{T})$ plus N times the Donsker and Varadhan functional of the chain $I+(T/N)Q$ at n/N. Every admissible face has an explicit power of T in its largest probability for large $T \le N^{1/3}$, and a constant given by a finite-dimensional variational formula. For $T = \tau \log N$ the face of the mode minimizes $\tau\lambda_F + |F| - 1$ at first order, with λ_F the exit rate. If two admissible faces differ by $\Delta \ge 1$ states and their exit rates by $\delta > 0$, we locate every crossing of their largest probabilities there up to $O(1/\log N)$. For faces whose states communicate, the crossing is unique for $T \le \varepsilon N$, with coefficient $1/2$ of $\log\log N$. Near a facet, the continuous-time occupation density is the facet density times an explicit kernel, with relative error $O(1/T)$ at times $x \le cT$ off the facet, with c small. The counts obey a lattice form with extra error T^2/N. The kernel is Bessel for $x = o(T)$ at the facet profiles where the entry and exit laws of the removed state agree. At a profile where the rate has a first correction, of order $m \ge 3$, it is Bessel if and only if $x = o(T^{1-2/m})$. On the complete graph with the uniform start, each face crosses a vertex exactly once.

Keywords: occupation times of Markov chains, mode, local limit theorem, principal Dirichlet eigenvalue, Donsker and Varadhan functional, Bessel kernel

MSC Classification: Primary 60J10; Secondary 60J27 , 60F10 , 60F05 , 15B48

1 Introduction

Let Q be a generator on d states, and run the chain for N steps with transition matrix $I+(T/N)Q$. Its path follows the continuous-time chain with generator Q up to time T, while the transition matrix tends to the identity as N grows. So the chain degenerates. Its spectral gap is T/N times that of Q, and for fixed T the visit fractions converge in law to the occupation fractions of the continuous-time chain at time T, which charge the faces of the simplex. We let T range from 1 to h_*N, for a fixed h_* with $h_* \max_i q_i < 1$, and describe the law of the occupation counts $n_i = \#\{t \le N : X_t = i\}$. The support $F = \{i : n_i > 0\}$ of the count vector is a face of the simplex.

Our main result is a local limit theorem for the count vector on each face, with an explicit constant and uniformly in T (Theorem A). Up to $O(1)$, $-\log P_N(n)$ is $(|F|-1)\log(N/\sqrt{T})$ plus N times the Donsker and Varadhan functional of the discrete chain at n/N. So each dimension of the face costs a factor N up to a power of $\log N$ when T is of order $\log N$, and a factor of order $\sqrt{N}$ when T is of order N. The functional is one functional of the whole chain, and on each face it is that of the chain killed on leaving the face. Two consequences follow. First, the law selects the support of the most likely count vector, the mode, and it locates the values of T at which two faces have equal largest probabilities (Theorem B). Second, near a facet the occupation density of the continuous-time chain on a face is the density on the facet times an explicit kernel, which is Bessel in a range of the time off the facet that we determine, and the counts follow the same law with a lattice kernel. The complete graph serves as a worked example (Theorem C).

To state the law, let J_V be the Donsker and Varadhan rate function of the empirical measure of the continuous-time chain with generator Q, given in (3). For $h = T/N$ let $hJ_V^{(h)}$ be the same rate for the chain with transition matrix $I + hQ$ (Section 3.3). It satisfies $J_V^{(h)} \ge J_V$ and tends to J_V as $h \to 0$. Say that a face F satisfies (H) if the chain can move between any two states of F without leaving F and the start gives F positive mass. Write λ_F for the exit rate of F and M_F for the largest probability of a count vector with support F (Definition 2.1).

Theorem A (the law on a face) *Fix the rates q_{ij}, the law of X_1 and h_* with $h_* \max_i q_i < 1$. Let F satisfy* (H)*, with $k = |F| \ge 2$ states, and let $\mathcal{K}$ be a compact subset of the open face. Uniformly for* $\operatorname{supp} n = F$, $n/N \in \mathcal{K}$ *and* $1 \le T \le h_*N$,

$$-\log P_N(n) = (k-1)\log N + T\, J_V^{(h)}(n/N) - \frac{k-1}{2}\log T - \log C_F^{(h)}(n/N) + O\Big(\frac{1}{T}\Big),$$

where $C_F^{(h)} > 0$ is an explicit function that tends to the function C_F of Definition 5.1 as $h \to 0$. Uniformly for $1 \le T \le h_*N$,

$$M_F = N^{-(k-1)}\, T^{(k-1)/2} K_F^{(h)}\, (1 - h\lambda_F)^N \big(1 + O(1/T)\big),$$

where $K_F^{(h)} = C_F^{(h)}(\alpha_F^*)$ and α_F^* is the quasi-ergodic law of F. For $k = 1$ the second display holds exactly, with $K_F^{(h)} = \mu_i/(1 - hq_i)$ for $F = \{i\}$. For $1 \le T \le N^{1/3}$ both displays hold with J_V, C_F, $K_F = C_F(\alpha_F^*)$ and $e^{-T\lambda_F}$ in place of $J_V^{(h)}$, $C_F^{(h)}$, $K_F^{(h)}$ and $(1 - h\lambda_F)^N$, and with the error $O(1/T + T^2/N)$. The second display then holds for every admissible face F, one that a path of positive probability can fill, with $k = |F|$, for $T_1 \le T \le N^{1/3}$, and with

$T^{(k-1)/2}$ *and* K_F *replaced by an explicit power* T^{p_F} *and a constant* $K_F^{\mathrm{red}} > 0$ *given by a finite-dimensional variational formula.*

Theorem A is Theorems 5.6, 5.7 and 4.5. Good's formula for the counts is an integral over a torus (Lemma A.2). In the variable of the torus the integrand is analytic in h up to $h = 0$, where it is the integrand of the continuum density of the chain killed on leaving F. So one Laplace argument with h as a parameter gives the law for $1 \le T \le h_* N$ (Theorem 5.4). Only for bounded T do we compare with the continuum, to keep the ratio bounded below. The rate of the killed chain is the restriction of $J_V^{(h)}$ to the face (Proposition 3.5). Its least value is $-\frac{1}{h}\log(1-h\lambda_F)$, attained only at α_F^* for every h (Proposition 3.6). With the continuum rate the error T^2/N cannot be avoided, since $TJ_V^{(h)} - TJ_V$ is T^2/N times a function of n/N that vanishes at most at one point, up to $O(T^3/N^2)$. On faces with (H) the continuum form follows from the discrete one for $T \le N^{1/2}$. We state it for $T \le N^{1/3}$, the range of a second proof through the continuum density, which needs $T^2/N \le 1/T$ to locate the most likely counts and which we make uniform over a class of chains. The factor $T^{(k-1)/2}$ appears because the visit fractions fluctuate on the scale $T^{-1/2}$ in each of the $k-1$ free directions. For reversible faces K_F involves the square root of a weighted count of spanning trees (Proposition 5.2). The face maxima hold with a field v that weights a composition by $e^{Tv\cdot n/N}$, uniformly for v in a compact set, and uniformly over a compact class of reversible chains (Proposition 5.12). If the states of F do not all communicate, the time goes to the classes of least exit rate and least size k^*, and to one of them when $k^* \ge 2$. Every other class C gets time of order one and costs $(T/N)^{|C|-1}$. Each switch from one class to the next costs T/N too. This gives the power $p_F = (k^*-1)/2 + |F| - k^*$.

1.1 The mode and its crossings

When $T = \tau \log N$ the cost $e^{-T\lambda_F}$ of staying in F and the cost $N^{-(|F|-1)}$ of spreading over the count vectors of F are both powers of N. Put $\Phi_\tau(F) = \tau\lambda_F + |F| - 1$. A face is admissible if a path of positive probability has exactly that support.

Theorem B (the support of the mode) *Fix the rates* q_{ij} *and the law of* X_1.

(i) Let $T = \tau \log N$ *with* $\tau > 0$ *fixed. Every admissible face* F *has* $M_F = N^{-\Phi_\tau(F)+o(1)}$. *For large* N *the support of every most likely composition is an admissible face that minimizes* Φ_τ *among the admissible faces. If each such minimizing face satisfies* (H), *then the visit fractions of every most likely composition are within* $O(1/\log N)$ *of the quasi-ergodic law of its support.*

(ii) Let F *and* G *satisfy* (H), *and suppose that* F *has* $\Delta \ge 1$ *more states than* G *and an exit rate smaller by* $\delta > 0$. *For large* N *the equation* $M_F = M_G$ *has exactly one root* T *in* $[1, h_* N]$, *for any fixed* h_* *with* $h_* \max_i q_i < 1$, *and it satisfies*

$$\delta T + \frac{\Delta}{2}\log T = \Delta \log N - \log\frac{K_F}{K_G} + O\Big(\frac{1}{\log N}\Big).$$

If F *and* G *are only admissible, every root in* $[T_1, N^{1/3}]$ *satisfies this equation with* $(p_F - p_G)\log T$ *in place of* $\frac{\Delta}{2}\log T$ *and* K^{red} *in place of* K.

Part (i) is Theorem 4.2(ii)–(iv) and part (ii) is Theorem 5.8 and Corollary 4.6. The face maxima of Theorem A keep every root in a window of order $\log N$, and there the uniqueness comes from a derivative of the local law in T (Lemma A.5). Theorem 4.4 reads the winners in (i) off the lower-left boundary of the convex hull of the points $(\lambda_F, |F| - 1)$. So as τ grows the winning size can only grow. Solving the equation in (ii) gives $T = \frac{\Delta}{\delta}[\log N - \frac{1}{2}\log\log N + O(1)]$. The coefficient $\frac{1}{2}$ of $\log\log N$ holds for every such pair, whatever the graph, the rates, the sizes or the start, and without reversibility. For admissible faces it is $(p_F - p_G)/\Delta$, which is 1 and then 0 in Example 5.11. If Q has symmetric support or μ full support, and at each switch of (i) only two values of $(\lambda_F, |F|)$ compete, Corollary 5.10 locates every change of the mode for T of order $\log N$ up to $O(1/\log N)$. When more compete, Theorem 5.9 describes the window of T in which they do.

1.2 The boundary layer

Section 6 follows the occupation density of the continuous-time chain from the inside of a face F up to a facet $G = F \setminus \{i\}$. Let x be the time spent in i, $S = T - x$, and β the profile of the time on G, a point of the open facet. Let π_β^{in} be the law of the state from which the chain tilted to β enters i, and π_β^{out} the law of the state into which it leaves i. The Bessel kernel is $e^{-\gamma x}\sqrt{\omega S/x}\, I_1(2\sqrt{\omega S x})$, where ω and γ are explicit and $\gamma - q_i$ is $\omega/2$ times the effective resistance between π_β^{in} and π_β^{out} (Proposition 6.6). For every irreducible face whose facet satisfies (H), Theorem 6.1 writes the density on F as the density on G times an explicit generalized kernel, with relative error $O(1/T)$ for x up to a small multiple of T. Proposition 6.2 gives the range of the Bessel form at each β. At the β with $\pi_\beta^{\mathrm{in}} = \pi_\beta^{\mathrm{out}}$, which are all β when Q is reversible and some β for some chains that are not, the Bessel kernel holds with relative error $O(1/T + \sqrt{x/T})$, so for $x = o(T)$. At a β where the rate has a term beyond the Bessel exponent, in powers of $\sqrt{x/S}$, let $m \geq 3$ be the order of the first one. There the relative Bessel form holds if and only if $x = o(T^{1-2/m})$. We do not know whether such a term exists whenever $\pi_\beta^{\mathrm{in}} \neq \pi_\beta^{\mathrm{out}}$. For the cycle on three states with rates 2 in one direction and 1 in the other, at the uniform profile on a facet of 2 states, $m = 3$, and the logarithm of the ratio to the Bessel form is $\frac{49}{64\sqrt{\omega}}\, x^{3/2} S^{-1/2}$ to leading order. For a pair face the kernel is the facet form of Pedler's formula, which we prove with an explicit error for all x (Proposition 6.5). The proof rests on a local law for a weighted chain on the facet with a complex parameter (Theorem 6.8). We prove it on a torus through a complex critical point, with a Laplace lemma for a complex Hessian. The counts follow the same law. With n_i visits to i, $P_N(n)$ is the probability of the counts on G in $N - n_i$ steps times a lattice kernel, the generalized kernel with the Laguerre polynomials of Section 8 in place of the Bessel function, with the extra relative error T^2/N for T up to a small multiple of $N^{1/2}$ (Theorem 6.12). The lattice kernel is T/N times the generalized kernel up to $O(T^2/N)$, so the Bessel forms hold for the counts as well. For a pair face the discrete Pedler formula is exact, and we compare it with the lattice kernel at every T (Theorem 6.11). For three or more states the counts are open when T^2/N is of order one.

1.3 An example, the complete graph

On the complete graph with the uniform start, $Q = J - dI$ with J the all-ones matrix, every face ties at first order and the constants decide. A face of k states has $K_F = \frac{1}{d} k^{k+1/2} (4\pi)^{-(k-1)/2}$, and the k^{k-2} spanning trees of K_k enter through it. Put $a_k = \frac{1}{2}\log(4\pi) - \frac{k+1/2}{k-1}\log k$, write V_N for the probability of a vertex, and call a composition balanced if its positive coordinates differ by at most 1. The next theorem collects results of our preprint [39], which this paper supersedes. Here they come out of the general theory where it applies, and Table 1 says which parts use the exact arguments of the preprint.

Theorem C (the complete graph [39]) *Let $Q = J - dI$ and let the start be uniform.*

(i) Let F be a face of $2 \le k \le d$ states. For every $N \ge k$ the equation $M_F = V_N$ has exactly one root $T_{N,k}$ in $(0, N/(d-1))$, and the probability $T_{N,k}/N$ of a switch to a given state decreases strictly in N. As $N \to \infty$,

$$T_{N,k} + \tfrac{1}{2}\log T_{N,k} = \log N + a_k + O\Big(\frac{1}{\log N}\Big).$$

(ii) For every N and every $0 < T \le N/d$, the most likely compositions with a given support are exactly the balanced ones.

(iii) Since $a_2 > a_3 > \cdots > a_d$, for large N and T up to any fixed multiple of $\log N$ the mode changes exactly once, from the d vertices straight to the balanced compositions with d positive coordinates.

Part (i) is Theorem 7.4, part (ii) is Theorem 7.5 and part (iii) is Corollary 7.6. The uniform start matters in (ii). For $d = 2$, $N = 4$ and the start in state 1, the most likely composition with support $\{1, 2\}$ is $(3, 1)$ at $T = \frac{1}{2}$, 1 and $\frac{3}{2}$, since the three compositions $(3, 1)$, $(2, 2)$ and $(1, 3)$ have the probabilities $63, 57, 49$ over 512 at $T = \frac{1}{2}$, $15, 13, 9$ over 64 at $T = 1$ and $165, 147, 75$ over 512 at $T = \frac{3}{2}$. Figure 1 shows the window of (iii).

1.4 Related work

The exact law of the count vector goes back to Whittle [44], Dawson and Good [16], Goodman [24] and Good [23]. For two states it is the Markov binomial law of Gabriel [22]. Dobrushin [19] classified the limit laws of the number of visits of a two-state chain whose switch probabilities scale with the length, and for switch probabilities of order $1/N$ the limit is not Gaussian. This is the case $d = 2$ of our regime. Čekanavičius and Vellaisamy [11] give compound Poisson approximations to the Markov binomial law and survey the work since Dobrushin. Stadje [43] finds the joint laws of the numbers of visits and their limits when one state becomes nearly absorbing, with the transition probabilities out of it of order $1/N$. Here every row of the transition matrix degenerates at the same rate. Good proposed a saddle-point analysis of his formula (10) and did not carry it out [23, pp. 42–43], and Theorem 5.4(a) is that analysis, uniform in T and with its constant. Part (b) does the same for his

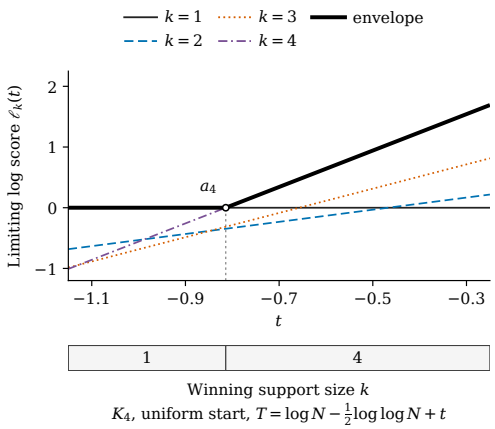


Fig. 1 Support selection on the complete graph with four states, unit rates and uniform start, in the window $T = \log N - \frac{1}{2}\log\log N + t$. Thin lines are limiting log face maxima relative to a vertex, and the bold curve is their upper envelope. The winner jumps from one state to all four at $t = a_4$. The scores are derived in Section 7, and the marked tie concerns the limit, not a finite-N equality

discrete formula (3), uniformly in $T/N \le h_*$. Polettini [42] and Carugno, Vivo and Coghi [10] derive the exact law from the BEST and matrix-tree theorems.

For a fixed chain the occupation counts form a Markov additive process, and its large deviations are classical. Miller [33] and Ney and Nummelin [34, 35] obtain the rates from a Perron eigenvalue, Kontoyiannis and Meyn [30, Theorem 6.3] give exact large deviation asymptotics for a real functional, and Hervé and Ledoux [26] prove a local limit theorem for densities of the additive component. These results concern a fixed transition matrix, and their constants depend on it. For instance, Theorems 6.3 and 6.5 of [30] give the exact asymptotics of $\mathbb{P}\{S_n \ge nc\}$ as $n \to \infty$, where S_n is the partial sum of a real functional of one geometrically ergodic chain, in the non-lattice and the lattice case, with no rate in n and no uniformity over kernels. Here the matrix $I + hQ$ tends to the identity and its spectral gap T/N tends to 0, so these results give no bound uniform in N. Theorem 5.4(b) is an exact local large deviation estimate for the whole count vector of the chain killed on leaving F, with an explicit constant, and the new point is that it is uniform as $I + hQ \to I$. At $h = 0$ it joins Theorem 5.4(a), the same estimate for the occupation times of the continuous-time chain. We found no statement in these sources that covers either, and none that is uniform as the chain degenerates.

In continuous time, Brydges, van der Hofstad and König [9] give an explicit formula for the joint density of the local times on the event that the range is a given set, and use it for large deviation upper bounds beyond the exponential scale. On the event that the range up to time T is F, Lemma 2.4 gives this density, mixed over the start and written as Good's run expansion, and Theorem 5.4(a) gives its exact asymptotics on compact subsets of the open face. Huang, Kious, Sidoravicius and Tarrès [27] write the joint density of the local times, the last exit tree and the cycling numbers of a chain with symmetric rates as a product of Bessel functions, one for each edge. Pedler's occupation-time density for two-state chains [37] and the Bessel forms of Kovchegov,

Meredith and Nir [31] for the occupation time of one state are exact formulas for a fixed chain too. In the boundary layer the Bessel kernel arises differently. It is the large-T form of the contribution of the visits to one state, after the facet is integrated out by a local law, and without $\pi^{\mathrm{in}} = \pi^{\mathrm{out}}$ it can fail beyond x of order $T^{1/3}$.

The rate inside a face is the Donsker and Varadhan functional of the killed chain [12, 20, 21]. Guillin, Nectoux and Wu [25] prove a large deviation principle for the occupation measure of a process conditioned to stay in a set. The quasi-ergodic law goes back to Darroch and Seneta [14, 15], and Collet, Martínez and San Martín [13] is the standard reference. Dietz and Sethuraman [17, 18], Bouguet and Cloez [8] and Benaïm, Bravo and Faure [2] study chains whose switching decays along the trajectory, while here the rate is fixed during each path and changes with N. Bertini, Gabrielli and Landim [6], Landim [32] and Kim and Landim [29] expand the rate functions of a fixed chain in a parameter, the last for the joint rate of the empirical measure and the flow [4, 5]. Here the functional enters through its restriction to each face, and faces of different dimension compete.

The two-state walk is treated at every N in [38]. Our preprint [39] on the complete graph is superseded by this paper. Table 1 lists the results of the preprint that we reprove, where, which part, and how. The other results of [39] are not reproved here, namely Proposition 2.1, Proposition 3.7, Sections 4 and 5 on a start and a field, Theorem 6.2, Remark 6.3 and Appendix C. A companion paper [41] asks what the Donsker and Varadhan functional and the modal supports determine about the generator.

Organization. Section 2 sets up the exact law and Section 3 the killed functional, in continuous and in discrete time. Section 4 proves first-order face selection and gives the face maxima of reducible faces. It uses the local law and the face maxima of Section 5, whose proofs do not use Section 4. Section 6 treats the boundary layer, Sections 7 and 8 the complete graph, and Section 9 the checks and the open questions. The arXiv version of this paper [40] numbers its sections and statements as here, and it has every proof. Where a proof is only there, the text says so in one sentence.

2 Setting, notation and the exact law

2.1 Setting and notation

The chain. Fix $d \geq 2$ and a generator Q on $[d] = \{1, \dots, d\}$. Its rates are $q_{ij} \geq 0$ for $i \neq j$, the exit rates are $q_i = \sum_{j \neq i} q_{ij}$, $Q_{ii} = -q_i$, and $q_{\max} = \max_i q_i$. We also write q for the vector (q_i). The start μ is a probability law on $[d]$, and $\mu(F) = \sum_{i \in F} \mu_i$. We write $V = [d]$ for the face of all states. For $N \geq 2$ and $T > 0$ put

$$h = \frac{T}{N}, \qquad L = \log N, \qquad \tau = \frac{T}{L}.$$

The chain $X_1, \dots, X_N$ has $X_1 \sim \mu$ and transition matrix $I + hQ$. So at each step it jumps from i to j with probability hq_{ij}, and N steps look like the continuous-time chain with generator Q run for time T. We assume $hq_{\max} \leq \frac{1}{2}$ unless we say otherwise. The *composition* is the vector of visit counts $n = (n_1, \dots, n_d)$, with $n_i = \#\{t \leq N :$

Table 1 Results of the preprint [39] reproved here. "General" means that the proof is a case of the general theory of Sections 2 to 5, and "as in [39]" that it is the exact argument of the preprint

Item of [39]	Result here	Part covered	Method
Prop. 2.2	Lemma C.1, (2), (20)	all	as in [39], with the run form from Lemma 2.3
Thm. 3.1	Cor. 7.3, Thm. 7.4(ii), (iv)	(ii) to (iv), and (i) without the term $-(k-1)/(8T)$	general, and (iv) as in [39]
Thm. 3.2	Thm. 7.4(i), (iii)	all but the term $1/(8T_{N,k})$ and the third-order expansion	(i) as in [39], and (iii) general
Thm. 3.3	Thm. 5.6, Ex. 7.2	the local law (7), not the root (8)	general
Lemma 3.5, Prop. 3.6	Lemma 7.1, Prop. 5.2(b), Ex. 7.2	all but Prop. 3.6(5)	general, and Lemma 3.5 as in [39]
Thm. 3.8	Thm. 7.5	all	as in [39]
Cor. 3.9	Cor. 7.6	all	as in [39], with Thm. 7.4(ii)
Ex. 3.10	Rem. 7.7	all	as in [39]
Lemma 6.1	Lemma 8.1	all	as in [39]
Thm. 6.4	Thm. 8.2	all	as in [39]

$X_t = i\}$, so that $n_1 + \cdots + n_d = N$. We write $P_N(n)$ for the probability that the chain has composition n, $\operatorname{supp} n = \{i : n_i > 0\}$ for its support, and $\alpha = n/N$ for the visit fractions.

Faces. A face is a nonempty set $F \subseteq [d]$, of size $k = k_F = |F|$. Let Δ_F be the set of probability vectors on $[d]$ that vanish off F, and Δ_F° the set of those that are positive on F. We number the states of F as $1, \dots, k$ and use the coordinates $(\alpha_2, \dots, \alpha_k)$ on Δ_F and $(t_2, \dots, t_k)$ on $\{t \in \mathbb{R}^F : t_1 + \cdots + t_k = T\}$. We write $x \cdot y = \langle x, y\rangle = \sum_{i\in F} x_i y_i$ and $x \circ y = (x_i y_i)_{i\in F}$. Let $A_F = (q_{ij})_{i,j\in F}$ with zero diagonal, and $Q_F = A_F - \operatorname{diag}(q_i)_{i\in F}$. Since q_i includes the rates of leaving F, Q_F generates the chain killed when it leaves F. The switching graph G_F has an edge $i \to j$ whenever $i, j \in F$, $i \neq j$ and $q_{ij} > 0$.

Perron roots. A square matrix B is Metzler if its off-diagonal entries are nonnegative, and then its Perron root $\varrho(B)$ is its largest real eigenvalue. If B is irreducible, $\varrho(B)$ is simple and has positive left and right eigenvectors [3].

Definition 2.1 (faces) Let F be a face. We call F *irreducible* if G_F is strongly connected. In particular every singleton is irreducible. We call F *admissible* if for some N some path $x_1 \cdots x_N$ of positive probability has support exactly F (for $0 < hq_{\max} < 1$ this does not depend on h). We say that F satisfies (H) if F is irreducible and $\mu(F) > 0$. The *exit rate* of F is $\lambda_F = -\varrho(Q_F)$, so $\lambda_{\{i\}} = q_i$. If F is irreducible, r_F and l_F are right and left eigenvectors of Q_F for $-\lambda_F$ with positive entries and $\langle l_F, r_F\rangle = 1$, and $\alpha_F^* = l_F \circ r_F$ is the *quasi-ergodic law* of F [14, 15]. The *face maximum* is $M_F = \max\{P_N(n) : \operatorname{supp} n = F\}$, with $\max \emptyset = 0$.

We call λ_F the exit rate for the following reason. If F is irreducible, then $e^{TQ_F} r_F = e^{-T\lambda_F} r_F$ with r_F positive. So the chance that the continuous-time chain started in F stays in F up to time T lies between $e^{-T\lambda_F} \min r_F / \max r_F$ and $e^{-T\lambda_F} \max r_F / \min r_F$. If F is reducible, we order the strongly connected components $C_1, \dots, C_r$ of G_F so that every edge between components goes forward. Then Q_F is block triangular with diagonal blocks Q_{C_j}, and $\lambda_F = \min_j \lambda_{C_j}$. In all cases $0 \le \lambda_F \le \min_{i \in F} q_i$, since the rows of Q_F have sums at most 0 and a Perron root is at least every diagonal entry. For a face F satisfying (H), α^*_F is the long-run law of the visit fractions of the chain conditioned to stay in F.

Conventions. Unless we say otherwise, Q, μ and the faces under discussion are fixed, and every constant, every implied constant in $O(\cdot)$ and every threshold N_0 depends only on them and on the compact set $\mathcal{K}$ when one appears. A bound that holds "uniformly for $1 \le T \le N^{1/3}$" holds for all $N \ge N_0$ and all T in that range with one implied constant. Some results fix h_* with $h_* q_{\max} < 1$ and hold uniformly for $1 \le T \le h_* N$. There $h \le h_*$ replaces $h q_{\max} \le \frac{1}{2}$, and the constants also depend on h_*. Subscripts as in $O_{\mathcal{K}}$ stress a dependence.

2.2 Admissible faces

Lemma 2.2 (admissible faces) *Let F be a face and $C_1, \dots, C_r$ the strongly connected components of G_F.* (a) *F is admissible if and only if the components can be numbered so that $\mu(C_1) > 0$ and, for each $j < r$, $q_{xy} > 0$ for some $x \in C_j$ and $y \in C_{j+1}$.* (b) *An irreducible face is admissible if and only if $\mu(F) > 0$. So* (H) *holds exactly for the irreducible admissible faces.* (c) *If Q has symmetric support, that is $q_{ij} > 0$ exactly when $q_{ji} > 0$, then every admissible face is irreducible.*

Proof We translate paths into walks in G_F. Take a path of positive probability with support F and list the states of its maximal runs. This list is a walk in G_F that starts at a state of $\operatorname{supp} \mu$ and visits every state of F. Conversely, every such walk $v_1, \dots, v_J$ is the list of runs of a path of positive probability with support F, as soon as $N \ge J$. The path follows the walk and then stays in its last state, and a stay has positive probability. So F is admissible exactly when G_F has a walk that starts in $\operatorname{supp} \mu$ and visits every state of F.

(a) Suppose that there is such a walk. Once it has left a component C for a state $y \notin C$, it cannot come back to C. Otherwise y could be reached from C and C from y, and y would belong to C. Hence the walk visits each component in one block of consecutive times, and it passes from one block to the next along a single edge. We number the components in the order of these blocks. The walk starts in C_1, so $\mu(C_1) > 0$, and the edge from block j to block $j+1$ is an edge $x \to y$ with $x \in C_j$ and $y \in C_{j+1}$. This is the condition in (a).

Conversely, suppose that the components are numbered as in (a), with edges $x_j \to y_{j+1}$ from C_j to C_{j+1} for $j < r$. Take $y_1 \in C_1$ with $\mu_{y_1} > 0$ and any $x_r \in C_r$. Since C_j is strongly connected, there is a walk inside C_j from y_j to x_j that visits every state of C_j. We join these r walks by the edges $x_j \to y_{j+1}$. The result is a walk that starts in $\operatorname{supp} \mu$ and visits every state of F.

(b) This is (a) with $r = 1$.

(c) If Q has symmetric support, an edge from one component to another would come with an edge back, and the two components would be the same. So no edge joins two components, and the condition in (a) forces $r = 1$. □

So $M_F > 0$ for large N if F is admissible, and $M_F = 0$ otherwise.

2.3 The exact law

For $m \in \mathbb{Z}^F_{\geq 1}$ put $M = \sum_{i \in F} m_i$ and $W(m) = \sum_c \mu_{c_1} \prod_{j<M} q_{c_j c_{j+1}}$, the sum over the words $c = c_1 \cdots c_M$ in F with $c_j \neq c_{j+1}$ in which each $i \in F$ occurs m_i times. Such a word lists the states of the runs of a path, and m_i is the number of runs in state i.

Lemma 2.3 (run expansion) *Let $hq_{\max} \leq 1$ and let n have support F. Then, with $\binom{n_i-1}{m_i-1} = 0$ for $m_i > n_i$ and $0^0 = 1$,*

$$P_N(n) = \sum_{m \in \mathbb{Z}^F_{\geq 1}} W(m)\, h^{M-1} \prod_{i \in F} \binom{n_i - 1}{m_i - 1} (1 - hq_i)^{n_i - m_i}.$$

This is Good's coefficient identity [23, (3)], with the exponents $n_i - 1$, written for the chain killed on leaving F.

Proof The states $c_1 \cdots c_M$ of the maximal runs of a path with composition n form a word as in $W(m)$. The path starts at c_1, of probability μ_{c_1}, it has $M - 1$ switches, of probabilities $hq_{c_j c_{j+1}}$, and $n_i - m_i$ stays in state i, of probability $1 - hq_i$ each. The number of paths with the word c and composition n is $\prod_i \binom{n_i-1}{m_i-1}$, the number of ways to write each n_i as an ordered sum of m_i positive run lengths. Summing over the words c with given m gives $W(m)$, and summing over m gives the claim. □

For $F = \{i\}$ the only path is i^N, so uniformly for $1 \leq T \leq N^{1/2}$

$$M_{\{i\}} = \mu_i (1 - hq_i)^{N-1} = \mu_i e^{-q_i T} \big(1 + O(T^2/N)\big). \tag{1}$$

For $t \in (0, \infty)^F$ let $H(t) = \sum_m W(m) \prod_{i \in F} t_i^{m_i - 1}/(m_i - 1)!$, which is finite since $W(m) \leq k^M q_{\max}^{M-1}$. In the continuum limit the binomial coefficients of Lemma 2.3 become the powers $t_i^{m_i-1}/(m_i - 1)!$, and Lemma 5.5 below shows that the discrete law is close to the following density for $T \leq N^{1/2}$.

Lemma 2.4 (the continuum density) *Let $(Y_s)_{s \geq 0}$ be the continuous-time chain with generator Q and $Y_0 \sim \mu$, let $\mathcal{O}_i = \int_0^T \mathbf{1}\{Y_s = i\}\, ds$, and let E_F be the event that Y stays in F during $[0, T]$ and visits every state of F. Then $\mathbb{P}(E_F,\ \mathcal{O} \in \cdot\,)$ has the density $e^{-\langle q, t\rangle} H(t)$ in $(t_2, \ldots, t_k)$ on $\{t \in (0, \infty)^F : \sum_i t_i = T\}$, and $\mathbb{P}(E_F,\ \mathcal{O}/T \in \cdot\,)$ has the density*

$$f_T(\alpha) = T^{k-1} e^{-T\langle q, \alpha\rangle} H(T\alpha)$$

in $(\alpha_2, \ldots, \alpha_k)$. For n with support F and $t = hn$, $N^{-(k-1)} f_T(n/N) = h^{k-1} e^{-\langle q, t\rangle} H(t)$. For $F = \{i\}$, $f_T = \mathbb{P}(E_F) = \mu_i e^{-q_i T}$.

The density is Good's formula [23, (10)] for the killed chain, whose diagonal keeps the full exit rates q_i. The proof checks the exponents $m_i - 1$.

Proof Up to a null set, E_F is the disjoint union of the events E_c that Y visits exactly $c_1, \dots, c_M$ in this order during $[0,T]$, over the words c in the sums $W(m)$. Fix c, and let s_j be the time that Y spends in its j-th run, so that $s_1 + \cdots + s_M = T$. For $j < M$ the run has the density $q_{c_j} e^{-q_{c_j} s_j}$ and ends with a jump to c_{j+1} of probability $q_{c_j c_{j+1}}/q_{c_j}$, and the last run survives with probability $e^{-q_{c_M} s_M}$. So on $\{s_j > 0 \ \forall j\}$ the event E_c has the density $\mu_{c_1} \prod_{j<M} q_{c_j c_{j+1}}\, e^{-\langle q, \mathcal{O}\rangle}$ in $(s_1, \dots, s_{M-1})$, where $\mathcal{O}_i$ is the sum of the s_j with $c_j = i$. For each $i \neq c_M$ we replace the last holding time in state i by $\mathcal{O}_i$, a triangular change of variables with unit diagonal. With $\mathcal{O} = t$ fixed, the $m_i - 1$ other holding times in state i are positive with sum less than t_i, so they range over a simplex of volume $t_i^{m_i-1}/(m_i-1)!$. Integrating them out gives the density $\mu_{c_1} \prod_{j<M} q_{c_j c_{j+1}}\, e^{-\langle q,t\rangle} \prod_i t_i^{m_i-1}/(m_i-1)!$ in the coordinates t_i with $i \neq c_M$, and any $k-1$ of the coordinates give the same Lebesgue measure on $\{\sum_i t_i = T\}$. Summing over c and m gives $e^{-\langle q,t\rangle} H(t)$, and $t = T\alpha$ has Jacobian T^{k-1}. For $F = \{i\}$ the event E_F says that Y starts at i and does not jump before time T. □

Example 2.5 (two models) (i) For $d = 2$, $q_{12} = q_{21} = 1$ and uniform μ the chain gives the directions of the persistent random walk on $\mathbb{Z}$ of [38], which repeats its previous step with probability $p = 1 - h$. So $T = N(1-p)$, and the faces $\{1\}$ and $\{2\}$ are the endpoints. (ii) For $Q = J - dI$, with J the all-ones matrix, the chain switches to each other state with probability h and keeps its state with probability $p = 1 - (d-1)h$. With uniform μ it is the model of [39], whose switch probability r is our h. A word $x_1 \cdots x_N$ then has probability $\frac{1}{d} h^{N-1} (p/h)^{\#\{s<N:\, x_s = x_{s+1}\}}$, so the path law is the one-dimensional d-state Potts chain with free boundaries [45]. Here $q_i = d - 1$ for every i, so Lemma 2.3 reads

$$P_N(n) = p^{N-1} \sum_{a \in F} \mu_a S_n^{(a)}(u), \qquad u = \frac{h}{p}, \tag{2}$$

for $p > 0$ and $\operatorname{supp} n = F$, where $S_n^{(a)}(u)$ is the sum of u^j over the words with the counts n that start with a and have j switches. Indeed such a word has probability $\mu_a p^{N-1-j} h^j = \mu_a p^{N-1} u^j$. In both models Q has symmetric support and μ full support, so all admissible faces satisfy (H) by Lemma 2.2.

Figure 2 shows exact support competition for the three-state path. There λ_F is 1 for an end, $(3 - \sqrt{5})/2$ for a pair of neighbors and 0 for all three states, so to first order the mode moves at $\tau = (1 + \sqrt{5})/2$ and at $\tau = (3 + \sqrt{5})/2$, that is, at $T = 7.3$ and $T = 11.8$ for $N = 90$. At $N = 90$ it moves near $T = 4.9$ and $T = 8.5$. Corrections of this size are the subject of Section 5.

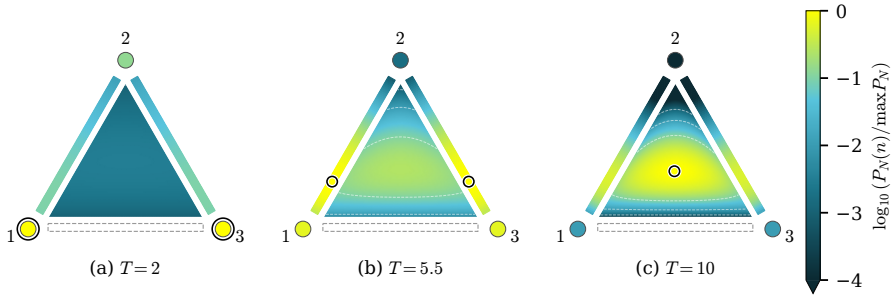
3 The killed functional

Every rate in this paper is the restriction of one functional, the Donsker and Varadhan occupation functional of the chain.

3.1 The restriction identity

For a probability vector α on $[d]$ let

$$J_V(\alpha) = \sup_{u \in (0,\infty)^d} \sum_{i \in [d]} \alpha_i \Big(-\frac{(Qu)_i}{u_i} \Big) = \sup_{u \in (0,\infty)^d} \sum_{i \in [d]} \alpha_i \Big(q_i - \sum_{j \neq i} q_{ij} \frac{u_j}{u_i} \Big), \tag{3}$$



(a) $T=2$ (b) $T=5.5$ (c) $T=10$

Fig. 2 The exact law of the composition for the path $1-2-3$ (rate 1 on each edge, uniform start, $N=90$) on the simplex of visit fractions, with the faces drawn apart. The triangle holds the compositions with three states, each band those with two, and each disk a vertex. The color is the probability relative to the mode, the thin lines are spaced by a factor of ten, and the rings mark the modes. The mode is (a) at an end of the path, (b) on a pair of neighbors and (c) inside. The dashed band $\{1,3\}$ has probability zero

the Donsker and Varadhan rate of the empirical measure of the continuous-time chain with generator Q [20]. For a face F and $\alpha \in \Delta_F$ let

$$J_F(\alpha) = \sup_{u\in(0,\infty)^F} \sum_{i\in F} \alpha_i \Big(-\frac{(Q_F u)_i}{u_i}\Big) = \sup_{u\in(0,\infty)^F} \sum_{i\in F} \alpha_i \Big(q_i - \sum_{j\in F\setminus\{i\}} q_{ij}\frac{u_j}{u_i}\Big),$$

the same functional for the chain killed on leaving F. Let $q_i^{\mathrm{out}}(F) = \sum_{j\notin F} q_{ij}$. Taking $u = \mathbf{1}$ gives $J_F(\alpha) \geq \sum_i \alpha_i q_i^{\mathrm{out}}(F)$, and dropping the terms with q_{ij} gives $J_F(\alpha) \leq \sum_i \alpha_i q_i$.

Proposition 3.1 (the restriction identity) *For every face F and every $\alpha \in \Delta_F$, $J_F(\alpha) = J_V(\alpha)$. Hence $J_F = J_{F'}$ on $\Delta_{F'}$ for every nonempty $F' \subseteq F$.*

Proof Fix $\alpha \in \Delta_F$ and $u \in (0,\infty)^d$. The terms of (3) with $i \notin F$ vanish, since $\alpha_i = 0$ there. For $i \in F$ the terms with $j \notin F$ are nonnegative and are subtracted, so

$$\sum_{i\in F} \alpha_i \Big(q_i - \sum_{j\neq i} q_{ij}\frac{u_j}{u_i}\Big) \leq \sum_{i\in F} \alpha_i \Big(q_i - \sum_{j\in F\setminus\{i\}} q_{ij}\frac{u_j}{u_i}\Big) \leq J_F(\alpha).$$

So $J_V(\alpha) \leq J_F(\alpha)$. Conversely, let $w \in (0,\infty)^F$ and let u equal w on F and ε off F. The terms with $j \notin F$ are then $-q_{ij}\varepsilon/w_i$, which tend to zero as $\varepsilon \downarrow 0$, and the supremum over w gives $J_V(\alpha) \geq J_F(\alpha)$. The second claim is the first for F and for F'. □

So the cost of a point of Δ_F does not depend on the face in which we read it.

3.2 The variational formula

For a face F and a field $v \in \mathbb{R}^F$ let

$$\lambda_F(v) = -\varrho\big(Q_F + \operatorname{diag} v\big), \tag{4}$$

so that $\lambda_F(0) = \lambda_F$. If F is irreducible, let r and l be positive right and left Perron vectors of $Q_F + \operatorname{diag} v$ with $l \cdot r = 1$, and put $\alpha_F^*(v) = l \circ r$, so that $\alpha_F^*(0) = \alpha_F^*$. Since $\sum_i \alpha_i v_i = v \cdot \alpha$,

$$J_F(\alpha) - v \cdot \alpha = \sup_{u \in (0,\infty)^F} \sum_{i \in F} \alpha_i \Big(- \frac{((Q_F + \operatorname{diag} v)u)_i}{u_i} \Big). \tag{5}$$

So a field replaces the diagonal q_i by $q_i - v_i$ and changes nothing else.

Lemma 3.2 (monotonicity in the face) *If $F \subsetneq G$ are faces and Q_G is irreducible, then $\lambda_G(v_G) < \lambda_F(v_F)$ for every field $v \in \mathbb{R}^G$. Now let Q be irreducible. Then $\lambda_V(v) < \lambda_F(v_F)$ for every proper face F, and for every fixed $v \in \mathbb{R}^d$ there is $\gamma(v) > 0$ with $\lambda_F(v_F) \geq \lambda_V(v) + \gamma(v)$ for all proper F. So the score $\Phi_{\tau,v}(F) = |F| - 1 + \tau \lambda_F(v_F)$ is least at V for $\tau > (d-1)/\gamma(v)$.*

The last two claims need Q irreducible. For $d = 2$, $q_{12} = 1$ and $q_{21} = 0$ we have $\lambda_V = 0 = \lambda_{\{2\}}$.

Proof Choose c with $B = Q_G + \operatorname{diag} v_G + cI \geq 0$. Its principal submatrix on F is $Q_F + \operatorname{diag} v_F + cI$, since the diagonal of a killed generator keeps the full exit rates. Let $\widetilde{B}$ be this submatrix padded with zeros to $G \times G$. Then $0 \leq \widetilde{B} \leq B$, and $\widetilde{B} \neq B$ because the irreducible matrix B has a positive entry in a row of F and a column off F. So $\varrho(\widetilde{B}) < \varrho(B)$ [3, Ch. 2], and $\varrho(\widetilde{B})$ is the Perron root of the submatrix on F, since the padding adds only the eigenvalue 0. This holds whether or not F is irreducible, and subtracting c gives the first claim. If Q is irreducible, the first claim with $G = V$ gives the second. For fixed v there are finitely many proper faces, so $\gamma(v)$, the least of the numbers $\lambda_F(v_F) - \lambda_V(v)$, is positive. Finally $\Phi_{\tau,v}(F) - \Phi_{\tau,v}(V) = |F| - d + \tau(\lambda_F(v_F) - \lambda_V(v)) \geq -(d-1) + \tau\gamma(v)$. □

Proposition 3.3 (the rate on a face) *Let F be irreducible with $k = |F| \geq 2$, and let $v \in \mathbb{R}^F$.*

(a) On Δ_F, $J_F(\alpha) - v \cdot \alpha \geq \lambda_F(v)$, with equality exactly at $\alpha = \alpha_F^*(v)$. In particular $J_F \geq \lambda_F$, with equality exactly at $\alpha_F^* = l_F \circ r_F$.

(b) The rate J_F is convex on Δ_F and strictly convex on Δ_F°. For $\alpha \in \Delta_F^\circ$ the supremum defining $J_F(\alpha)$ is attained at a vector $u = r_\alpha$, which is unique up to a positive factor, and α is the stationary law of the generator on F with off-diagonal entries $q_{ij} r_{\alpha,j} / r_{\alpha,i}$.

(c) If there is $\nu \in (0,\infty)^F$ with $\nu_i q_{ij} = \nu_j q_{ji}$ for all $i, j \in F$, then for every $\alpha \in \Delta_F$

$$J_F(\alpha) = \sum_{\{i,j\} \subseteq F,\ i \neq j} \big(\sqrt{\alpha_i q_{ij}} - \sqrt{\alpha_j q_{ji}} \big)^2 + \sum_{i \in F} \alpha_i q_i^{\mathrm{out}}(F).$$

(d) For every nonempty face $F' \subsetneq F$ and every $\alpha \in \Delta_{F'}$, $J_F(\alpha) - v \cdot \alpha = J_{F'}(\alpha) - v \cdot \alpha \geq \lambda_{F'}(v_{F'}) > \lambda_F(v)$.

For symmetric Q_F, $l_F = r_F$ and α_F^* is the entrywise square of the Perron vector normalized in ℓ^2. On a face that is not reversible the formula in (c) is only an upper bound, by its proof.

Proof Put $B = Q_F + \operatorname{diag} v$, an irreducible Metzler matrix with the switching graph G_F and $\lambda_F(v) = -\varrho(B)$.

Setup. With $u = e^{\psi}$, (5) reads $J_F(\alpha) - v \cdot \alpha = \sup_{\psi} \Psi_\alpha(\psi)$ with

$$\Psi_\alpha(\psi) = \sum_i \alpha_i (q_i - v_i) - \sum_{i \neq j} \alpha_i q_{ij} e^{\psi_j - \psi_i},$$

where i, j run over F. It is concave in ψ, affine in α, and unchanged when a constant is added to ψ. Let G^{ψ} be the conservative generator on F with off-diagonal entries $q_{ij} e^{\psi_j - \psi_i}$. It has the graph G_F, so it has exactly one stationary law, and it does not involve v. Since $\nabla_\psi \Psi_\alpha = -\alpha^\top G^{\psi}$ and a differentiable concave function attains its maximum exactly at its critical points, ψ attains the supremum if and only if α is the stationary law of G^{ψ}. We call this the criterion.

A lower bound on every face. Let F' be any face, irreducible or not, and $B' = Q_{F'} + \operatorname{diag} v_{F'}$. We show $J_{F'}(\alpha) - v \cdot \alpha \geq -\varrho(B')$ on $\Delta_{F'}$. Let $s > \varrho(B')$ and write $B' = B_0 - cI$ with $B_0 \geq 0$. Then $s + c > \varrho(B_0)$, so $u = (sI - B')^{-1}\mathbf{1} = \sum_{\ell \geq 0} (s+c)^{-\ell-1} B_0^{\ell} \mathbf{1}$ converges and is positive. Since $(sI - B')u = \mathbf{1}$, we get $-(B'u)_i / u_i = -s + 1/u_i > -s$ at every state, and letting $s \downarrow \varrho(B')$ gives the claim.

The value at $\alpha^*_F(v)$. Take $\psi = \log r$ and $R = \operatorname{diag} r$. Then $G^{\psi} = R^{-1}(B - \varrho(B)I)R$, since the off-diagonal entries agree and $Br = \varrho(B) r$. So $(l \circ r)^\top G^{\psi} = l^\top (B - \varrho(B)I) R = 0$, and by the criterion ψ attains the supremum at $\alpha^*_F(v)$. Every ratio $-(Br)_i / r_i$ equals $\lambda_F(v)$, so the value is $\lambda_F(v)$.

(b) Convexity holds because J_F is a supremum of affine functions. Let $\alpha \in \Delta_F^\circ$ and take $v = 0$.

The supremum is attained. Normalize $\min_i \psi_i = 0$ and put $\bar{\psi} = \max_i \psi_i$. Let a be the least positive rate q_{ij} in F. A shortest path in G_F from a state where $\psi = 0$ to a state where $\psi = \bar{\psi}$ has at most $k - 1$ edges, so one of its edges $i \to j$ has $\psi_j - \psi_i \geq \bar{\psi}/(k-1)$. Keeping only this edge gives $\Psi_\alpha(\psi) \leq \sum_i \alpha_i q_i - a(\min_i \alpha_i) e^{\bar{\psi}/(k-1)}$, which tends to $-\infty$ as $\bar{\psi} \to \infty$. So the continuous function Ψ_α attains its supremum on a bounded set. Write $r_\alpha = e^{\psi}$ for a maximizer. The last claim of (b) is the criterion.

The maximizer is unique up to a factor. If ψ and $\psi + \delta$ both attain the supremum, Ψ_α is constant on the segment between them. There it is a constant minus a sum of the convex functions $\alpha_i q_{ij} e^{\psi_j - \psi_i + s(\delta_j - \delta_i)}$, so each of them is affine in s, and $\delta_j = \delta_i$ on every edge of G_F. As G_F is strongly connected, δ is constant.

Strict convexity. Let $\alpha^0 \neq \alpha^1$ lie in Δ_F°, put $\alpha^s = (1-s)\alpha^0 + s\alpha^1$ with $0 < s < 1$, and let ψ attain $J_F(\alpha^s)$. Since $\Psi_\alpha(\psi)$ is affine in α, $J_F(\alpha^s) = (1-s)\Psi_{\alpha^0}(\psi) + s\Psi_{\alpha^1}(\psi) \leq (1-s)J_F(\alpha^0) + sJ_F(\alpha^1)$. Equality would make α^0 and α^1 both stationary for G^{ψ} by the criterion, which is impossible.

(d) The equality is Proposition 3.1. The lower bound on F' gives $J_{F'}(\alpha) - v \cdot \alpha \geq \lambda_{F'}(v_{F'})$, and Lemma 3.2 with $G = F$ gives $\lambda_{F'}(v_{F'}) > \lambda_F(v)$.

(a) The lower bound and the value at $\alpha^*_F(v)$ are proved above. Let $\alpha \neq \alpha^*_F(v)$. If α lies on the boundary of Δ_F, then (d) with $F' = \operatorname{supp} \alpha$ gives $J_F(\alpha) - v \cdot \alpha > \lambda_F(v)$. If $\alpha \in \Delta_F^\circ$ and equality held, strict convexity would give a value below $\lambda_F(v)$ at $\frac{1}{2}(\alpha + \alpha^*_F(v))$.

(c) Since $q_i = \sum_{j \in F \setminus \{i\}} q_{ij} + q_i^{\mathrm{out}}(F)$, the function inside the supremum is $\sum_i \alpha_i q_i^{\mathrm{out}}(F)$ plus, for each pair $\{i, j\}$, the term $\alpha_i q_{ij}(1 - u_j/u_i) + \alpha_j q_{ji}(1 - u_i/u_j)$. By the AM–GM inequality this term is at most $(\sqrt{\alpha_i q_{ij}} - \sqrt{\alpha_j q_{ji}})^2$, which gives the upper bound. For $\alpha \in \Delta_F^\circ$ the vector $u_i = \sqrt{\alpha_i / \nu_i}$ gives $\alpha_i q_{ij} u_j / u_i = \alpha_j q_{ji} u_i / u_j$ by $\nu_i q_{ij} = \nu_j q_{ji}$, so there is equality for every pair at once. If some α_i vanish, take $u_i = \varepsilon$ for these i and let $\varepsilon \to 0$. A pair with $\alpha_i = 0 < \alpha_j$ then has the term $\alpha_j q_{ji}(1 - \varepsilon/u_j) \to \alpha_j q_{ji}$, which is its bound, and a pair with $\alpha_i = \alpha_j = 0$ has the term 0. □

Proposition 3.4 (the variational formula) *Let F be irreducible. For every $v \in \mathbb{R}^F$,*

$$\lambda_F(v) = \min_{\alpha \in \Delta_F} \big(J_F(\alpha) - v \cdot \alpha \big),$$

and the minimum is attained only at $\alpha_F^*(v)$. Equivalently, $-\lambda_F(v)$ is the Legendre transform of J_F at v, and J_F is the Legendre transform of $v \mapsto -\lambda_F(v)$ on Δ_F°. The map $v \mapsto \lambda_F(v)$ is concave and real-analytic, and $\lambda_F(v + c\mathbf{1}) = \lambda_F(v) - c$. If Q_F is reversible with respect to ν, let $S_F = D_{\sqrt{\nu}}(-Q_F)D_{\sqrt{\nu}}^{-1}$. Then with $\psi_i = \sqrt{\alpha_i}$,

$$J_F(\alpha) - v \cdot \alpha = \sum_{i \in F} (q_i - v_i)\alpha_i - \sum_{i \neq j \in F} \sqrt{q_{ij} q_{ji}}\, \psi_i \psi_j$$

is the Rayleigh quotient of the symmetric matrix $S_F - \operatorname{diag} v$ at the unit vector ψ, and $\lambda_F(v) = \lambda_{\min}(S_F - \operatorname{diag} v)$.

The formula is the finite form of the Donsker and Varadhan formula for the principal eigenvalue [20], close to the characterization of Collatz and Wielandt [3, Ch. 2].

Proof The formula and the unique minimizer are Proposition 3.3(a). As a minimum of affine functions of v, $\lambda_F(v)$ is concave. It is real-analytic because the Perron root of an irreducible Metzler matrix is a simple eigenvalue, and the shift is clear.

Legendre duality. The formula says $-\lambda_F(v) = \max_\alpha (v \cdot \alpha - J_F(\alpha))$. Conversely $J_F(\alpha) \geq v \cdot \alpha + \lambda_F(v)$ for all α and v. Let $\alpha \in \Delta_F^\circ$, take r_α from Proposition 3.3(b), and put $v = q - A_F r_\alpha / r_\alpha$, with the quotient taken entrywise. Then $Q_F + \operatorname{diag} v$ has the off-diagonal entries of Q_F and the diagonal $-A_F r_\alpha / r_\alpha$, so $r_\alpha > 0$ is a null vector of it. Only the Perron root has a positive eigenvector, so $\lambda_F(v) = 0$. Also $J_F(\alpha) = \langle q - A_F r_\alpha / r_\alpha, \alpha \rangle = v \cdot \alpha$, so equality holds at this v.

The reversible form. Reversibility gives $\sqrt{\nu_i}\, q_{ij} / \sqrt{\nu_j} = \sqrt{q_{ij} q_{ji}}$, so S_F is symmetric with diagonal q_i and off-diagonal entries $-\sqrt{q_{ij} q_{ji}}$. Expanding the squares in Proposition 3.3(c) gives $J_F(\alpha) = \langle \psi, S_F \psi \rangle$, and subtracting $v \cdot \alpha = \langle \psi, \operatorname{diag}(v) \psi \rangle$ gives the display. Every unit vector $\psi \geq 0$ arises from $\alpha = \psi \circ \psi$, and the off-diagonal entries of $S_F - \operatorname{diag} v$ are nonpositive, so passing to the entrywise modulus does not increase the quadratic form. Hence the minimum over Δ_F is $\lambda_{\min}(S_F - \operatorname{diag} v)$. □

3.3 The functional of the discrete chain

When T is a fixed fraction of N the counts follow the functional of the discrete chain itself. Let $0 < h < 1/q_{\max}$ and $p_i = 1 - h q_i$, so that $I + hQ$ is a stochastic matrix with positive diagonal. For a face F and $\alpha \in \Delta_F$ let

$$J_F^{(h)}(\alpha) = \frac{1}{h} \sup_{u \in (0,\infty)^F} \sum_{i \in F} \alpha_i \log \frac{u_i}{((I + hQ_F)u)_i},$$

and put $J_F^{(0)} = J_F$. For $F = V$, $h J_V^{(h)}$ is the Donsker and Varadhan rate of the empirical measure of the chain with transition matrix $I + hQ$ [20]. For $\psi \in \mathbb{R}^F$ put $\theta_i(\psi) = \sum_{j \in F \setminus \{i\}} q_{ij} e^{\psi_j - \psi_i}$, and let $L_h(y) = \int_0^1 y\, ds / (1 - shy)$, which is $-\log(1 -$

$hy)/h$ for $h > 0$ and y for $h = 0$. With $u = e^{\psi}$ we have $((I+hQ_F)u)_i/u_i = p_i + h\theta_i(\psi)$, so

$$J_F^{(h)}(\alpha) = \sup_{\psi \in \mathbb{R}^F} G_h(\alpha, \psi), \qquad G_h(\alpha, \psi) = \sum_{i \in F} \alpha_i L_h\big(q_i - \theta_i(\psi)\big),$$

and at $h = 0$ this is the supremum that defines J_F. Since $L_h(y) \geq y$, we have $J_F^{(h)} \geq J_F$. We say that a function on $[0, h_*] \times U$ is real-analytic up to $h = 0$ if near each point it is the restriction of a real-analytic function on an open set.

Proposition 3.5 (the discrete functional) *Fix h_* with $h_* q_{\max} < 1$.*

(a) For every face F, every $0 < h < 1/q_{\max}$ and every $\alpha \in \Delta_F$, $J_F^{(h)}(\alpha) = J_V^{(h)}(\alpha)$. Hence $J_F^{(h)} = J_{F'}^{(h)}$ on $\Delta_{F'}$ for every nonempty $F' \subseteq F$.

(b) Let F be irreducible with $k \geq 2$, and let $(h, \alpha) \in [0, h_*] \times \Delta_F^\circ$. The supremum defining $J_F^{(h)}(\alpha)$ is attained at exactly one $\psi = \psi^{(h)}(\alpha)$ with $\psi_1 = 0$, and there the Hessian of $G_h(\alpha, \cdot)$ in $(\psi_2, \dots, \psi_k)$ is negative definite. The maps $\psi^{(h)}(\alpha)$, $\theta^{(h)}(\alpha) = \theta(\psi^{(h)}(\alpha))$ and $J_F^{(h)}(\alpha)$ are real-analytic on $[0, h_*] \times \Delta_F^\circ$ up to $h = 0$, and $\theta^{(0)}(\alpha) = A_F r_\alpha / r_\alpha$ with r_α from Proposition 3.3(b). The function $J_F^{(h)}$ is convex on Δ_F and strictly convex on Δ_F°.

(c) Let F be irreducible with $k \geq 2$, $0 \leq h \leq h_*$, $\alpha \in \Delta_F^\circ$ and $\psi \in \mathbb{R}^F$, and put $\theta = \theta(\psi)$. Then ψ attains the supremum if and only if the vector $(\alpha_i/(p_i + h\theta_i))_{i \in F}$ is proportional to $l \circ e^{\psi}$, where l is a positive left null vector of $A_F - \operatorname{diag} \theta$.

Proof (a) As in the proof of Proposition 3.1, the terms $hq_{ij}u_j$ with $j \notin F$ only increase $((I + hQ)u)_i$, which gives $J_V^{(h)} \leq J_F^{(h)}$ on Δ_F, and $u = \varepsilon$ off F with $\varepsilon \downarrow 0$ gives the converse, since the diagonal of $I + hQ_F$ keeps the full exit rates. The second claim is the first for F and for F'.

(c) We have $\partial\theta_i/\partial\psi_m = q_{im} e^{\psi_m - \psi_i}$ for $m \neq i$ and $\partial\theta_i/\partial\psi_i = -\theta_i$, and $L_h'(y) = 1/(1 - hy)$. So with $w_i = \alpha_i/(p_i + h\theta_i)$ the gradient of $G_h(\alpha, \cdot)$ is $-w^\top G^\psi$, where G^ψ is the conservative generator of the proof of Proposition 3.3, with off-diagonal entries $q_{ij} e^{\psi_j - \psi_i}$. The function $G_h(\alpha, \cdot)$ is concave by the proof of (b) below. So ψ attains the supremum if and only if w is proportional to the stationary law of G^ψ. With $U = \operatorname{diag}(e^{\psi})$ we have $G^\psi = U^{-1}(A_F - \operatorname{diag} \theta)U$, whose stationary law is proportional to $l \circ e^{\psi}$.

(b) *Concavity.* At $h = 0$ the Hessian of $-G_0$ in a direction δ is $\sum_i \alpha_i \sum_{j \neq i} q_{ij} e^{\psi_j - \psi_i} (\delta_j - \delta_i)^2$. For $h > 0$,

$$-hG_h(\alpha, \psi) = \sum_{i \in F} \alpha_i \log\Big(p_i + \sum_{j \in F \setminus \{i\}} hq_{ij} e^{\psi_j - \psi_i}\Big).$$

The i-th term is α_i times the logarithm of a sum of exponentials of the affine functions 0 and $\psi_j - \psi_i$, with the positive weights p_i and hq_{ij}. Its second derivative in a direction δ is α_i times the variance of the values 0 and $\delta_j - \delta_i$ under the weights proportional to p_i and $hq_{ij} e^{\psi_j - \psi_i}$. As $p_i > 0$, it vanishes only if $\alpha_i = 0$ or $\delta_j = \delta_i$ on every edge $i \to j$ of G_F. So for every $h \in [0, h_*]$ the function $G_h(\alpha, \cdot)$ is concave, and for $\alpha \in \Delta_F^\circ$ its Hessian is negative definite on $\{\delta_1 = 0\}$, since G_F is strongly connected. This is where $h_* q_{\max} < 1$ is used.

The supremum is attained. Let $\psi_1 = 0$ and let $\bar{\psi}$ be the largest difference of two entries of ψ. As in the proof of Proposition 3.3(b), some edge $i \to j$ has $\psi_j - \psi_i \geq \bar{\psi}/(k - 1)$, so

$\theta_i(\psi) \geq a e^{\bar{\psi}/(k-1)}$ with a the least positive rate in F. Since L_h is increasing and tends to $-\infty$ at $-\infty$, the term of i tends to $-\infty$ as $\bar{\psi} \to \infty$, while every term is at most $\alpha_i L_h(q_i) \leq q_{\max}/p_{\min}$, with $p_{\min} = 1 - h_* q_{\max}$. So the supremum is attained, and by strict concavity at exactly one ψ with $\psi_1 = 0$.

Analyticity. The function $L_h(y)$ is real-analytic in (h, y) where $1 - hy > 0$, including $h = 0$ and small negative h, and $1 - h(q_i - \theta_i) \geq p_i > 0$ for $h \in [0, h_*]$. So G is real-analytic near every point (h, α, ψ) with $h \in [0, h_*]$. At the maximizer the gradient in $(\psi_2, \dots, \psi_k)$ vanishes and the Hessian is invertible. The implicit function theorem gives a real-analytic solution near (h, α), which is $\psi^{(h)}(\alpha)$ for $h \geq 0$ by uniqueness. Hence $\theta^{(h)}(\alpha)$ and $J_F^{(h)}(\alpha) = G_h(\alpha, \psi^{(h)}(\alpha))$ are real-analytic up to $h = 0$. At $h = 0$ the maximizer is $\log r_\alpha$, normalized.

Convexity. As a supremum of affine functions, $J_F^{(h)}$ is convex. Let $\alpha^0 \neq \alpha^1$ lie in Δ_F°, let $0 < s < 1$, and let ψ attain $J_F^{(h)}(\alpha^s)$ at $\alpha^s = (1-s)\alpha^0 + s\alpha^1$. Equality in $J_F^{(h)}(\alpha^s) \leq (1-s)J_F^{(h)}(\alpha^0) + sJ_F^{(h)}(\alpha^1)$ would make ψ attain the supremum at α^0 and at α^1. By (c) both would then be proportional to $((p_i + h\theta_i) l_i e^{\psi_i})_{i \in F}$, and as probability vectors they would be equal. □

For a field $v \in \mathbb{R}^F$, $J_F^{(h)}(\alpha) - v \cdot \alpha$ is $1/h$ times the supremum of $\sum_i \alpha_i \log(u_i/(Mu)_i)$ with $M = e^{h \operatorname{diag} v}(I + hQ_F)$, since $-hv_i\alpha_i = \alpha_i \log e^{-hv_i}$. For any face F' and $v \in \mathbb{R}^{F'}$ we put $\lambda_{F'}^{(h)}(v) = -\frac{1}{h} \log \varrho\big(e^{h \operatorname{diag} v}(I + hQ_{F'})\big)$, which is defined since the matrix is nonnegative with a positive diagonal.

Proposition 3.6 (the discrete exit rate) *Let F be irreducible with $k \geq 2$, let $0 < h < 1/q_{\max}$ and $v \in \mathbb{R}^F$, and put $M = e^{h \operatorname{diag} v}(I + hQ_F)$. Let r and l be positive right and left Perron vectors of M with $l \cdot r = 1$, and put $\alpha_F^{*(h)}(v) = l \circ r$.*

(a) On Δ_F, $J_F^{(h)}(\alpha) - v \cdot \alpha \geq \lambda_F^{(h)}(v)$, with equality only at $\alpha_F^{*(h)}(v)$, which lies in Δ_F°. For every nonempty face $F' \subsetneq F$ and every $\alpha \in \Delta_{F'}$, $J_F^{(h)}(\alpha) - v \cdot \alpha \geq \lambda_{F'}^{(h)}(v_{F'}) > \lambda_F^{(h)}(v)$.

(b) At $v = 0$, $\varrho(M) = 1 - h\lambda_F$ and $\alpha_F^{*(h)}(0) = \alpha_F^*$, so $\lambda_F^{(h)}(0) = -\frac{1}{h}\log(1 - h\lambda_F)$ for every h.

(c) Fix h_* with $h_* q_{\max} < 1$, and put $\lambda_F^{(0)}(v) = \lambda_F(v)$ and $\alpha_F^{*(0)}(v) = \alpha_F^*(v)$. Then $\lambda_F^{(h)}(v)$ and $\alpha_F^{*(h)}(v)$ are real-analytic on $[0, h_*] \times \mathbb{R}^F$ up to $h = 0$.

Proof Put $B = (M - I)/h$. Its off-diagonal entries $e^{hv_i} q_{ij}$ are nonnegative and positive exactly on the edges of G_F, so B is an irreducible Metzler matrix. Its entries are real-analytic in (h, v) up to $h = 0$, and $B = Q_F + \operatorname{diag} v$ at $h = 0$. Let $\kappa = \varrho(B)$. Then $\varrho(M) = 1 + h\kappa$, and M and B have the same Perron vectors. As κ is a simple eigenvalue of B, it and the normalized Perron vectors are real-analytic in (h, v) up to $h = 0$, and so is $\lambda_F^{(h)}(v) = -\kappa \int_0^1 ds/(1 + sh\kappa)$. This gives (c). At $v = 0$ we have $B = Q_F$ for every h, which gives (b).

(a) *A lower bound on every face.* Let $F' \subseteq F$ be nonempty, irreducible or not, and let M' be the principal submatrix of M on F', which is $e^{h \operatorname{diag} v_{F'}}(I + hQ_{F'})$ since the diagonal keeps the full exit rates. For $\alpha \in \Delta_{F'}$, part (a) of Proposition 3.5 gives $J_F^{(h)}(\alpha) = J_{F'}^{(h)}(\alpha)$, so $J_F^{(h)}(\alpha) - v \cdot \alpha$ is $1/h$ times the supremum of $\sum_{i \in F'} \alpha_i \log(u_i/(M'u)_i)$ over $u \in (0, \infty)^{F'}$. Let $s > \varrho(M')$. As $M' \geq 0$, the vector $u = (sI - M')^{-1}\mathbf{1} = \sum_{j \geq 0} s^{-j-1} M'^j \mathbf{1}$ is positive, and $M'u = su - \mathbf{1}$. Also $M'u > 0$, since M' has a positive diagonal. So $u_i/(M'u)_i > 1/s$ for

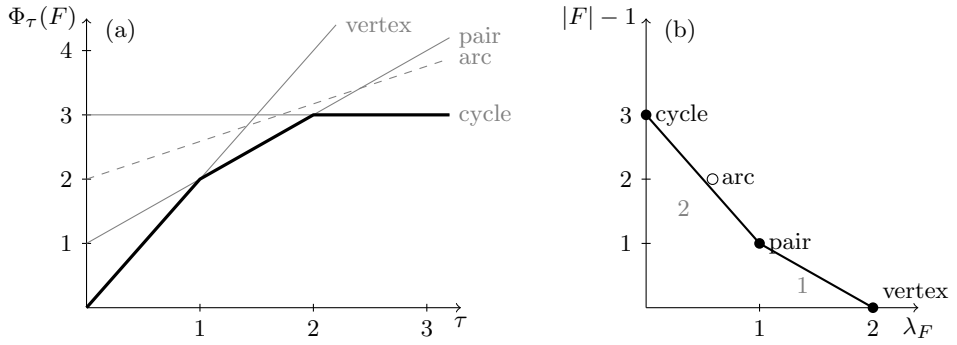


Fig. 3 The 4-cycle. (a) The exponents $\Phi_\tau(F) = \tau\lambda_F + |F| - 1$ of the four kinds of faces against τ, and their lower envelope (bold). The mode sits on the face whose line is lowest, namely a vertex for $\tau < 1$, an adjacent pair for $1 < \tau < 2$ and the whole cycle for $\tau > 2$. The dashed line of the arc of three states is never lowest. (b) The same faces as points $(\lambda_F, |F| - 1)$, with the lower-left boundary of their convex hull. The gray numbers are minus the slopes of its edges, the values of τ where the winner changes (Theorem 4.4). The arc (open) lies above the boundary

every i, and letting $s \downarrow \varrho(M')$ gives $J_F^{(h)}(\alpha) - v \cdot \alpha \geq \lambda_{F'}^{(h)}(v_{F'})$. If $F' \neq F$, then M' padded with zeros is at most M and differs from it, since the irreducible M has a positive entry in a row of F' and a column off F'. So $\varrho(M') < \varrho(M)$ [3, Ch. 2], which gives the second claim. The case $F' = F$ is the lower bound.

Equality. With $u = r$ every ratio $r_i/(Mr)_i$ is $1/\varrho(M)$. The function $\psi \mapsto \sum_i \alpha_i(\psi_i - \log(Me^\psi)_i)$ is concave, and at $\psi = \log r$ its derivative in ψ_m is $\alpha_m - \sum_i \alpha_i M_{im} r_m/(\varrho(M) r_i)$. For $\alpha = l \circ r$ this is $\alpha_m - (l^\top M)_m r_m/\varrho(M) = 0$. So $u = r$ attains the supremum at $\alpha_F^{*(h)}(v)$, and equality holds there. This point lies in Δ_F° since $r, l > 0$.

Uniqueness. On the boundary of Δ_F the inequality is strict by the second claim. In Δ_F°, $J_F^{(h)}(\alpha) - v \cdot \alpha$ is strictly convex by Proposition 3.5(b), so it has only one minimizer. □

4 First-order face selection

4.1 An example with four states on a cycle

Take $d = 4$ states on a cycle, with rate 1 between neighbors and the uniform start. Up to symmetry the admissible faces are a vertex, an adjacent pair, an arc of three states and the whole cycle, with exit rates 2, 1, $2 - \sqrt{2}$ and 0. For the arc, $-Q_F = 2I - A$ with A the adjacency matrix of the path $1-2-3$, whose largest eigenvalue is $\sqrt{2}$. The chain stays in a face F up to time T with probability about $e^{-T\lambda_F}$, and this mass is shared by about $(N/\sqrt{T})^{|F|-1}$ compositions. Hence $M_F \approx N^{-(|F|-1)} T^{(|F|-1)/2} e^{-T\lambda_F}$, and with $T = \tau \log N$ this is $N^{-\Phi_\tau(F)}$ up to powers of $\log N$. Figure 3(a) plots the four exponents. The vertices win for $\tau < 1$, the pairs for $1 < \tau < 2$ and the whole cycle for $\tau > 2$. The arcs never win, since they beat the pairs only for $\tau > 1 + \sqrt{2}$, where the cycle beats them. The whole cycle carries almost all the mass for every $\tau > 0$, and it carries the mode only for $\tau > 2$, because it spreads its mass over about N^3 compositions. Theorems 4.2 and 5.7 make this rule exact, and for a pair Corollary 5.3 gives the constant $1/\sqrt{2\pi}$ in front.

4.2 Face selection

We also need an upper bound on a whole face, including near its boundary, where the local law does not apply. For an irreducible face F put $V_F = \{\theta \in \mathbb{R}^F : \varrho(A_F - \operatorname{diag}\theta) = 0\}$. Each $\theta \in V_F$ gives an exponential bound with the linear rate $\alpha \mapsto \langle q-\theta, \alpha\rangle$, and J_F is the supremum of these linear rates. Indeed, for $u \in (0,\infty)^F$ the entrywise quotient $\theta = A_F u/u$ lies in V_F, since u is a positive null vector of the irreducible Metzler matrix $A_F - \operatorname{diag}\theta$ and only its Perron root has a positive eigenvector. Then $\langle q-\theta, \alpha\rangle$ is the function of u inside the supremum that defines $J_F(\alpha)$. Conversely every $\theta \in V_F$ arises in this way from a positive Perron vector of $A_F - \operatorname{diag}\theta$. For a field $v \in \mathbb{R}^F$ the vector $\theta = q - v - \lambda_F(v)\mathbf{1}$ lies in V_F, since $A_F - \operatorname{diag}\theta = Q_F + \operatorname{diag} v + \lambda_F(v) I$.

Lemma 4.1 (a uniform upper bound) *Let F be irreducible with $k = |F| \geq 2$, and let $\theta \in V_F$. There is C_θ, depending on θ, Q and μ and bounded for θ in compact subsets of V_F, such that for every $n \in \mathbb{Z}^F_{\geq 0}$ with $\operatorname{supp} n = F$ and every h with $hq_{\max} \leq \frac{1}{2}$, the probability that $X_1, \dots, X_{|n|}$ have composition n is at most $C_\theta h^{k-1} \exp\bigl(-h\langle n, q-\theta\rangle\bigr)$.*

The proof is in Appendix A. It writes the probability as an integral over a torus and bounds the integrand by its value at the real point. The lemma holds for a path of any length $|n|$, so it also applies to the blocks of a path. With $\theta = q - v - \lambda_F(v)\mathbf{1}$ and $|n| = N$ it gives

$$P_N(n) e^{Tv\cdot n/N} \leq C_\theta h^{k-1} e^{-T\lambda_F(v)} \tag{6}$$

for every n with support F. The next theorem uses the local law and the face maxima of Section 5, which are proved there without it. Its part (i) is a weak form of Theorem 5.6(b), and its part (ii) is the only part that covers reducible faces.

Theorem 4.2 (first-order face selection) *Fix Q and μ.*

(i) Let F satisfy (H), let $k = |F|$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact. There is N_0 such that, uniformly for $N \geq N_0$, $1 \leq T \leq N^{1/3}$ and $\operatorname{supp} n = F$ with $n/N \in \mathcal{K}$,

$$\log P_N(n) = -(k-1)L - T\, J_F(n/N) + O_{\mathcal{K}}(1 + \log T).$$

(ii) Let F be admissible with components $C_1, \dots, C_r$, and let k^* be the least size of a component C_j with $\lambda_{C_j} = \lambda_F$. There are κ, T_0 and N_0 such that for $N \geq N_0$ and $T_0 \leq T \leq N^{1/3}$

$$\log M_F \geq -(k_F - 1)L + (k_F - k^*)\log T - T\lambda_F - \kappa,$$
$$\log M_F \leq -(k_F - 1)L + (k_F - 1)\log T - T\lambda_F + \kappa.$$

If F is not admissible, $M_F = 0$.

(iii) Let $T = \tau L$ with $\tau > 0$ fixed, and let $\mathcal{F}_\tau$ be the set of admissible faces that minimize Φ_τ. Then $\log \max_n P_N(n) = -L \min_F \Phi_\tau(F) + O(\log L)$, with the minimum over admissible faces, and for large N every global mode $\hat{n}$ has $\operatorname{supp}\hat{n} \in \mathcal{F}_\tau$. This holds for every start μ.

(iv) If in (iii) every face in $\mathcal{F}_\tau$ is irreducible, then every global mode $\hat{n}$ satisfies $\hat{n}/N = \alpha^*_F + O(1/L)$ with $F = \operatorname{supp}\hat{n}$. This is the case when Q has symmetric support or μ has full support (Theorem 4.4(c)).

Proof (i) This is Theorem 5.6(b), since $J_V = J_F$ on Δ_F, $\log C_F$ is bounded on $\mathcal{K}$ and $J_{\{i\}} = q_i$.

(ii) This follows from Theorem 4.5 below, whose proof in Appendix E does not use the present theorem. After T_1 and N_1 are enlarged, that theorem gives $\log M_F = -(k_F-1)L + p_F \log T - T\lambda_F + O(1)$ for $N \ge N_1$ and $T_1 \le T \le N^{1/3}$. Since $T \ge 1$ and $k_F - k^* \le p_F \le k_F - 1$, both bounds hold with $T_0 = T_1$ and $N_0 = N_1$. If F is not admissible, no path has support F, so $M_F = 0$. Appendix A.7 of the arXiv version of this paper [40] gives a direct proof by explicit paths, which does not use the local law.

(iii) For large N we have $T_0 \le T = \tau L \le N^{1/3}$, and both bounds in (ii) read $\log M_F = -L\,\Phi_\tau(F) + O(\log L)$ for every admissible face, while $M_F = 0$ for the others. Since $\max_n P_N(n) = \max_F M_F$ over finitely many faces, the first claim follows. Let $m_\tau = \min_F \Phi_\tau(F)$. An admissible $F \notin \mathcal{F}_\tau$ has $\Phi_\tau(F) \ge m_\tau + \varepsilon_0$ for a fixed $\varepsilon_0 > 0$, so $M_F \le N^{-m_\tau - \varepsilon_0 + o(1)}$, which is smaller than $\max_n P_N(n) = N^{-m_\tau + o(1)}$ for large N. A global mode has positive probability, so its support is admissible, and hence it lies in $\mathcal{F}_\tau$.

(iv) By (iii), for large N a global mode $\hat{n}$ maximizes P_N on its face $F \in \mathcal{F}_\tau$, which satisfies (H) by Lemma 2.2(b). So Theorem 5.7(b) with $T = \tau L \le N^{1/3}$ gives $\hat{n}/N = \alpha_F^* + O(1/L)$. □

On a reducible face the power of T can differ from $(k_F - 1)/2$, as in Example 5.11. Theorem 4.5 below gives the power and the constant for every admissible face.

Remark 4.3 (a field) Let $v \in \mathbb{R}^d$, put $M_F(T, v) = \max\{P_N(n) e^{Tv\cdot n/N} : \operatorname{supp} n = F\}$ and $\Phi_{\tau,v}(F) = |F| - 1 + \tau\lambda_F(v_F)$. On an admissible face, $Q_F + \operatorname{diag} v_F$ is block triangular in the order of the components, so $\lambda_F(v_F) = \min_j \lambda_{C_j}(v_{C_j})$. The upper bound in (ii) holds for $M_F(T, v)$ with $\lambda_F(v_F)$ in place of λ_F. The proof in Appendix A.7 of the arXiv version of this paper [40] applies with (6) on each block and the exact weight $e^{hv_i N_j}$ on a block $\{i\}$. For F satisfying (H), Theorem 5.7 gives the matching lower bound. If μ has full support, a reducible admissible face F has a component C with $\lambda_C(v_C) = \lambda_F(v_F)$, which is admissible with fewer states, so $\Phi_{\tau,v}(C) \le \Phi_{\tau,v}(F) - 1$. So if μ has full support, then for fixed τ and v, (iii) and (iv) hold for the weighted law with $\Phi_{\tau,v}$ and $\alpha_F^*(v)$, every face that minimizes $\Phi_{\tau,v}$ is irreducible, and a strict minimizer carries every maximizer of the weighted law for large N.

4.3 Reading the winners off a convex hull

For an admissible face F put $P_F = (\lambda_F, |F| - 1)$, and let $\mathcal{S}$ be the set of these points. Then $\Phi_\tau(F) = \tau x + y$ at $(x, y) = P_F$. Picture a line of slope $-\tau$ moving up from far below. The first points of $\mathcal{S}$ that it touches are the winners at τ. For each $\tau > 0$ the points of $\operatorname{conv}\mathcal{S}$ where $\tau x + y$ is least form a vertex or an edge of $\operatorname{conv}\mathcal{S}$, and we call the union of these over all $\tau > 0$ the lower-left boundary of $\operatorname{conv}\mathcal{S}$. Figure 3(b) shows it for the 4-cycle.

Theorem 4.4 (geometry of the winners) *Let $\mathcal{F}_\tau$ be the set of admissible faces that minimize Φ_τ.*

(a) The lower-left boundary of $\operatorname{conv}\mathcal{S}$ *is a path whose vertices lie in $\mathcal{S}$. Read from right to left, they are $v_i = (\lambda^{(i)}, y^{(i)})$, $0 \le i \le m$, with $\lambda^{(0)} > \cdots > \lambda^{(m)}$ and $y^{(0)} < \cdots < y^{(m)}$. Put $\tau_i = (y^{(i)} - y^{(i-1)})/(\lambda^{(i-1)} - \lambda^{(i)})$ for $1 \le i \le m$, $\tau_0 = 0$ and $\tau_{m+1} = \infty$. Then $\tau_1 < \cdots < \tau_m$. For $\tau_i < \tau < \tau_{i+1}$, $\mathcal{F}_\tau$ is the set of faces F with $P_F = v_i$. At $\tau = \tau_i$ it is the set of faces whose point lies on the closed edge $[v_{i-1}, v_i]$.*

(b) *If $0 < \tau < \tau'$, $F \in \mathcal{F}_\tau$ and $G \in \mathcal{F}_{\tau'}$, then $\lambda_F \geq \lambda_G$ and $|F| \leq |G|$. Both inequalities are strict when $\mathcal{F}_\tau \cap \mathcal{F}_{\tau'} = \emptyset$.*

(c) *If Q has symmetric support, or μ has full support, then every face in $\mathcal{F}_\tau$ is irreducible, for every $\tau > 0$.*

Proof (a) and (b) The proofs, through the kinks of the concave function $\tau \mapsto \min_{(x,y)\in\mathcal{S}}(\tau x + y)$, are in Section 4 of the arXiv version of this paper [40].

(c) With symmetric support every admissible face is irreducible by Lemma 2.2(c). Now let μ have full support, and let F be admissible and reducible. Take a component C of F with $\lambda_C = \lambda_F$. It is irreducible with $\mu(C) > 0$, so it is admissible by Lemma 2.2(b). It has the same exit rate as F and fewer states, so $\Phi_\tau(C) \leq \Phi_\tau(F) - 1$. Hence $F \notin \mathcal{F}_\tau$. □

A face with small λ_F for its size is a set of states that the chain rarely leaves. As τ grows the exit rate counts for more, and the mode moves to larger such sets. It need not grow around its first vertex, and it can move to a disjoint set. Faces of one size share the term $|F| - 1$, so only the faces with the least exit rate of each size can win.

4.4 Reducible faces

Let F be admissible, with components $C_1, \dots, C_r$ numbered as in Lemma 2.2(a). A path of positive probability with support F visits them in this order, each in one block of consecutive times. The order is unique, since consecutive blocks are joined by an edge and the components form an acyclic graph, in which at most one path passes through every vertex. Put $k_j = |C_j|$ and $\lambda_j = \lambda_{C_j}$, so $\lambda_F = \min_j \lambda_j$. Let k^* be the least k_j with $\lambda_j = \lambda_F$, let S^* be the set of these j with $k_j = k^*$, and put

$$a_F = \frac{k^* - 1}{2}, \qquad p_F = a_F + k_F - k^*, \qquad R_j = (q_{xy})_{x\in C_j,\, y\in C_{j+1}}.$$

For an irreducible C with $|C| \geq 2$, states $y, x \in C$ and $t \in (0,\infty)^C$ let $\Phi_C(t)_{yx} = e^{-\langle q,t\rangle} H_{A_C,e_y,e_x}(t)$, in the notation before Lemma A.3. As in Lemma 2.4, this is the density of the occupation times of the chain started at y on the event that it stays in C up to time $|t|$, visits every state of C and ends at x. For $C = \{i\}$ put $\Phi_C(t) = e^{-q_i t}$. For $j \in S^*$ let r and l be the vectors r_{C_j} and l_{C_j} of Definition 2.1, with $l \cdot r = 1$, and let

$$A_j = \sup\, e^{\lambda_F |t^<|}\, \mu_{C_1}^\top \Phi_{C_1}(t^{(1)}) R_1 \cdots \Phi_{C_{j-1}}(t^{(j-1)}) R_{j-1}\, r,$$
$$B_j = \sup\, e^{\lambda_F |t^>|}\, l^\top R_j \Phi_{C_{j+1}}(t^{(j+1)}) R_{j+1} \cdots \Phi_{C_r}(t^{(r)})\, \mathbf{1},$$

over all $t^{(m)} \in (0,\infty)^{C_m}$ for $m < j$, respectively $m > j$, where $|t^<|$ and $|t^>|$ are their total times, $A_1 = \mu_{C_1} \cdot r$ and $B_r = l \cdot \mathbf{1}$. Finally let

$$K_F^{\mathrm{red}} = \max_{j\in S^*} \frac{A_j B_j}{(2\pi)^{a_F} \sqrt{\det \Sigma_{C_j}}},$$

where Σ_{C_j} is Σ_α of Definition 5.1 for the face C_j at $\alpha = \alpha^*_{C_j}$, and the denominator is 1 if $k^* = 1$. If F is irreducible, then $p_F = (k_F - 1)/2$ and $K_F^{\mathrm{red}} = K_F$.

Theorem 4.5 (face maxima on reducible faces) *Let F be admissible. Then $0 < K_F^{\mathrm{red}} < \infty$, and there are T_1, N_1 and C such that for $N \geq N_1$ and $T_1 \leq T \leq N^{1/3}$*

$$\Big| \frac{M_F}{N^{-(k_F-1)} T^{p_F} e^{-T\lambda_F}} - K_F^{\mathrm{red}} \Big| \leq C\Big(\frac{1}{T} + \frac{T^2}{N}\Big).$$

In particular the ratio lies between $\frac{1}{2}K_F^{\mathrm{red}}$ and $2K_F^{\mathrm{red}}$ after T_1 and N_1 are enlarged.

So the power of T is p_F, between the two powers of Theorem 4.2(ii). The time goes to the components of least exit rate and least size k^*, and to one of them when $k^* \geq 2$. Such a component contributes $T^{-(k^*-1)/2}$ through its local law, and each other component C gets time of order one and contributes $h^{|C|-1}$. Each switch from one component to the next contributes one more factor h. The constants A_j and B_j are maxima over these short blocks. For irreducible F the theorem is Theorem 5.7(b) at $v = 0$. The proof is in Appendix E.

Corollary 4.6 (crossings of admissible faces) *Let F and G be admissible, with $\Delta = k_F - k_G \geq 1$ and $\delta = \lambda_G - \lambda_F > 0$. Put $\pi = p_F - p_G$, $\kappa = \log(K_F^{\mathrm{red}}/K_G^{\mathrm{red}})$ and $D_N = \log M_F - \log M_G$, and let T_1 serve for both faces in Theorem 4.5.*

(i) For large N, D_N has a zero in $[\frac{\Delta}{2\delta}L, \frac{2\Delta}{\delta}L]$.

(ii) For large N, every zero $T_N \in [T_1, N^{1/3}]$ of D_N satisfies $\delta T_N + \pi \log T_N = \Delta L - \kappa + O(1/L)$.

(iii) All such zeros lie in one interval of length $O(1/L)$, and

$$T_N = \frac{\Delta}{\delta}\Big[L - \frac{\pi}{\Delta}\log L + b\Big] + O\Big(\frac{\log L}{L}\Big), \qquad b = -\frac{\pi}{\Delta}\log\frac{\Delta}{\delta} - \frac{\kappa}{\Delta}.$$

The coefficient of $\log\log N$ is π/Δ. It is $\frac{1}{2}$ when F and G are irreducible, which gives Theorem 5.8(i) to (iii) for zeros in $[T_1, N^{1/3}]$. In Example 5.11, $p_{\{1\}} = 0$, $p_{\{1,2\}} = 1$ and $p_{\{1,2,3\}} = 1$, and $K^{\mathrm{red}}_{\{1,2\}} = b$ from $A_2 = \sup_t be^{(1-b)t} = b$ and $B_2 = 1$. So the corollary gives the coefficients 1 and 0 of parts (ii) and (iii) there, with their constants. We do not claim that the zero is unique.

5 The second-order theory

To locate a crossing up to $o(1)$ we need the power of T and the constant in front of $N^{-(k-1)}e^{-T\lambda_F}$ for every face that satisfies (H). In this section gradients and Hessians in α are taken in the coordinates $\alpha' = (\alpha_2, \dots, \alpha_k)$, $|\cdot|$ is the Euclidean norm, and $q = (q_i)_{i\in F}$.

5.1 The tilted chain and the constant

To find the probability of a composition with fractions α we change the rates so that α becomes typical. The optimal vector r_α in the definition of $J_F(\alpha)$ gives a new generator $\widehat{Q}_\alpha$ on F, with the same switching graph, whose stationary law is exactly α. Under it the visit fractions satisfy a central limit theorem with covariance Γ_α/T, and the price of the change of rates is $e^{-TJ_F(\alpha)}$ times boundary factors for the start and the end of the path. The constant $C_F(\alpha)$ collects these factors and the Gaussian normalization.

Definition 5.1 (tilt and constants) Let F be irreducible with $k \geq 2$. For $\alpha \in \Delta_F^\circ$ let r_α attain the supremum defining $J_F(\alpha)$ (Proposition 3.3(b)), and put $\theta(\alpha) = A_F r_\alpha / r_\alpha$ and $l_\alpha = \alpha / r_\alpha$ (entrywise). The *tilted generator* is $\widehat{Q}_\alpha = R^{-1}(A_F - \operatorname{diag}\theta(\alpha))R$ with $R = \operatorname{diag} r_\alpha$. With $D = \operatorname{diag}\alpha$ let

$$Z_\alpha = (\mathbf{1}\alpha^\top - \widehat{Q}_\alpha)^{-1} - \mathbf{1}\alpha^\top, \qquad \Gamma_\alpha = DZ_\alpha + Z_\alpha^\top D,$$

and let Σ_α be Γ_α with the row and column of state 1 deleted. Finally let

$$C_F(\alpha) = \frac{(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})}{(2\pi)^{(k-1)/2}\sqrt{\det \Sigma_\alpha}}, \qquad K_F = C_F(\alpha_F^*).$$

For a single state we put $K_{\{i\}} = \mu_i$. Nothing in Definition 5.1 changes when r_α is scaled, and nothing in it involves the diagonal of Q_F. The objects depend on A_F and μ only, so a field does not change them. For $v \in \mathbb{R}^F$ we put $K_F(v) = C_F(\alpha_F^*(v))$, so that $K_F(0) = K_F$, and $K_{\{i\}}(v) = \mu_i$. Also $l_\alpha \cdot r_\alpha = 1$ and $J_F(\alpha) = \langle q - \theta(\alpha), \alpha\rangle$. Since r_α minimizes $u \mapsto \sum_{i\neq j} \alpha_i q_{ij} u_j / u_i$, its derivative in u_j, which is $\sum_i l_{\alpha,i} q_{ij} - l_{\alpha,j}\theta_j(\alpha)$, vanishes at r_α. Hence

$$l_\alpha^\top \big(A_F - \operatorname{diag}\theta(\alpha)\big) = 0. \tag{7}$$

Also $(A_F - \operatorname{diag}\theta(\alpha))r_\alpha = 0$ with $r_\alpha > 0$, so $\theta(\alpha) \in V_F$ [3]. So $\widehat{Q}_\alpha$ is an irreducible generator with the same off-diagonal zero pattern as A_F, and by (7) its stationary law is α. If $(\mathbf{1}\alpha^\top - \widehat{Q}_\alpha)v = 0$, multiplying by $\alpha^\top$ gives $\alpha^\top v = 0$, so $\widehat{Q}_\alpha v = 0$. Hence v is a multiple of $\mathbf{1}$, and $\alpha^\top v = 0$ gives $v = 0$. Thus Z_α exists, and

$$Z_\alpha \mathbf{1} = 0, \qquad \alpha^\top Z_\alpha = 0, \qquad -\widehat{Q}_\alpha Z_\alpha = I - \mathbf{1}\alpha^\top. \tag{8}$$

So $\Gamma_\alpha \mathbf{1} = 0$. As Γ_α is symmetric, its adjugate is then a multiple of $\mathbf{1}\mathbf{1}^\top$, so deleting another state instead of state 1 gives the same $\det \Sigma_\alpha$. At α_F^* the vector $u = r_F$ gives the value $\lambda_F = J_F(\alpha_F^*)$, so $r_{\alpha_F^*}$ is a multiple of r_F, $\theta(\alpha_F^*) = q - \lambda_F \mathbf{1}$, and $l_{\alpha_F^*}$ is the matching multiple of l_F.

For conductances $c_{ij} = c_{ji}$ on F let $\operatorname{tree}(c) = \sum_{\mathcal{T}} \prod_{\{i,j\}\in\mathcal{T}} c_{ij}$, the sum over the spanning trees $\mathcal{T}$ of the complete graph on F.

Proposition 5.2 (the constant) *Let F be irreducible with $k \geq 2$.*

(a) The maps $\alpha \mapsto \theta(\alpha)$, $J_F(\alpha)$, Σ_α, r_α and l_α (with $\sum_i r_{\alpha,i} = 1$) are real-analytic on Δ_F°, Σ_α is positive definite, and $\nabla^2 J_F(\alpha) = \Sigma_\alpha^{-1}$. Hence $\det \Sigma_\alpha \cdot \det \nabla^2 J_F(\alpha) = 1$ and $C_F(\alpha) = (\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})\sqrt{\det \nabla^2 J_F(\alpha)}/(2\pi)^{k-1}$.

(b) *Suppose that* $\nu_i q_{ij} = \nu_j q_{ji}$ *for* $i, j \in F$ *and some* $\nu \in (0,\infty)^F$. *Then* r_α *is a multiple of* $\sqrt{\alpha/\nu}$ *and* l_α *of* $\sqrt{\alpha\nu}$, *and* $\widehat{Q}_\alpha$ *is reversible with conductances* $c_{ij}(\alpha) = \sqrt{\alpha_i\alpha_j}\, q_{ij}\sqrt{\nu_i/\nu_j}$. *Also* $J_F(\alpha) = \sum_i q_i\alpha_i - \sum_{i\neq j} c_{ij}(\alpha)$, *and*

$$\det \Sigma_\alpha = \frac{2^{k-1}(\prod_i \alpha_i)^2}{\operatorname{tree}(c(\alpha))}, \qquad C_F(\alpha) = \frac{(\mu\cdot r_\alpha)(l_\alpha\cdot \mathbf{1})}{(4\pi)^{(k-1)/2}}\,\frac{\sqrt{\operatorname{tree}(c(\alpha))}}{\prod_i \alpha_i}.$$

Proof The proof of (a) is in Appendix A and uses Lemma A.1.

(b) We guess the vector r_α and check that the guess is a critical point. Put $u = \sqrt{\alpha/\nu}$ and $l = \alpha/u = \sqrt{\alpha\nu}$ entrywise. As we saw before (7), the derivative of $\sum_{i\neq j}\alpha_i q_{ij} u_j/u_i$ in u_j is $\sum_i l_i q_{ij} - l_j(A_F u)_j/u_j$. Here

$$\sum_i l_i q_{ij} = \sum_i \sqrt{\alpha_i\nu_i}\, q_{ij} \qquad \text{and} \qquad \frac{l_j(A_F u)_j}{u_j} = \nu_j \sum_i q_{ji}\sqrt{\alpha_i/\nu_i} = \sum_i \sqrt{\alpha_i\nu_i}\, q_{ij},$$

where the last step uses $\nu_j q_{ji} = \nu_i q_{ij}$. So the derivative vanishes at u. The function is convex in $\log u$, so its critical point u is a minimizer, and the minimizer is unique up to a factor (Proposition 3.3(b)). Hence r_α is a multiple of u and l_α is the matching multiple of l. Then $\alpha_i(\widehat{Q}_\alpha)_{ij} = l_i q_{ij} u_j = c_{ij}(\alpha)$ for $i \neq j$, and $c_{ij}(\alpha) = c_{ji}(\alpha)$ because $q_{ij}\sqrt{\nu_i/\nu_j} = q_{ji}\sqrt{\nu_j/\nu_i}$. So $\widehat{Q}_\alpha$ is reversible with these conductances. Also $\alpha_i\theta_i(\alpha) = l_i(A_F u)_i = \sum_{j\neq i} c_{ij}(\alpha)$, and $J_F(\alpha) = \langle q - \theta(\alpha), \alpha\rangle$ gives the formula for J_F.

The formula for $\det\Sigma_\alpha$, and with it that for $C_F(\alpha)$, follows from the matrix-tree theorem [7] for the Laplacian $-D\widehat{Q}_\alpha$ of the conductances, and its proof is in Section 5.1 of the arXiv version of this paper [40]. □

Corollary 5.3 (symmetric faces) *Let* F *satisfy* (H) *with* $k \geq 2$, *and let* Q_F *be symmetric with constant row sums. Then* $\lambda_F = q_i - \sum_{j\in F\setminus\{i\}} q_{ij}$ *for every* $i \in F$, α_F^* *is uniform on* F, *and*

$$K_F = \mu(F)\, k^{(k+1)/2}\sqrt{\operatorname{tree}(q_F)}\,(4\pi)^{-(k-1)/2},$$

where $\operatorname{tree}(q_F)$ *is* $\operatorname{tree}(c)$ *for the conductances* $c_{ij} = q_{ij}$, $i, j \in F$.

Proof The positive vector $\mathbf{1}$ is a right and left eigenvector of Q_F, so it belongs to the Perron root. So λ_F is minus the row sum, r_F and l_F are multiples of $\mathbf{1}$, and $\alpha_F^* = \mathbf{1}/k$. At $\alpha = \mathbf{1}/k$, Proposition 5.2(b) with $\nu = \mathbf{1}$ gives $r_\alpha = \mathbf{1}$, $l_\alpha = \alpha$ and $(\mu\cdot r_\alpha)(l_\alpha\cdot\mathbf{1}) = \mu(F)$. Also $c_{ij}(\alpha) = q_{ij}/k$, so $\operatorname{tree}(c(\alpha)) = k^{-(k-1)}\operatorname{tree}(q_F)$, and $\prod_i \alpha_i = k^{-k}$. □

5.2 The local law and the face maxima

A field $v \in \mathbb{R}^F$ enters the continuum density as an exact factor. By Lemma 2.4, $f_T(\alpha)e^{Tv\cdot\alpha}$ is the density f_T of the killed matrix $Q_F + \operatorname{diag} v$, which has the same A_F and μ and the diagonal $q - v$ in place of q. On the lattice the factor is $e^{Tv\cdot n/N} = e^{h\langle v,n\rangle}$.

The discrete constant. Let F be irreducible with $k \geq 2$, fix h_* with $h_* q_{\max} < 1$, and let $(h,\alpha) \in [0,h_*] \times \Delta_F^\circ$. Let $\theta = \theta^{(h)}(\alpha)$ be the vector of Proposition 3.5(b), let r and l be positive right and left null vectors of $A_F - \operatorname{diag}\theta$ with $l\cdot r = 1$, put $c_h(\alpha) = \sum_i \alpha_i/(1 - hq_i + h\theta_i)$, and let

$$C_F^{(h)}(\alpha) = c_h(\alpha)\,(\mu\cdot r)(l\cdot\mathbf{1})\sqrt{\det\nabla^2 J_F^{(h)}(\alpha)/(2\pi)^{k-1}}.$$

Lemma A.6 shows that $\nabla^2 J_F^{(h)}$ is positive definite and that $C_F^{(h)}$ is positive and real-analytic up to $h = 0$. At $h = 0$ we have $c_0 = 1$ and $\theta^{(0)}(\alpha) = \theta(\alpha)$, so $C_F^{(0)} = C_F$ by Proposition 5.2(a). For $v \in \mathbb{R}^F$ put $K_F^{(h)}(v) = C_F^{(h)}(\alpha_F^{*(h)}(v))$ and $K_F^{(h)} = K_F^{(h)}(0)$, and for a single state $K_{\{i\}}^{(h)}(v) = \mu_i/(1 - hq_i)$. At α_F^* the maximizer of Proposition 3.5(b) is $\log r_F$ up to a constant, for every h, by part (c) of that proposition. So $\theta^{(h)}(\alpha_F^*) = q - \lambda_F \mathbf{1}$ and $c_h(\alpha_F^*) = 1/(1 - h\lambda_F)$.

Theorem 5.4 (local law) *Let F satisfy (H) with $k \geq 2$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact.*

(a) Uniformly for $\alpha \in \mathcal{K}$, $T \geq 1$ and $v \in \mathbb{R}^F$,

$$f_T(\alpha) e^{Tv\cdot\alpha} = T^{(k-1)/2} C_F(\alpha) e^{-T(J_F(\alpha) - v\cdot\alpha)} \big(1 + O_{\mathcal{K}}(1/T)\big).$$

(b) Fix h_* with $h_* q_{\max} < 1$. Uniformly for $1 \leq T \leq h_* N$, $\operatorname{supp} n = F$ with $\alpha = n/N \in \mathcal{K}$, and $v \in \mathbb{R}^F$,

$$P_N(n) e^{Tv\cdot\alpha} = N^{-(k-1)} T^{(k-1)/2} C_F^{(h)}(\alpha) e^{-T(J_F^{(h)}(\alpha) - v\cdot\alpha)} \big(1 + O_{\mathcal{K}}(1/T)\big).$$

For $F = \{i\}$, $f_T = \mu_i e^{-q_i T}$ and $P_N(Ne_i) = \mu_i (1 - hq_i)^{N-1}$.

The field cancels from the two sides, so the case $v = 0$ is the whole content, and the error does not depend on v. In (b) the ratio of the two sides lies between two positive constants for all T in the range, so the error is relative.

Idea of the proof. The proof is in Appendix A. The generating function of the words in H is a matrix resolvent, so Cauchy's formula writes H as an integral over a torus with the radii $\theta_2(\alpha), \dots, \theta_k(\alpha)$ (Lemma A.2). All the coefficients are nonnegative, so the integrand is largest at the real point, and a Laplace lemma with a parameter (Lemma B.1) gives the expansion. The Hessian at the real point comes from the Perron root of a perturbed generator (Lemma A.1), which brings in Σ_α. Also $f_T(\alpha)$ is continuous and positive in (T, α), as H is a convergent power series with some $W(m) > 0$ by (H). For (b) the same torus carries Good's discrete formula, with $e^{t_i z_i}$ replaced by $(1 - hq_i + hz_i)^{n_i - 1}$ (Lemma A.2). In the variable z the integrand is analytic in h up to $h = 0$, where it is the integrand of (a), so one Laplace argument with (h, α) in the compact set $[0, h_*] \times \mathcal{K}$ covers $1 \leq T \leq h_* N$. Only for bounded T does the proof use the comparison of Lemma 5.5, to keep the ratio bounded below.

Lemma 5.5 (discrete and continuum) *Let F satisfy (H) with $k \geq 2$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact. Uniformly for $\operatorname{supp} n = F$ with $n/N \in \mathcal{K}$ and $1 \leq T \leq N^{1/2}$, $P_N(n) = N^{-(k-1)} f_T(n/N)(1 + O_{\mathcal{K}}(T^2/N))$.*

The proof is in Appendix A. A field multiplies both sides by the same factor $e^{Tv\cdot n/N}$. Theorem 5.4(b) makes this transfer unnecessary for the local law, but the proofs of Theorem 5.4(b) for bounded T and of Lemma A.5 use it. Together with the restriction identities the local law gives one display for all faces.

Theorem 5.6 (the stratified local law) *Fix Q, a start μ and $h_* $ with $h_* q_{\max} < 1$. Let F satisfy (H) with $|F| = k \geq 2$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact.*

(a) Uniformly for $\operatorname{supp} n = F$, $n/N \in \mathcal{K}$ *and* $1 \leq T \leq h_* N$,

$$-\log P_N(n) = (k-1)L + T\, J_V^{(h)}(n/N) - \frac{k-1}{2}\log T - \log C_F^{(h)}(n/N) + O_{\mathcal{K}}\Big(\frac{1}{T}\Big).$$

(b) Uniformly for $\operatorname{supp} n = F$, $n/N \in \mathcal{K}$ *and* $1 \leq T \leq N^{1/3}$,

$$-\log P_N(n) = (k-1)L + T\, J_V(n/N) - \frac{k-1}{2}\log T - \log C_F(n/N) + O_{\mathcal{K}}\Big(\frac{1}{T} + \frac{T^2}{N}\Big).$$

For a singleton $-\log P_N(Ne_i) = -\log \mu_i - (N-1)\log(1-hq_i)$, which is $Tq_i - \log \mu_i + O(T^2/N)$. With $T = \tau L$ the exponent of N in (b) is $(k-1) + \tau J_V(n/N)$, the lattice entropy of the face plus τ times the Donsker and Varadhan cost of the point.

Proof (a) This is Theorem 5.4(b) with $v = 0$ and Proposition 3.5(a), which replaces $J_F^{(h)}$ by $J_V^{(h)}$ on Δ_F. (b) By Proposition 3.5(b) and Lemma A.6, $J_F^{(h)}$ and $\log C_F^{(h)}$ are real-analytic in (h, α) up to $h = 0$, so $TJ_F^{(h)} = TJ_F + O(Th) = TJ_F + O(T^2/N)$ and $\log C_F^{(h)} = \log C_F + O(T/N)$ uniformly on $\mathcal{K}$. Put this into (a) and replace J_F by J_V on Δ_F by Proposition 3.1. Part (b) also follows from Theorem 5.4(a) and Lemma 5.5. The singleton case is the path i^N and (1). □

For a field $v \in \mathbb{R}^d$ we write $\lambda_F(v)$, $\alpha_F^*(v)$, $K_F(v)$, $\lambda_F^{(h)}(v)$, $\alpha_F^{*(h)}(v)$ and $K_F^{(h)}(v)$ for the quantities of v_F, and

$$M_F(T, v) = \max\{P_N(n) e^{Tv\cdot n/N} : \operatorname{supp} n = F\},$$

so that $M_F(T, 0) = M_F$.

Theorem 5.7 (face maxima) *Let F satisfy (H), let $\mathcal{V} \subset \mathbb{R}^d$ be compact, and fix h_* with $h_* q_{\max} < 1$.*

(a) If $k \geq 2$, then uniformly for $v \in \mathcal{V}$ and $1 \leq T \leq h_* N$,

$$M_F(T, v) = N^{-(k-1)} T^{(k-1)/2} K_F^{(h)}(v) e^{-T\lambda_F^{(h)}(v)}\big(1 + O_{\mathcal{V}}(1/T)\big),$$

and every maximizer $\hat{n}$ satisfies $\hat{n}/N = \alpha_F^{*(h)}(v) + O_{\mathcal{V}}(1/T)$. At $v = 0$ this reads $M_F = N^{-(k-1)} T^{(k-1)/2} K_F^{(h)} (1 - h\lambda_F)^N (1 + O(1/T))$ with $\hat{n}/N = \alpha_F^* + O(1/T)$. For $k = 1$, $M_{\{i\}}(T, v) = e^{Tv_i}\mu_i (1 - hq_i)^{N-1} = K_{\{i\}}^{(h)}(v) e^{-T\lambda_{\{i\}}^{(h)}(v)}$ exactly.

(b) If $k \geq 2$, then uniformly for $v \in \mathcal{V}$ and $1 \leq T \leq N^{1/3}$,

$$M_F(T, v) = N^{-(k-1)} T^{(k-1)/2} K_F(v) e^{-T\lambda_F(v)}\big(1 + O_{\mathcal{V}}(1/T + T^2/N)\big),$$

and every maximizer $\hat{n}$ satisfies $\hat{n}/N = \alpha_F^*(v) + O_{\mathcal{V}}(1/T)$. For $k = 1$, $M_{\{i\}}(T, v) = e^{Tv_i} M_{\{i\}}$ with $M_{\{i\}}$ given by (1). In both cases, uniformly for $v \in \mathcal{V}$ and $1 \leq T \leq N^{1/3}$,

$$\log M_F(T, v) = -(k-1)L + \tfrac{k-1}{2}\log T + \log K_F(v) - T\lambda_F(v) + O_{\mathcal{V}}(1/T + T^2/N). \quad (9)$$

The constants depend on $\mathcal{V}$ only through the compact set $\{\alpha_F^{*(h)}(v) : 0 \le h \le h_*,\ v \in \mathcal{V}\}$, the bounds of $\lambda_F^{(h)}$ and $K_F^{(h)}$ on $[0,h_*]\times\mathcal{V}$, and finitely many vectors of V_F.

Proof (a) For $k = 1$ the only composition is Ne_i. Let $k \ge 2$. For $(h,v) \in [0,h_*] \times \mathcal{V}$ put $J^{h,v}(\alpha) = J_F^{(h)}(\alpha) - v\cdot\alpha - \lambda_F^{(h)}(v)$, which is nonnegative on Δ_F and vanishes only at $\alpha_F^{*(h)}(v)$, by Proposition 3.6(a) for $h > 0$ and Proposition 3.3(a) for $h = 0$. Put $X = N^{-(k-1)}T^{(k-1)/2}K_F^{(h)}(v)e^{-T\lambda_F^{(h)}(v)}$, the value that (a) claims, and write $x = \alpha' - \alpha_F^{*(h)}(v)'$ for the displacement of α from $\alpha_F^{*(h)}(v)$. Below, constants depend only on Q, μ, h_* and $\mathcal{V}$.

The balls. By Proposition 3.6(c) the set $\mathcal{A} = \{\alpha_F^{*(h)}(v) : 0 \le h \le h_*,\ v \in \mathcal{V}\}$ is a compact subset of Δ_F°, and $\lambda_F^{(h)}(v)$ and $\log K_F^{(h)}(v)$ are bounded. By Proposition 3.5(b) and Lemma A.6, $J_F^{(h)}$ and $\log C_F^{(h)}$ are real-analytic in (h,α) up to $h = 0$, with $\nabla^2 J_F^{(h)}$ positive definite, and $J^{h,v}$ has a critical point at $\alpha_F^{*(h)}(v)$. Taylor's theorem with bounds on the derivatives of order up to three on $[0,h_*]$ times a compact neighborhood of $\mathcal{A}$ gives $\rho_0 > 0$ and positive constants c_U, C_U, C_1 and C_3 with the following properties. The set $\mathcal{K}_\mathcal{V}$ of points within ρ_0 of $\mathcal{A}$ lies in Δ_F°, and for $(h,v) \in [0,h_*]\times\mathcal{V}$ the ball $U(h,v) = \{|x| \le \rho_0\}$ satisfies

$$c_U|x|^2 \le J^{h,v} \le C_U|x|^2, \qquad \bigl|\log(C_F^{(h)}/K_F^{(h)}(v)) - g\cdot x\bigr| \le C_1|x|^2$$

on $U(h,v)$, where $g = \nabla \log C_F^{(h)}(\alpha_F^{*(h)}(v))$ has $|g| \le C_1$. Also $\log C_F^{(h)}$ is C_3-Lipschitz on $\mathcal{K}_\mathcal{V}$.

Outside $U(h,v)$. For $\beta \in \Delta_F$ the definition of $J_F^{(h)}$ and $((I+hQ_F)u)_i/u_i = 1 - hq_i + h\theta_i$ with $\theta = A_F u/u$ give $J_F^{(h)}(\beta) = \sup_{\theta\in V_F}\sum_i \beta_i L_h(q_i-\theta_i)$, and for fixed $\theta \in V_F$ the sum is continuous in $(h,\beta) \in [0,h_*]\times\Delta_F$, since $1 - h(q_i-\theta_i) \ge 1 - h_* q_{\max}$. The set $\mathcal{E}$ of triples $(h,v,\beta) \in [0,h_*]\times\mathcal{V}\times\Delta_F$ with $\beta \notin U(h,v)^\circ$ is compact, and $J^{h,v}(\beta)$ is positive and lower semicontinuous on it. So $J^{h,v} \ge 3\eta$ on $\mathcal{E}$ for some $\eta > 0$. For each point of $\mathcal{E}$ the supremum gives $\theta \in V_F$ with $\sum_i \beta_i L_h(q_i-\theta_i) - v\cdot\beta > \lambda_F^{(h)}(v) + 2\eta$, and the same θ works near that point. By compactness finitely many vectors $\theta_j \in V_F$ suffice. Now let $\operatorname{supp} n = F$ with $(h,v,n/N) \in \mathcal{E}$, and take j that works there. The bound (A5) with $\theta = \theta_j$, multiplied by $e^{h\langle v,n\rangle}$, gives, since $(1 - hq_i + h\theta_{j,i})^{n_i} = e^{-hn_i L_h(q_i-\theta_{j,i})}$ with $hn_i = T\beta_i$ for $\beta = n/N$,

$$P_N(n)e^{Tv\cdot n/N} \le C(T/N)^{k-1}e^{-T(\lambda_F^{(h)}(v)+2\eta)} = X\cdot O\bigl(T^{(k-1)/2}e^{-2\eta T}\bigr).$$

Inside $U(h,v)$. By Theorem 5.4(b) with $\mathcal{K} = \mathcal{K}_\mathcal{V}$, for $\alpha = n/N \in U(h,v)$,

$$P_N(n)e^{Tv\cdot n/N} = X\,\frac{C_F^{(h)}(\alpha)}{K_F^{(h)}(v)}\,e^{-TJ^{h,v}(\alpha)}(1+E), \qquad |E| \le C_2/T, \tag{10}$$

and $1+E$ lies between two positive constants. For $T \ge 2C_1/c_U$ the logarithm of the two middle factors is at most

$$g\cdot x + C_1|x|^2 - c_U T|x|^2 \le g\cdot x - \tfrac{c_U T}{2}|x|^2 \le \frac{|g|^2}{2c_U T},$$

the maximum of a concave quadratic. So $P_N(n)e^{Tv\cdot n/N} \le X(1+O(1/T))$ on $U(h,v)$. The bound outside $U(h,v)$ is smaller for large T, and hence $M_F(T,v) \le X(1+O(1/T))$. For the lower bound let n^0 have support F and $|n_i^0 - N\alpha_F^{*(h)}(v)_i| < 1$ for all i, which exists for $N \ge N_0$. Then $|x| \le k/N$, $C_F^{(h)}(n^0/N) \ge K_F^{(h)}(v)e^{-kC_3/N}$ and $TJ^{h,v}(n^0/N) \le C_U k^2 T/N^2 \le C_U k^2 h_*/N$. Since $1/N \le h_*/T$, (10) gives $P_N(n^0)e^{Tv\cdot n^0/N} \ge X(1-O(1/T))$.

Bounded T. Fix T_2 such that the two bounds give $M_F(T,v)/X \in [\frac12,\frac32]$ for $T \ge T_2$. For $1 \le T \le T_2$ the same three steps, with $1+E$ between two positive constants, give

$M_F(T,v)/X \in [c,C]$ with $0<c<C$. This proves the first display, and Proposition 3.6(b) gives the form at $v=0$.

The maximizer. Let $\hat n$ be a maximizer with displacement $\hat x$. For $T \ge T_3$ the bound outside $U(h,v)$ is below $X/2 \le P_N(n^0)e^{Tv\cdot n^0/N}$, so $\hat n/N \in U(h,v)$, and $|E| \le \frac12$ in (10). Taking logarithms in the comparison of $\hat n$ with n^0, and using $\log(C_F^{(h)}/K_F^{(h)}(v)) \le C_3|\hat x|$ and $TJ^{h,v} \ge c_U T|\hat x|^2$, we get

$$c_U T|\hat x|^2 \le C_3|\hat x| + \frac{C_4}{T}$$

for all $T_3 \le T \le h_*N$. Solving the quadratic inequality gives $|\hat x| \le \big(C_3 + \sqrt{C_3^2 + 4c_UC_4}\big)/(2c_UT)$. For $1 \le T < T_3$ the claim holds since $|\hat x|$ is bounded.

(b) By Proposition 3.6(c) and Lemma A.6, uniformly for $v \in \mathcal{V}$, $K_F^{(h)}(v) = K_F(v)(1+O(h))$, $\lambda_F^{(h)}(v) = \lambda_F(v) + O(h)$ and $\alpha_F^{*(h)}(v) = \alpha_F^*(v) + O(h)$. For $T \le N^{1/3}$ we have $h \le T^2/N \le 1/T$ and $Th = T^2/N \le 1$. So the value X in (a) is $N^{-(k-1)}T^{(k-1)/2}K_F(v)e^{-T\lambda_F(v)}(1+O(T^2/N))$, and every maximizer is within $O(1/T)$ of $\alpha_F^*(v)$. For $k=1$ the formula is the path i^N, and (9) is the logarithm of the two displays.

A second proof of (b), the continuum route, runs the proof of (a) with J_F, C_F, Theorem 5.4(a) with Lemma 5.5, and Lemma 4.1, and it is in the proof of this theorem in the arXiv version of this paper [40]. □

The range of part (b). The proof of Theorem 5.7(b) from part (a) uses only $Th \le 1$ and $h \le 1/T$, so it gives the display and the location of the maximizer for $1 \le T \le N^{1/2}$. The same lines give Theorem 5.6(b) for $T \le N^{1/2}$. We state both for $T \le N^{1/3}$, the range of the continuum route, which needs $T^2/N \le 1/T$ to locate the maximizer and is the route that Proposition 5.12 makes uniform. The error T^2/N is not an artifact of the proof. By the envelope theorem and $L_h(y) = y + hy^2/2 + O(h^2)$ for bounded y, uniformly on $\mathcal{K}$,

$$TJ_F^{(h)}(\alpha) = TJ_F(\alpha) + \frac{T^2}{2N}\sum_{i\in F}\alpha_i\big(q_i - \theta_i(\alpha)\big)^2 + O\Big(\frac{T^3}{N^2}\Big),$$

and $-N\log(1-h\lambda_F) = T\lambda_F + T^2\lambda_F^2/(2N) + O(T^3/N^2)$. The sum vanishes only where $\theta(\alpha) = q$. Since θ determines α, as the stationary law of $R^{-1}(A_F - \operatorname{diag}\theta)R$ with R the diagonal of the Perron vector, this happens only when $\lambda_F = 0$ and $\alpha = \alpha_F^*$. So the continuum exponent TJ_V fails when T is of order $N^{1/2}$, while the discrete functional absorbs every power of T/N and has no threshold below h_*N. Theorems 4.2(ii) and 4.5, which cover reducible faces, and Theorem 7.4(ii) on the complete graph are stated for $T \le N^{1/3}$, and the tie windows of Theorem 5.9 for T of order L. We have not extended them.

5.3 Crossings and tie windows

Theorem 5.8 (the crossing equation) *Let F and G satisfy (H), with $\Delta = k_F - k_G \ge 1$ and $\delta = \lambda_G - \lambda_F > 0$, and let $D_N(T) = \log M_F - \log M_G$.*

(i) For large N, D_N has a zero in $[\frac{\Delta}{2\delta}L, \frac{2\Delta}{\delta}L]$.

(ii) Fix h_* with $h_* q_{\max} < 1$. For large N, every zero $T_N \in [1, h_* N]$ of D_N satisfies

$$\delta T_N + \frac{\Delta}{2} \log T_N = \Delta L - \log \frac{K_F}{K_G} + O\Big(\frac{1}{L}\Big). \tag{11}$$

(iii) All such zeros lie in one interval of length $O(1/L)$, and

$$T_N = \frac{\Delta}{\delta}\Big[L - \frac{1}{2}\log L + b_{FG}\Big] + O\Big(\frac{\log L}{L}\Big), \qquad b_{FG} = -\frac{1}{2}\log\frac{\Delta}{\delta} - \frac{1}{\Delta}\log\frac{K_F}{K_G}.$$

(iv) For large N, D_N has exactly one zero in $[1, h_* N]$.

Proof Where the zeros lie. We show that for large N every zero of D_N in $[1, h_* N]$ lies in $W = [\frac{\Delta}{2\delta} L, \frac{2\Delta}{\delta} L]$. By Theorem 5.7(a) with $v = 0$ for F, and for G too if $k_G \geq 2$, uniformly for $1 \leq T \leq h_* N$,

$$D_N(T) = -\Delta L + \Xi_N(T) + \log \frac{K_F^{(h)}}{K_G^{(h)}} + O\Big(\frac{1}{T}\Big), \qquad \Xi_N(T) = N \log \frac{1 - h\lambda_F}{1 - h\lambda_G} + \frac{\Delta}{2}\log T.$$

If $G = \{j\}$ this holds by the exact formula $M_G = K_G^{(h)}(1 - h\lambda_G)^N$ of Theorem 5.7(a). Since $\lambda_F < \lambda_G \leq q_{\max} < 1/h$,

$$\Xi_N'(T) = \frac{\lambda_G}{1 - h\lambda_G} - \frac{\lambda_F}{1 - h\lambda_F} + \frac{\Delta}{2T} = \frac{\delta}{(1 - h\lambda_F)(1 - h\lambda_G)} + \frac{\Delta}{2T} \geq \delta,$$

and the first term of Ξ_N vanishes at $T = 0$, so $\Xi_N(T) \geq \delta T$ for $T \geq 1$. The constants $K_F^{(h)}$ and $K_G^{(h)}$ are continuous and positive in $h \in [0, h_*]$, so the logarithm of their ratio is bounded. So a zero T_N has $\Xi_N(T_N) = \Delta L + O(1)$, hence $\delta T_N \leq \Delta L + O(1)$ and $T_N < \frac{2\Delta}{\delta} L$. For $T \leq \frac{2\Delta}{\delta} L$ we have $Th = O(L^2/N)$, so $\Xi_N(T) = \delta T + \frac{\Delta}{2}\log T + O(L^2/N)$, and $\Xi_N(\frac{\Delta}{2\delta} L) = \frac{\Delta}{2} L + O(\log L)$. As Ξ_N is increasing, $T_N > \frac{\Delta}{2\delta} L$ for large N. Also $W \subset [1, N^{1/3}]$ for large N.

On W. We subtract (9) for G from (9) for F. Uniformly on $[1, N^{1/3}]$,

$$D_N(T) = -\Delta L + \Xi(T) + \log \frac{K_F}{K_G} + O\Big(\frac{1}{T} + \frac{T^2}{N}\Big), \qquad \Xi(T) = \delta T + \frac{\Delta}{2}\log T.$$

The function Ξ is increasing, with $\Xi' \geq \delta$.

(i) At $T = \frac{\Delta}{2\delta} L$ the display gives $D_N = -\frac{\Delta}{2} L + O(\log L) < 0$, and at $T = \frac{2\Delta}{\delta} L$ it gives $D_N = \Delta L + O(\log L) > 0$. For fixed N, M_F is a maximum of finitely many polynomials in h, and it is positive because F is admissible. The same holds for G. So D_N is continuous and has a zero in between.

(ii) Let $T_N \in [1, h_* N]$ be a zero. By the first step $T_N \in W$, so T_N is of order L, the error term of the display is $O(1/T_N + T_N^2/N) = O(1/L)$, and $D_N(T_N) = 0$ is (11).

(iii) Let T° be the solution of $\Xi(T^\circ) = \Delta L - \log(K_F/K_G)$. By (ii) every zero has $\Xi(T_N) = \Xi(T^\circ) + O(1/L)$, and $\Xi' \geq \delta$, so every zero is within $O(1/L)$ of T°. To solve for T°, we use $T^\circ = \frac{\Delta}{\delta} L + O(\log L)$, so $\log T^\circ = \log L + \log\frac{\Delta}{\delta} + O(\log L/L)$. Putting this into $\delta T^\circ = \Delta L - \frac{\Delta}{2}\log T^\circ - \log(K_F/K_G)$ and dividing by δ gives the explicit form.

(iv) By the first step, for large N every zero in $[1, h_* N]$ lies in W. We show that D_N is strictly increasing on W. For fixed N, M_F is the maximum of the finitely many polynomials $T \mapsto P_N(n)$ with $\operatorname{supp} n = F$ that are not identically zero. By Lemma 2.3 each term of such a polynomial is either zero for all h or positive for all $0 < h < 1/q_{\max}$, so these polynomials are positive on W. So $\log M_F$ has at each $T \in W$ a right derivative, the largest $\partial_T \log P_N(n)$ over the maximizers n at T, and a left derivative, the least of them. By Theorem 5.7 every

maximizer has $n/N = \alpha_F^* + O(1/L)$ on W. So for large N it lies in a closed ball $\mathcal{K} \subset \Delta_F^\circ$ around α_F^*, and $J_F(n/N) = \lambda_F + O(1/L)$, since J_F is smooth on $\mathcal{K}$ with $J_F(\alpha_F^*) = \lambda_F$. Lemma A.5 then gives both one-sided derivatives of $\log M_F$ as $-\lambda_F + O(1/L)$ on W. The same holds for G if $k_G \geq 2$. If $G = \{j\}$, then $\log M_G = \log \mu_j + (N-1)\log(1-hq_j)$ has the derivative $-(N-1)q_j/(N(1-hq_j)) = -\lambda_G + O(L/N)$. So both one-sided derivatives of D_N are $\delta + O(1/L) \geq \delta/2$ on W for large N. A continuous function whose right derivative is positive at every point of an interval is strictly increasing there. So D_N has at most one zero in W, and by (i) it has one. □

On the 4-cycle at $N = 10^{14}$ the explicit form in (iii) differs from the root of (11) without its error term by 0.035 for a vertex against a pair and by 0.087 for a pair against the cycle, the two crossings that carry the mode. So computations should solve (11).

At a switch value τ^* of Theorem 4.4 several faces minimize Φ_{τ^*}, sometimes more than two, as all faces do at $\tau^* = 1$ for the complete graph (Corollary 7.3). Then the competition plays out in a window $T = \tau^*(L - \frac{1}{2}\log L + t)$ with t of order 1. In this window each tied face has a line $\ell_F(t)$, and the mode follows the highest line.

Theorem 5.9 (tie windows) *Fix $\tau^* > 0$. Let m^* be the least value of Φ_{τ^*} over the admissible faces and $\mathcal{F}^*$ the set of minimizers. Assume that every face in $\mathcal{F}^*$ is irreducible. By Theorem 4.4(c) this holds if Q has symmetric support or μ has full support. Put $T = \tau^*(L - \frac{1}{2}\log L + t)$, and for $F \in \mathcal{F}^*$ let $\ell_F(t) = \log K_F + \frac{k_F - 1}{2}\log \tau^* - \tau^* \lambda_F t$ and $E(t) = \max_{G \in \mathcal{F}^*} \ell_G(t)$. The function E is convex and piecewise linear. We call the finitely many points where its slope changes the* kinks *of E, and we write t_j for them.*

(i) Uniformly for t in compact sets, for $F \in \mathcal{F}^*$, $\log M_F = -m^* L + \frac{m^*}{2}\log L + \ell_F(t) + O(\log L/L)$.

(ii) There are $\varepsilon_1, \varepsilon_0 > 0$ such that for large N and $|T/L - \tau^*| \leq \varepsilon_1$, every admissible $F \notin \mathcal{F}^*$ has $M_F \leq N^{-\varepsilon_0} \max_{G \in \mathcal{F}^*} M_G$.

(iii) There are $\varepsilon, t_0 > 0$ such that the following holds for large N. If $t_0 \leq t \leq \varepsilon L$, then every global mode lies on a face $F \in \mathcal{F}^*$ with $\ell_F = E$ on $[t_0, \infty)$. If $-\varepsilon L \leq t \leq -t_0$, then every global mode lies on a face $F \in \mathcal{F}^*$ with $\ell_F = E$ on $(-\infty, -t_0]$. A face of the first kind has the least λ_F in $\mathcal{F}^*$, and a face of the second kind has the largest.

The numbers ε_0, ε_1, ε and t_0 depend only on Q, μ and τ^*. Consequently the following holds for every $\eta > 0$ and large N. If $|t| \leq \varepsilon L$ and t is at distance at least η from every kink of E, then every global mode lies on a face whose line equals E near t. So for $|t| \leq \varepsilon L$ the line of the face that carries the global mode changes only at $T = \tau^*(L - \frac{1}{2}\log L + t_j) + o(1)$, where t_j runs over the kinks of E, and it changes near each kink. Coincident lines do not distinguish their faces at this order.

The proof, which puts $T = \tau^*(L - \frac{1}{2}\log L + t)$ into (9) for the tied faces and uses Theorem 4.2(ii) for the others, is in Section 5.3 of the arXiv version of this paper [40].

Corollary 5.10 (where the mode changes) *Let Q have symmetric support or μ full support. Let $v_0, \dots, v_m$ and $\tau_1 < \cdots < \tau_m$ be the vertices and switch values of Theorem 4.4(a), and assume that no point of $\mathcal{S}$ other than v_{i-1} and v_i lies on the edge $[v_{i-1}, v_i]$, for $1 \leq i \leq m$. Put $\Delta_i = y^{(i)} - y^{(i-1)}$ and $\delta_i = \lambda^{(i-1)} - \lambda^{(i)}$, let $K^{(i)}$ be the largest K_F over the faces F with*

$P_F = v_i$, and let $T_{N,i}$ be the root of $\delta_i T + \frac{\Delta_i}{2}\log T = \Delta_i L - \log(K^{(i)}/K^{(i-1)})$. Fix $0 < a < b$. There are C and N_0 such that for $N \ge N_0$ and every $T \in [aL, bL]$ with $|T - T_{N,j}| \ge C/L$ for all j, every global mode $\hat{n}$ has $P_{\operatorname{supp}\hat{n}} = v_i$, where i is the number of j with $T_{N,j} < T$.

The proof, by (9) and Theorem 5.9(ii) near each switch value and by Theorem 4.2(ii) between them, is in Section 5.3 of the arXiv version.

The equation for $T_{N,i}$ is (11) for a face of largest constant on each side, so Theorem 5.8(iii) gives $T_{N,i}$ explicitly. If three or more points of $\mathcal{S}$ lie on one edge, Theorem 5.9 locates the changes only up to $o(1)$. At a kink of E where the line of G gives way to that of F, $\Delta = \tau^*\delta$ and $t_j = b_{FG}$, so the kink is where Theorem 5.8(iii) puts the crossing of F and G. The coefficient $\frac{1}{2}$ of $\log\log N$ comes from the power $T^{(k-1)/2}$ in Theorem 5.7, so it needs only (H).

5.4 A reducible face

The best composition of a reducible face can visit a state only once, with no Gaussian window in that direction. Then the coefficient $\frac{1}{2}$ of Theorem 5.8 can fail, and in the next example it is 1 and then 0, as Corollary 4.6 predicts.

Example 5.11 (a directed 3-cycle) Let $d = 3$, $q_{12} = b > 2$, $q_{23} = q_{31} = 1$, all other rates 0, and $\mu_1 = 1$. From state 1 the chain moves only along $1 \to 2 \to 3 \to 1$, so the admissible faces are $\{1\}$, $\{1,2\}$ (reducible, $\lambda = 1$) and $\{1,2,3\}$ (irreducible, $\lambda = 0$).

(i) Exactly $M_{\{1\}} = (1-bh)^{N-1}$ and $M_{\{1,2\}} = bh(1-h)^{N-2}$, and the latter is attained only at $n = (1, N-1, 0)$. For the full face, $\alpha^* = (1, b, b)/(1+2b)$ and $K_{\{1,2,3\}} = (1+2b)^2/(2\sqrt{3}\,\pi b)$.

(ii) The equation $M_{\{1\}} = M_{\{1,2\}}$ has exactly one root $T_N^{(1)}$ with $bh < 1$, and $T_N^{(1)} = \frac{1}{b-1}[L - \log L + \log\frac{b-1}{b}] + O(\log L/L)$. The coefficient of $\log\log N$ in the bracket is 1.

(iii) Every root $T \in [L/2, 2L]$ of $M_{\{1,2\}} = M_{\{1,2,3\}}$ satisfies $T = L + \log\frac{2\sqrt{3}\,\pi b^2}{(1+2b)^2} + O(1/L)$, so the coefficient of $\log\log N$ is 0.

The proof, which computes $\det \Sigma_{\alpha^*} = 3b^2/(1+2b)^4$ from the Hessian of J_V, is in Section 5.4 of the arXiv version of this paper [40].

5.5 Face maxima uniformly over a class

The constants of Theorem 5.7 depend on Q and μ. We make them uniform over a compact class of reversible chains on a fixed graph. This is a forward statement about the face maxima, and the companion paper [41] uses it to make its finite-horizon recovery uniform over such a class.

Fix a connected graph $\mathcal{G} = ([d], \mathcal{E})$, numbers $0 < w_* \le R$ and $0 < \mu_* \le 1/d$, and a compact set $\mathcal{V} \subset \mathbb{R}^d$ of fields. A generator Q is *on* $\mathcal{G}$ if $q_{ij} > 0$ exactly when $\{i,j\} \in \mathcal{E}$. For a reversible irreducible Q with stationary law π put $w_{ij} = q_{ij}q_{ji}$ and $S_V = D_{\sqrt{\pi}}(-Q)D_{\sqrt{\pi}}^{-1}$, the matrix S_F of Proposition 3.4 for $F = V$. It is symmetric, with diagonal q_i and the entries $-\sqrt{w_{ij}}$ on edges. Let $\mathcal{C}$ be the set of reversible irreducible

generators on $\mathcal{G}$ with $\|S_V\|_2 \le R$ and $\sqrt{w_{ij}} \ge w_*$ on every edge, and let $\mathcal{M}$ be the set of starts with $\min_i \mu_i \ge \mu_*$. We use $M_F(T,v)$, $\alpha_F^*(v)$ and $K_F(v)$ as defined before Theorem 5.7.

Proposition 5.12 (uniform face maxima on the class) *Let $\mathcal{G}$ induce a connected graph on the face F, with $k = |F| \ge 2$. There are N_0 and C, depending only on $\mathcal{G}$, R, w_*, μ_* and $\mathcal{V}$, such that for all $Q \in \mathcal{C}$, $\mu \in \mathcal{M}$ and $v \in \mathcal{V}$ the following hold.*

(a) For all $N \ge 2$ and $T > 0$ with $TR \le N/2$, $M_F(T,v) \le C(T/N)^{k-1} e^{-T\lambda_F(v) + CT^2/N}$.

(b) For $N \ge N_0$ and $1 \le T \le N^{1/3}$, we have $C^{-1} \le K_F(v) \le C$ and

$$M_F(T,v) = K_F(v)\, N^{-(k-1)} T^{(k-1)/2} e^{-T\lambda_F(v)}(1+E), \qquad |E| \le C\Big(\frac{1}{T} + \frac{T^2}{N}\Big).$$

For a singleton, $M_{\{i\}}(T,v) = \mu_i(1 - Tq_i/N)^{N-1} e^{Tv_i}$ exactly.

The proof, which makes the constants of the continuum route to Theorem 5.7(b) and of (6) uniform over the compact set $\mathcal{C} \times \mathcal{M} \times \mathcal{V}$, is in Section 5.5 of the arXiv version of this paper [40].

6 The boundary layer at a facet

The local law of Theorem 5.4(a) holds on compact subsets of the open face. This section follows the occupation density of the continuous-time chain from the inside of a face F up to a facet $G = F \setminus \{i\}$, where the time spent in i is small. We state the main results for the density in Section 6.2, after the setting, and prove them in Sections 6.3 to 6.8. Section 6.9 gives the same law for the counts, with a lattice kernel.

6.1 Setting

Let F be an irreducible face with $k = |F| \ge 2$, let $i \in F$, and put $G = F \setminus \{i\}$. We assume that G satisfies (H), so G is irreducible and $\mu(G) > 0$. The start mass μ_i may be zero. For $j \in G$ put

$$c_j^- = q_{ji}, \qquad c_j^+ = q_{ij},$$

the rates into i and out of i. Both vectors are nonnegative and nonzero, since F is irreducible. For $z \in \mathbb{C}$ let

$$Q_G(z) = Q_G + z\, c^-(c^+)^\top, \qquad \mu^{(z)} = \mu_G + z\mu_i c^+, \qquad e^{(z)} = \mathbf{1} + zc^-,$$

and write $Q_G(z) = A(z) - \operatorname{diag} d(z)$ with $A(z)$ the off-diagonal part and $d(z) = q_G - z\, c^- \circ c^+$. A jump $j \to j'$ of $Q_G(z)$ of rate $zc_j^- c_{j'}^+$ stands for a visit to i that is entered from j and left to j'.

Occupation coordinates. Let E_F be the event of Lemma 2.4. On E_F let $s \in (0,\infty)^G$ be the times spent in the states of G, $S = \sum_{j \in G} s_j$, and $x = T - S$ the time spent in i. We write

$$f_F(s,x) = e^{-\langle q_G, s\rangle - q_i x} H_F(s,x), \qquad f_G(s) = e^{-\langle q_G, s\rangle} H_G(s),$$

where H_F and H_G are the sums H of Lemma 2.4 for the faces F and G. By that lemma f_F is the density of the occupation times on E_F in any $k-1$ of the coordinates, and $f_G(s)$ is the same density for G at total time S. In this notation Theorem 5.4(a) for the face G reads

$$f_G(S\beta) = S^{-(k-2)/2} C_G(\beta) e^{-S J_G(\beta)} \big(1 + O_{\mathcal{K}}(1/S)\big) \tag{13}$$

uniformly for β in a compact $\mathcal{K} \subset \Delta_G^\circ$ and $S \geq 1$, with $C_G(\beta) = \mu_j$ and $J_G = q_j$ when $G = \{j\}$. The profile on the facet is $\beta = s/S \in \Delta_G^\circ$.

The weighted chain. For real $z \geq 0$ the matrix $Q_G(z)$ is Metzler and irreducible. We let $J_G(\beta, z)$ be its rate, the supremum in Proposition 3.3 with $Q_G(z)$ in place of Q_F. Its local-law constant $C_G(\beta, z)$ is Definition 5.1 for the off-diagonal part $A(z)$, with the start $\mu^{(z)}$ in place of μ and the end vector $e^{(z)}$ in place of $\mathbf{1}$. Namely

$$C_G(\beta, z) = \frac{(\mu^{(z)} \cdot r_\beta(z))(l_\beta(z) \cdot e^{(z)})}{(2\pi)^{(k-2)/2} \sqrt{\det \Sigma_\beta(z)}}.$$

So $J_G(\beta, 0) = J_G(\beta)$ and $C_G(\beta, 0) = C_G(\beta)$.

The facet constants. Let $r_\beta = r_\beta(0)$ and $l_\beta = \beta / r_\beta$ be the vectors of Definition 5.1 for the face G, so $l_\beta \cdot r_\beta = 1$. Put

$$a_\beta = c^- \cdot l_\beta, \qquad b_\beta = c^+ \cdot r_\beta, \qquad \omega_\beta = a_\beta b_\beta, \qquad \pi^{\mathrm{in}}_{\beta,j} = \frac{c_j^- l_{\beta,j}}{a_\beta}, \qquad \pi^{\mathrm{out}}_{\beta,j} = \frac{c_j^+ r_{\beta,j}}{b_\beta}.$$

Only ω_β is intrinsic, since a_β and b_β move in opposite ways when r_β is scaled. The law π^{in}_β is the law of the state from which the tilted chain enters i, and π^{out}_β the law of the state into which it leaves i. Let $\mathcal{L}_\beta$ be the Laplacian on G with the conductances $\kappa_{jj'} = l_{\beta,j} q_{jj'} r_{\beta,j'} + l_{\beta,j'} q_{j'j} r_{\beta,j}$ for $j \neq j'$, which are twice those of the additive reversibilization $\frac{1}{2}(\widehat{Q}_\beta + \widehat{Q}^*_\beta)$ of the tilted generator $\widehat{Q}_\beta$, where $\widehat{Q}^*_\beta$ is its time reversal with respect to β. Its kernel is $\mathbb{R}\mathbf{1}$, since G is irreducible. Let $\mathcal{L}^+_\beta$ be its pseudo-inverse and

$$\gamma_\beta = q_i + \frac{\omega_\beta}{2} (\pi^{\mathrm{out}}_\beta - \pi^{\mathrm{in}}_\beta)^\top \mathcal{L}^+_\beta (\pi^{\mathrm{out}}_\beta - \pi^{\mathrm{in}}_\beta). \tag{14}$$

The quadratic form is the effective resistance of the network κ between the laws π^{in}_β and π^{out}_β. So $\gamma_\beta \geq q_i$, with equality exactly when the two laws agree. Finally put

$$\zeta = 2\sqrt{\omega_\beta S x}, \qquad z_* = \sqrt{\frac{x}{\omega_\beta S}}, \qquad \mathbf{B}_\beta(x, S) = e^{-\gamma_\beta x} \sqrt{\frac{\omega_\beta S}{x}}\, I_1(\zeta),$$

with the modified Bessel function I_1. We call $\mathbf{B}_\beta$ the Bessel kernel.

We extend the rate on F to occupation times by homogeneity of degree one, and write $\bar{J}_F(s, x) = T J_F((s, x)/T)$ with $T = S + x$. So $\bar{J}_F(S\beta, x) = S \bar{J}_F(\beta, x/S)$.

The generalized kernel. Fix a compact set $\mathcal{K} \subset \Delta_G^\circ$. Theorem 6.8 below extends $J_G(\beta, \cdot)$ and $C_G(\beta, \cdot)$ to holomorphic functions on a disc $|z| < 2z_0$, for $\beta \in \mathcal{K}$. With

these extensions we define, for $\beta \in \mathcal{K}$ and $x, S > 0$, the generalized kernel

$$\mathbf{K}_\beta(x,S) = e^{-q_i x}\,\frac{1}{2\pi \mathrm{i}} \oint_{|z|=\rho} e^{x/z}\,\frac{C_G(\beta,z)}{C_G(\beta,0)}\,e^{S(J_G(\beta)-J_G(\beta,z))}\,\frac{dz}{z^2},$$

with $0 < \rho < 2z_0$. The integrand is holomorphic on the punctured disc, so the integral does not depend on ρ.

6.2 The main results

Theorem 6.1 (the facet layer) *Let F be irreducible with $k = |F| \geq 2$, let $i \in F$, let $G = F \setminus \{i\}$ satisfy* (H)*, and let $\mathcal{K} \subset \Delta_G^\circ$ be compact. Then there are $\delta > 0$, T_0 and C such that the following hold uniformly for $\beta \in \mathcal{K}$, $T \geq T_0$ and $0 < x \leq \delta T$, with $S = T - x$ and $s = S\beta$.*

(a) $f_F(s,x) = f_G(s)\,\mathbf{K}_\beta(x,S)\,(1+E)$ *with* $|E| \leq C(T^{-2} + x^{1/2}T^{-3/2})$*, so* $|E| \leq 2C/T$*.*

(b) $f_F(s,x) = f_G(s)\,\mathbf{B}_\beta(x,S)\,e^E$ *with* $|E| \leq C\big(1/T + \sqrt{x/T}\,(1+x)\big)$*.*

(c) For the $\beta \in \mathcal{K}$ with $\pi_\beta^{\mathrm{in}} = \pi_\beta^{\mathrm{out}}$, and so for all $\beta \in \mathcal{K}$ if Q is reversible on F, we have $\gamma_\beta = q_i$, (b) holds with $|E| \leq C(1/T + \sqrt{x/T})$, and

$$f_F(s,x) = f_G(s)\,\mathbf{B}_\beta(x,S)\Big(1 + c_1(\beta)\,z_*\frac{I_0(\zeta)}{I_1(\zeta)} + E'\Big), \qquad |E'| \leq C\,\frac{1+x}{T},$$

where $c_1(\beta) = \partial_z \log C_G(\beta,z)$ at $z = 0$.

For $k = 2$, (a) holds with $E = 0$, and Proposition 6.5 gives (b) for all $x > 0$ with an explicit bound.

For reversible Q, (18) below gives $c_1(\beta)$ explicitly. Part (b) gives the relative Bessel form $f_F = f_G\mathbf{B}_\beta(1+o(1))$ for $x = o(T^{1/3})$, and beyond that a logarithmic form. The next proposition gives the exact range of the relative form. Put

$$r_\beta(\varepsilon) = \bar{J}_F(\beta,\varepsilon^2) - q_i\varepsilon^2 - J_G(\beta) + 2\sqrt{\omega_\beta}\,\varepsilon - (\gamma_\beta - q_i)\varepsilon^2.$$

So $S r_\beta(\sqrt{x/S})$ is the part of the rate $\bar{J}_F(s,x)$ that the exponent $SJ_G(\beta) + \gamma_\beta x - 2\sqrt{\omega_\beta S x}$ of $f_G\mathbf{B}_\beta$ does not capture.

Proposition 6.2 (the range of the Bessel form) *Let F, i, G and $\mathcal{K}$ be as in Theorem 6.1.*

(a) The function r_β is real-analytic near $\varepsilon = 0$, jointly with β, and $r_\beta(\varepsilon) = \varkappa_\beta\varepsilon^3 + O(\varepsilon^4)$ uniformly on $\mathcal{K}$, where

$$\varkappa_\beta = \frac{\partial_z^3 J_G(\beta,0)}{6\,\omega_\beta^{3/2}} + \frac{(\gamma_\beta - q_i)^2}{\omega_\beta^{1/2}}.$$

(b) There are $\delta > 0$, T_0 and C such that, uniformly for $\beta \in \mathcal{K}$, $T \geq T_0$ and $0 < x \leq \delta T$, with $S = T - x$ and $s = S\beta$,

$$\Big|\log\frac{f_F(s,x)}{f_G(s)\,\mathbf{B}_\beta(x,S)} + S\,r_\beta\Big(\sqrt{\frac{x}{S}}\Big)\Big| \leq C\Big(\frac{1}{T} + \sqrt{\frac{x}{T}}\Big).$$

In particular the logarithm is $-\varkappa_\beta x^{3/2}S^{-1/2} + O(1/T + \sqrt{x/T} + x^2/T)$.

(c) *If* $\pi_\beta^{\mathrm{in}} = \pi_\beta^{\mathrm{out}}$, *then* $r_\beta = 0$ *near* 0. *This holds for every* β *if* Q *is reversible on* F.

(d) *Fix* $\beta \in \mathcal{K}$ *with* $r_\beta \neq 0$, *and let* $m \geq 3$ *be the order of the first nonzero Taylor coefficient of* r_β *at* 0. *There is* $\delta_\beta > 0$ *such that for* $0 < x = x(T) \leq \delta_\beta T$, *the ratio* $f_F(s,x)/(f_G(s)\mathbf{B}_\beta(x,S))$ *tends to* 1 *as* $T \to \infty$ *if and only if* $x/T^{1-2/m} \to 0$. *If* $r_\beta = 0$ *near* 0, *the ratio tends to* 1 *whenever* $x/T \to 0$.

(e) *For the cycle with* $q_{12} = q_{23} = q_{31} = 2$ *and* $q_{21} = q_{32} = q_{13} = 1$, *a start with* $\mu_1 + \mu_2 > 0$, $i = 3$ *and* $\beta = (\frac{1}{2}, \frac{1}{2})$,

$$\omega_\beta = 2 + \tfrac{9}{4}\sqrt{2}, \qquad \partial_z^3 J_G(\beta, 0) = -\tfrac{1323}{64}\sqrt{2}, \qquad \varkappa_\beta = -\frac{49}{64\sqrt{\omega_\beta}} = -0.3363\ldots,$$

so $m = 3$ *there, and the ratio tends to infinity when* $x/T^{1/3} \to \infty$ *with* $x \leq \delta_\beta T$.

The condition in (c) is weaker than reversibility. Let G be the cycle of (e), with $q_{12} = q_{23} = q_{31} = 2$ and $q_{21} = q_{32} = q_{13} = 1$, and add a fourth state i with $q_{ji} = q_{ij} = c > 0$ for $j \in G$, and a start with $\mu(G) > 0$. The chain on $F = G \cup \{i\}$ is not reversible, since the products of the rates around the cycle are 8 and 1. At the uniform β, the criterion in the proof of Proposition 3.3 gives $r_\beta \propto \mathbf{1}$, since the uniform law is stationary for the cycle, and then $l_\beta \propto \beta$. So π_β^{in} and π_β^{out} are both uniform, and the relative Bessel form holds for $x = o(T)$ there. At other β the two laws differ.

Strategy of the proofs. The proofs have three steps. First, pulling the visits to i out of the run expansion writes f_F as a contour integral of the density $f_G^{(z)}$ of a weighted chain on G, in which z counts the visits to i (Lemma 6.3). Second, a local law for the weighted chain, uniform for complex z near 0, replaces $f_G^{(z)}(S\beta)$ by $S^{-(k-2)/2} C_G(\beta, z) e^{-S J_G(\beta,z)}$ (Theorem 6.8). Third, a saddle point on the circle $|z| = z_*$ evaluates the integral, and the expansion of $J_G(\beta, z)$ at $z = 0$ (Proposition 6.6) compares the result with the Bessel kernel. Sections 6.3 to 6.6 give the exact formulas, the constants and the matching of the prefactor, Section 6.7 the second step, and Section 6.8 the third.

6.3 The exact representation

Lemma 6.3 (the runs in i pulled out) *For* $z \in \mathbb{C}$ *and* $s \in (0,\infty)^G$ *let*

$$f_G^{(z)}(s) = e^{-\langle d(z), s\rangle} \sum_{m \in \mathbb{Z}_{\geq 1}^G} W_z(m) \prod_{j \in G} \frac{s_j^{m_j - 1}}{(m_j - 1)!},$$

$$W_z(m) = \sum_w \mu_{w_1}^{(z)} \prod_{l < |w|} A(z)_{w_l w_{l+1}}\, e_{w_{|w|}}^{(z)},$$

the inner sum over the words w *in* G *with no two equal neighbors and* m_j *letters* j. *Then* $z \mapsto f_G^{(z)}(s)$ *is entire,* $f_G^{(0)} = f_G$, *and for* $x > 0$ *and every* $\rho > 0$

$$f_F(s,x) = e^{-q_i x} \sum_{K \geq 1} \frac{x^{K-1}}{(K-1)!} [z^K] f_G^{(z)}(s) \tag{15}$$

$$= e^{-q_i x} \frac{1}{2\pi \mathrm{i}} \oint_{|z| = \rho} e^{x/z} f_G^{(z)}(s) \frac{dz}{z^2}. \tag{16}$$

So $f_G^{(z)}$ is the killed density on the facet of the chain in which each visit to i is replaced by an instantaneous jump $j \to j'$ of rate $z c_j^- c_{j'}^+$, and z counts the visits. The K visits have total length x, with density $q_i^K x^{K-1} e^{-q_i x}/(K-1)!$, and each return carries a factor $c_{j'}^+/q_i$. So the powers of q_i cancel.

Proof Fix a word c in F with no equal neighbors, every letter of F, and $K = m_i(c) \geq 1$ letters i. Its term in $H_F(s,x)$ is $\mu_{c_1} \prod q_{c_l c_{l+1}}$ times $x^{K-1}/(K-1)!$ times $\prod_{j \in G} s_j^{m_j - 1}/(m_j - 1)!$. The letters i are isolated in c. We delete them and mark the places where they were. The result is a marked word c' in G, in which two equal neighbors occur exactly where c had $j\,i\,j$. A mark between letters j and j' carries $q_{ji} q_{ij'} = c_j^- c_{j'}^+$. A mark before the first letter j' carries $\mu_i c_{j'}^+$, and a mark after the last letter j carries c_j^-. Two neighbors without a mark are different and carry $q_{jj'}$. So the weight of c is the product of these weights. Between two different letters of c' there is at most one mark, and between two equal letters exactly one.

We group the marked words by their reduced word w, obtained by merging each block of equal letters of c' into one letter, and by the numbers $n_1, n_2, \ldots$ of marks inside the blocks. If j forms ϱ blocks with n inner marks in all, it has $\varrho + n$ letters in c', and there are $\binom{n+\varrho-1}{\varrho-1}$ ways to spread the marks. Since $\binom{n+\varrho-1}{\varrho-1}/(n+\varrho-1)! = 1/(n!(\varrho-1)!)$,

$$\sum_{n \geq 0} \binom{n+\varrho-1}{\varrho-1} \frac{y^n s^{n+\varrho-1}}{(n+\varrho-1)!} = e^{ys} \frac{s^{\varrho-1}}{(\varrho-1)!}.$$

With $y = z c_j^- c_j^+$, one power of z for each mark, the inner marks give the factor $e^{z c_j^- c_j^+ s_j}$, which is the diagonal part of $d(z)$. The marks between different letters give $A(z)$, and the first and last marks give $\mu^{(z)}$ and $e^{(z)}$. Collecting the powers of z gives (15). The entries of $A(z)$, $\mu^{(z)}$ and $e^{(z)}$ are at most $q_{\max}(1 + |z| q_{\max})$, $1 + |z| q_{\max}$ and $1 + |z| q_{\max}$ in modulus. So the bound $W(m) \leq k^M q_{\max}^{M-1}$ of Section 2 holds for W_z with $q_{\max}$ replaced by $q_{\max}(1 + |z| q_{\max})$, up to a factor $(1 + |z| q_{\max})^2$, and the series for $f_G^{(z)}(s)$ converges locally uniformly in z. Hence it is entire. For $z = 0$ no word carries a mark, and $f_G^{(0)}$ is the sum of Lemma 2.4 for G. Finally we expand $e^{x/z} = \sum_{n \geq 0} x^n z^{-n}/n!$ and integrate term by term on the circle. This gives $\sum_{n \geq 0} (x^n/n!)[z^{n+1}] f_G^{(z)}(s)$, which is (15). □

The discrete law has the same structure. The only change is that the visits to i come in blocks of steps, whose lengths form a composition of n_i.

Lemma 6.4 (the discrete form) *Let $h q_{\max} \leq 1$, $\operatorname{supp} n = F$, $N' = N - n_i$, and let $\hat{n}$ be n restricted to G. For $z \in \mathbb{C}$ let*

$$P_{N'}^{(z)}(\hat{n}) = \sum_x \mu_{x_1}^{h,(z)} \prod_{t < N'} \left(I + h Q_G + z h^2 c^- (c^+)^\top\right)_{x_t x_{t+1}} e_{x_{N'}}^{h,(z)},$$

the sum over the words $x_1 \cdots x_{N'}$ in G with composition $\hat{n}$, where $\mu^{h,(z)} = \mu_G + z h \mu_i c^+$ and $e^{h,(z)} = \mathbf{1} + z h c^-$. Then $P_{N'}^{(0)}(\hat{n})$ is the probability of the composition $\hat{n}$ for the chain run for N' steps, and

$$P_N(n) = \sum_{K \geq 1} \binom{n_i - 1}{K - 1} (1 - h q_i)^{n_i - K} [z^K] P_{N'}^{(z)}(\hat{n}).$$

Proof We delete the maximal blocks of i from a path with composition n. What is left is a path $x_1 \cdots x_{N'}$ in G with composition $\hat{n}$. A block of length ℓ between $x_t = j$ and $x_{t+1} = j'$ replaces the factor $(I + hQ)_{jj'}$ by $hq_{ji}(1 - hq_i)^{\ell-1} hq_{ij'}$. A block at the start replaces μ_{x_1} by $\mu_i(1 - hq_i)^{\ell-1} hq_{ix_1}$, and a block at the end adds the factor $hq_{x_{N'}i}(1 - hq_i)^{\ell-1}$. We count the blocks by z. If there are K blocks, the factors $(1 - hq_i)^{\ell-1}$ give $(1 - hq_i)^{n_i - K}$, and the block lengths form one of the $\binom{n_i - 1}{K-1}$ compositions of n_i into K positive parts. □

6.4 Pair faces

For $k = 2$ the facet is one state and the weighted chain has no words to sum. The contour formula then gives the law in closed form.

Proposition 6.5 (pair faces) *Let $F = \{j, i\}$ with $q_{ji}, q_{ij} > 0$ and $\mu_j > 0$, and put $\omega = q_{ji}q_{ij}$ and $\zeta = 2\sqrt{\omega s x}$. For all $s, x > 0$*

$$f_F(s,x) = \mu_j e^{-q_j s - q_i x} \Big[\sqrt{\frac{\omega s}{x}}\, I_1(\zeta) + \Big(q_{ji} + \frac{\mu_i}{\mu_j} q_{ij} \Big) I_0(\zeta) + \frac{\mu_i}{\mu_j} \sqrt{\frac{\omega x}{s}}\, I_1(\zeta) \Big].$$

Here $\omega_\beta = \omega$, $\gamma_\beta = q_i$ and $f_G(s) = \mu_j e^{-q_j s}$, and for all $s, x > 0$

$$0 \le \frac{f_F(s,x)}{f_G(s)\mathbf{B}_\beta(x,s)} - 1 \le \Big(q_{ji} + \frac{\mu_i}{\mu_j} q_{ij} \Big) \Big(\frac{1}{\omega s} + \sqrt{\frac{x}{\omega s}} \Big) + \frac{\mu_i}{\mu_j}\, \frac{x}{s}.$$

This is the facet form of Pedler's formula for two-state chains [37]. For $x \le T/2$ the bound is $O(1/T + \sqrt{x/T})$, and the term $\sqrt{x/T}$ is attained, since the middle term of the formula has exactly this size.

Proof For $G = \{j\}$ the only word is j, so $f_G^{(z)}(s) = (\mu_j + z\mu_i q_{ij})(1 + z q_{ji}) e^{-(q_j - z\omega)s}$. With $z = z_* t$ and $z_* = \sqrt{x/(\omega s)}$ we have $x/z + \omega s z = \frac{\zeta}{2}(t + 1/t)$, and the generating function $e^{(\zeta/2)(t+1/t)} = \sum_{n \in \mathbb{Z}} I_n(\zeta) t^n$ [36, Eq. 10.35.1] gives

$$\frac{1}{2\pi \mathrm{i}} \oint e^{x/z + \omega s z} z^{n-2}\, dz = z_*^{n-1} I_{n-1}(\zeta) \qquad (n \in \mathbb{Z}). \tag{17}$$

We put this into (16) for $n = 0, 1, 2$ and use $I_{-1} = I_1$. This gives the formula. Each term is nonnegative, which gives the lower bound. For the upper bound we divide the last two terms by the first. This gives $(q_{ji} + \frac{\mu_i}{\mu_j} q_{ij}) z_* I_0(\zeta)/I_1(\zeta) + \frac{\mu_i}{\mu_j} x/s$. The lower bound $I_1(\zeta)/I_0(\zeta) \ge \zeta/(1 + \sqrt{1 + \zeta^2})$ of Amos [1] gives $I_0/I_1 \le (1 + \sqrt{1 + \zeta^2})/\zeta \le 1 + 2/\zeta$, and $z_*(1 + 2/\zeta) = \sqrt{x/(\omega s)} + 1/(\omega s)$. For a pair $\beta = (1)$, the vectors of the facet are $r_\beta = l_\beta = 1$, so $\omega_\beta = \omega$ and $\pi_\beta^{\mathrm{in}} = \pi_\beta^{\mathrm{out}}$, which gives $\gamma_\beta = q_i$. □

6.5 The constants for every irreducible chain

For $u \in (0, \infty)^G$ put

$$\Psi_\beta(u) = \sum_{j \in G} \beta_j \Big(q_j - \sum_{j' \in G \setminus \{j\}} q_{jj'} \frac{u_{j'}}{u_j} \Big), \qquad P_\beta(u) = (c^+ \cdot u) \sum_{j \in G} \frac{c_j^- \beta_j}{u_j}.$$

Since $\sum_j \beta_j (-(Q_G(z)u)_j / u_j) = \Psi_\beta(u) - z P_\beta(u)$, the rate of the weighted chain is $J_G(\beta, z) = \sup_u [\Psi_\beta(u) - z P_\beta(u)]$ for real $z \ge 0$.

Proposition 6.6 (the facet expansion of the rate) *Let F be irreducible, $i \in F$, and $G = F \setminus \{i\}$ irreducible.*

(i) For $s = S\beta$ with $\beta \in \Delta_G^\circ$ and $x > 0$,

$$\bar{J}_F(s,x) = q_i x + \sup_{z>0}\big[S\, J_G(\beta,z) - x/z\big].$$

(ii) The map $z \mapsto J_G(\beta,z)$ is real-analytic near $z = 0$, jointly with β, and

$$J_G(\beta,0) = J_G(\beta), \qquad \partial_z J_G(\beta,0) = -\omega_\beta, \qquad \partial_z^2 J_G(\beta,0) = 2\omega_\beta(\gamma_\beta - q_i).$$

(iii) There are $\delta > 0$ and C such that, uniformly for β in a compact subset of Δ_G° and $0 < x \le \delta S$,

$$\Big|\bar{J}_F(s,x) - \big(S\, J_G(\beta) + q_i x - 2\sqrt{\omega_\beta S x} + (\gamma_\beta - q_i)x\big)\Big| \le C\, x^{3/2} S^{-1/2}.$$

(iv) If $\nu_j q_{jj'} = \nu_{j'} q_{j'j}$ on F for some $\nu \in (0,\infty)^F$, then $\pi_\beta^{\mathrm{in}} = \pi_\beta^{\mathrm{out}}$, $\gamma_\beta = q_i$, $J_G(\beta,z) = J_G(\beta) - z\omega_\beta$ for all real $z \ge 0$, $\omega_\beta = \big(\sum_{j\in G}\sqrt{q_{ij}q_{ji}\beta_j}\big)^2$, and for all $s = S\beta$ and $x > 0$

$$\bar{J}_F(s,x) = S\, J_G(\beta) + q_i x - 2\sqrt{\omega_\beta S x}.$$

(v) The map $z \mapsto J_G(\beta,z)$ is affine on $[0,\infty)$ if and only if $\pi_\beta^{\mathrm{in}} = \pi_\beta^{\mathrm{out}}$. Then $J_G(\beta,z) = J_G(\beta) - z\omega_\beta$ for all real $z \ge 0$, and $\bar{J}_F(s,x) = S\, J_G(\beta) + q_i x - 2\sqrt{\omega_\beta S x}$ for all $s = S\beta$ and $x > 0$.

So in the visit fraction α_i the rate has a square-root singularity $-2\sqrt{\omega_\beta \alpha_i}$ at the facet, and the time in i is charged at the rate γ_β, not q_i.

Proof (i) In the supremum that defines $\bar{J}_F(s,x)$ we write $u = (u_G, \epsilon)$ with $\epsilon = u_i$. The terms with ϵ are $-\epsilon\sum_j s_j c_j^-/u_j - (x/\epsilon)(c^+ \cdot u_G)$, and both coefficients are positive. Their supremum over $\epsilon > 0$ is $-2\sqrt{xS\,P_\beta(u_G)}$. So $\bar{J}_F(s,x) = q_i x + \sup_u[S\Psi_\beta(u) - 2\sqrt{xSP_\beta(u)}]$. Now $2\sqrt{xSP} = \inf_{z>0}[SzP + x/z]$, and we exchange the two suprema.

(ii) Write $u = e^\psi$. For real $z \ge 0$ the function $\psi \mapsto \Psi_\beta(e^\psi) - zP_\beta(e^\psi)$ is concave, since $P_\beta(e^\psi)$ is a positive combination of the convex functions $e^{\psi_m - \psi_j}$, and it does not change when a constant is added to ψ. At $z = 0$ its maximizer is $\psi_0 = \log r_\beta$, unique up to a constant (Proposition 3.3(b)). The Hessian of $\sum_{j\neq j'}\beta_j q_{jj'} e^{\psi_{j'} - \psi_j}$ at ψ_0 is the Laplacian with the conductances $\kappa_{jj'}$, because $\beta_j q_{jj'} r_{\beta,j'}/r_{\beta,j} = l_{\beta,j} q_{jj'} r_{\beta,j'}$. So the Hessian of Ψ_β at ψ_0 is $-\mathcal{L}_\beta$, which is negative definite on the complement of the constants. By the implicit function theorem on that complement, the critical point $\psi(z)$ is real-analytic in (β,z) near $(\beta,0)$, and for real $z \ge 0$ it is the maximizer by concavity. So $J_G(\beta,z) = \Psi_\beta(\psi(z)) - zP_\beta(\psi(z))$ is real-analytic. The envelope theorem gives $\partial_z J_G = -P_\beta(\psi(z))$, so $\partial_z J_G(\beta,0) = -(c^+ \cdot r_\beta)(c^- \cdot l_\beta) = -\omega_\beta$. Differentiating the critical equation gives $-\mathcal{L}_\beta \psi'(0) = \nabla P$, with ∇P the gradient of $P_\beta(e^\psi)$ at ψ_0. Its entries are

$$\partial_{\psi_m} P = c_m^+ r_{\beta,m}\,(c^- \cdot l_\beta) - (c^+ \cdot r_\beta)\, c_m^- l_{\beta,m} = \omega_\beta(\pi_{\beta,m}^{\mathrm{out}} - \pi_{\beta,m}^{\mathrm{in}}),$$

which sum to zero. So $\partial_z^2 J_G(\beta,0) = -\nabla P \cdot \psi'(0) = \nabla P^\top \mathcal{L}_\beta^+ \nabla P$, and this is $2\omega_\beta(\gamma_\beta - q_i)$ by (14).

(iii) By the proof of (i), $\bar{J}_F(s,x) - q_i x = S\sup_\psi[\Psi_\beta(\psi) - 2\varepsilon P_\beta(\psi)^{1/2}]$ with $\varepsilon = \sqrt{x/S}$. The maximizer lies in a bounded set of ψ modulo constants, uniformly for β in a compact set, as in the proof of Proposition 3.3(b). Comparing its value with the value at ψ_0 gives

$\Psi_\beta(\psi_0) - \Psi_\beta(\psi^*) \le 2\varepsilon \sup P_\beta^{1/2}$ on that set. We normalize ψ and ψ_0 to sum to zero. The Hessian $-\mathcal{L}_\beta$ at ψ_0 is negative definite on these vectors and continuous in β, so Taylor's theorem gives $c, r > 0$ with $\Psi_\beta(\psi_0) - \Psi_\beta(\psi) \ge c|\psi - \psi_0|^2$ for $|\psi - \psi_0| \le r$, uniformly on the compact set of β. For $|\psi - \psi_0| > r$, concavity along the segment from ψ_0 to ψ gives $\Psi_\beta(\psi_0) - \Psi_\beta(\psi) \ge cr|\psi - \psi_0| \ge cr^2$. Hence $\Psi_\beta(\psi_0) - \Psi_\beta(\psi) \ge c\min(|\psi - \psi_0|^2, r^2)$, which gives $\psi^* - \psi_0 = O(\varepsilon^{1/2})$. The critical equation $\nabla\Psi_\beta(\psi^*) = \varepsilon\nabla P_\beta(\psi^*)/\sqrt{P_\beta(\psi^*)}$ has a right side of size $O(\varepsilon)$, and $\nabla^2\Psi_\beta$ is invertible near ψ_0 on the complement of the constants, so $\psi^* - \psi_0 = O(\varepsilon)$. We expand to second order around ψ_0. With $g = \nabla(P^{1/2}) = \nabla P/(2\sqrt{\omega_\beta})$, the maximizer is $\psi^* = \psi_0 - 2\varepsilon\mathcal{L}_\beta^+ g + O(\varepsilon^2)$, and the value is $J_G(\beta) - 2\varepsilon\sqrt{\omega_\beta} + 2\varepsilon^2 g^\top \mathcal{L}_\beta^+ g + O(\varepsilon^3)$. Here $2g^\top\mathcal{L}_\beta^+ g = \nabla P^\top \mathcal{L}_\beta^+ \nabla P/(2\omega_\beta) = \gamma_\beta - q_i$ by (ii). Multiplying by S gives (iii), since $S\varepsilon^3 = x^{3/2}S^{-1/2}$.

(iv) By Proposition 5.2(b) for the face G we may take $r_{\beta,j} = \sqrt{\beta_j/\nu_j}$ and $l_{\beta,j} = \sqrt{\beta_j\nu_j}$. Then $c_j^- l_{\beta,j} = q_{ji}\sqrt{\beta_j\nu_j} = \nu_i q_{ij}\sqrt{\beta_j/\nu_j} = \nu_i c_j^+ r_{\beta,j}$, so $\pi_\beta^{\text{in}} = \pi_\beta^{\text{out}}$ and $\gamma_\beta = q_i$. More is true. For every real $z \ge 0$ the matrix $Q_G(z)$ is reversible with respect to ν_G, since $\nu_j c_j^- c_{j'}^+ = \nu_i q_{ij} q_{ij'}$ is symmetric in j, j'. So by Proposition 5.2(b) its optimal vector is r_β for every z, and $J_G(\beta, z) = \Psi_\beta(r_\beta) - zP_\beta(r_\beta) = J_G(\beta) - z\omega_\beta$. Next $\omega_\beta = \nu_i\big(\sum_j q_{ij}\sqrt{\beta_j/\nu_j}\big)^2 = \big(\sum_j \sqrt{q_{ij}q_{ji}\beta_j}\big)^2$, using $q_{ji} = q_{ij}\nu_i/\nu_j$. Finally (i) gives $\sup_{z>0}[-Sz\omega_\beta - x/z] = -2\sqrt{\omega_\beta Sx}$.

(v) If the two laws agree, the gradient ∇P in the proof of (ii) vanishes. So ψ_0 is a critical point of the concave function $\psi \mapsto \Psi_\beta(e^\psi) - zP_\beta(e^\psi)$ for every real $z \ge 0$, hence a maximizer, and $J_G(\beta, z) = \Psi_\beta(r_\beta) - zP_\beta(r_\beta) = J_G(\beta) - z\omega_\beta$. Then (i) gives $\bar{J}_F$ as in (iv). Conversely, if $J_G(\beta, \cdot)$ is affine, then $\partial_z^2 J_G(\beta, 0) = 0$, so $\gamma_\beta = q_i$ by (ii), and the two laws agree, since the effective resistance in (14) vanishes only when they do. □

For $1/T \ll x \ll T$ the Bessel kernel is $(4\pi)^{-1/2}(\omega_\beta S)^{1/4}x^{-3/4}\exp(2\sqrt{\omega_\beta Sx} - \gamma_\beta x)(1 + O(1/\zeta))$, by $I_1(\zeta) = e^\zeta(2\pi\zeta)^{-1/2}(1 + O(1/\zeta))$ [36, Eq. 10.40.1]. So Proposition 6.6(iii) says that $f_G(s)\mathbf{B}_\beta(x, S)$ and the local law of F, which in occupation coordinates is $T^{-(k-1)/2}C_F(\alpha)e^{-TJ_F(\alpha)}$, have the same exponent up to $O(x^{3/2}S^{-1/2})$. For their powers of x and their constants to agree as well we need $C_F((1-\epsilon)\beta, \epsilon) \sim (4\pi)^{-1/2}\omega_\beta^{1/4}\epsilon^{-3/4}C_G(\beta)$ as $\epsilon \to 0$. The next proposition proves this for reversible Q.

6.6 Prefactor matching

Proposition 6.7 (prefactor matching) *Let Q be reversible on F with respect to ν, and let G satisfy (H), with $|G| \ge 1$. Uniformly for β in a compact subset of Δ_G°, as $\epsilon \to 0$,*

$$C_F((1-\epsilon)\beta, \epsilon) = (4\pi)^{-1/2}\,\omega_\beta^{1/4}\,\epsilon^{-3/4}\,C_G(\beta)\big(1 + O(\sqrt{\epsilon})\big),$$

with $C_G(\beta) = \mu_j$ when $G = \{j\}$.

The proof, by the tree formula of Proposition 5.2(b), is in Section 6.6 of the arXiv version of this paper [40].

The first amplitude correction. Put $c_1(\beta) = \partial_z \log C_G(\beta, z)$ at $z = 0$. For reversible Q,

$$c_1(\beta) = \frac{\mu_i b_\beta}{\mu_G \cdot r_\beta} + \frac{a_\beta}{l_\beta \cdot \mathbf{1}} + \frac{\omega_\beta}{2}\sum_{\{j,j'\}\subseteq G} R_{jj'}\pi_{\beta,j}\pi_{\beta,j'}, \tag{18}$$

with $\pi_\beta = \pi_\beta^{\mathrm{in}} = \pi_\beta^{\mathrm{out}}$ and $R_{jj'}$ the effective resistance for the conductances $l_{\beta,j} q_{jj'} r_{\beta,j'}$. The proof, by the tree formula and Kirchhoff's rule $\partial \log \mathrm{tree}(c)/\partial c_e = R_e$, is in Section 6.6 of the arXiv version.

6.7 The weighted facet chain

The layer needs the local law of the weighted chain $Q_G(z)$, uniformly for complex z near 0.

Theorem 6.8 (the weighted facet chain) *Let G satisfy* (H) *with $|G| = k-1 \geq 1$, let $i \notin G$ with $F = G \cup \{i\}$ irreducible, and let $\mathcal{K} \subset \Delta_G^\circ$ be compact. There are a number $z_0 > 0$, not given explicitly, and a constant C, both depending only on Q, μ, i and $\mathcal{K}$, and functions $J_G(\beta, z)$ and $C_G(\beta, z)$ on $\mathcal{K} \times \{|z| < 2z_0\}$, holomorphic in z, jointly continuous, real for real z, equal to the rate and the local-law constant of Section 6.1 for real $0 \leq z < 2z_0$, and with $C_G(\beta, 0) = C_G(\beta) > 0$, such that uniformly for $\beta \in \mathcal{K}$, $S \geq 1$ and complex $|z| \leq z_0$*

$$f_G^{(z)}(S\beta) = S^{-(k-2)/2}\, C_G(\beta, z)\, e^{-S\, J_G(\beta,z)}\, (1+\varepsilon), \qquad |\varepsilon| \leq C/S. \tag{19}$$

For $|G| = 1$ this holds with $\varepsilon = 0$.

At $z = 0$ the display is the local law (13), that is Theorem 5.4(a) for the face G. The proof inherits every constant from $z = 0$ by continuity, which is why z_0 is not explicit. The rate $J_G(\beta, z)$ is defined for all real $z \geq 0$, but for complex z the proof needs a critical point of the torus phase near the real one and a Hessian with positive definite real part, and neither need persist far from $z = 0$. The number δ of Theorem 6.1 is chosen so that the saddle radius z_* stays below $z_0/2$.

Lemma 6.9 (one and two states) *Theorem 6.8 holds when $|G| \leq 2$, with the functions J_G and C_G given in the proof.*

For $G = \{j\}$ the functions are $J_G(\beta, z) = q_j - z\omega$ and $C_G(\beta, z) = (\mu_j + z\mu_i q_{ij})(1 + zq_{ji})$, with $\varepsilon = 0$. For $G = \{1, 2\}$, $J_G(\beta, z) = \langle d(z), \beta \rangle - 2\sqrt{\beta_1 \beta_2 \varpi(z)}$ with $\varpi(z) = (q_{12} + zc_1^- c_2^+)(q_{21} + zc_2^- c_1^+)$. The proof, which sums the alternating words with the series of I_0 and I_1 and gives C_G for $G = \{1, 2\}$ in closed form, is in Section 6.7 of the arXiv version of this paper [40].

Facets with two or more states. Let $|G| = k-1 \geq 2$. We number the states of G as $1, \ldots, k-1$ and put $m = k-2$. For $z \in \mathbb{C}$ let $A'(z)$ be $A(z)$ without the row and column of state 1, and let X_z, g_z and b_z be the functions of Lemma A.3 on G for the weights $A(z)$, $\mu^{(z)}$ and $e^{(z)}$. Since c^-, c^+ and μ_i are nonnegative, the weights $A(|z|)$, $\mu^{(|z|)}$ and $e^{(|z|)}$ dominate these in the sense of that lemma. For fixed $\beta \in \Delta_G^\circ$ and $z' \in \mathbb{C}^m$ put

$$\Phi_z(z') = \beta_1 g_z(z') + \sum_{j \geq 2} \beta_j z_j .$$

Let $\theta(\beta)$ be the tilt of the face G at β, let $\theta(\beta, z)$ be that of $Q_G(z)$ for real $z \geq 0$, and let $\mathcal{H}(\beta)$ be the Hessian $\mathcal{H}$ of the proof of Theorem 5.4(a) for the face G at $\alpha = \beta$.

On $\mathbb{C}^m$ we use the norm $\|w\| = \max_j |w_j|$ with closed balls $\bar B(w,r)$, and $we^{\mathrm{i}\varphi}$ is the vector with entries $w_j e^{\mathrm{i}\varphi_j}$. The proof of Theorem 5.4(a) for G gives, for every $\beta \in \Delta_G^\circ$,

(P1) $\varrho(A'(0) - \operatorname{diag}\theta'(\beta)) < 0$ and $g_0(\theta'(\beta)) = \theta_1(\beta)$, by (A3),
(P2) $\nabla\Phi_0(\theta'(\beta)) = 0$, and $\nabla^2\Phi_0(\theta'(\beta)) = \mathcal{H}(\beta)$ is real, symmetric and positive definite,
(P3) $\operatorname{Re}\Phi_0(\theta'(\beta)e^{\mathrm{i}\varphi}) \le \Phi_0(\theta'(\beta)) - \sum_{j\ge 2}\beta_j\theta_j(\beta)(1-\cos\varphi_j)$ for $\varphi \in \mathbb{T}^m$,
(P4) $b_0(\theta'(\beta)) > 0$.

Also θ and $\mathcal{H}$ are continuous, by Proposition 5.2(a).

Lemma 6.10 (the complex critical point) *Let $|G| \ge 2$ and let $\mathcal{K} \subset \Delta_G^\circ$ be compact. There are $z_1 > 0$, $r_0 \ge r_1 > 0$, $\sigma_0 > 0$ and B_0, which depend only on Q, μ, i and $\mathcal{K}$, such that the following hold for every $\beta \in \mathcal{K}$.*

(a) $r_0 < \frac14 \min_j \theta_j(\beta)$, and $\varrho(A'(|z|) - \operatorname{diag} s') < 0$ for $|z| \le z_1$ and real s' with $\|s' - \theta'(\beta)\| \le 3r_0$.

(b) The functions $g_z(z')$ and $b_z(z')$ are holomorphic in (z,z') and bounded in modulus by B_0 on $\{|z| < z_1\} \times U_\beta$, where U_β is the set of z' with $|z_j| > \theta_j(\beta) - 3r_0$ for all j.

(c) For $|z| < z_1$ there is exactly one $\zeta'(\beta,z) \in \bar B(\theta'(\beta), r_1)$ with $\nabla\Phi_z(\zeta'(\beta,z)) = 0$. It is holomorphic in z and jointly continuous in (β,z), $\zeta'(\beta,0) = \theta'(\beta)$, $\|\zeta'(\beta,z) - \theta'(\beta)\| \le B_0|z|$, and $\zeta'(\beta,\bar z) = \overline{\zeta'(\beta,z)}$.

(d) The matrix $\mathcal{J}(\beta,z) = Z_c\nabla^2\Phi_z(\zeta'(\beta,z))Z_c$, where $Z_c = \operatorname{diag}\zeta'(\beta,z)$, is complex symmetric and jointly continuous, and $\mathcal{J}(\beta,0) = \Theta'\mathcal{H}(\beta)\Theta' \ge \sigma_0 I$, where $\Theta' = \operatorname{diag}\theta'(\beta)$.

(e) For real $0 \le z < z_1$ we have $\zeta'(\beta,z) = \theta'(\beta,z)$, $\Phi_z(\zeta'(\beta,z)) = \langle\beta,\theta(\beta,z)\rangle$, $\beta_1^2 \det\mathcal{J}(\beta,z) = \prod_{j\ge2}\theta_j(\beta,z)^2 \det\Sigma_\beta(z)$ and $\beta_1 b_z(\zeta'(\beta,z)) = (\mu^{(z)}\cdot r_\beta(z))(l_\beta(z)\cdot e^{(z)})$.

The proof, which finds $\zeta'(\beta,z)$ as the fixed point of the contraction $z' \mapsto z' - \mathcal{H}(\beta)^{-1}\nabla\Phi_z(z')$ of $\bar B(\theta'(\beta), r_1)$, is in Section 6.7 of the arXiv version of this paper [40].

Proof of Theorem 6.8 For $|G| = 1$ this is Lemma 6.9. Let $|G| \ge 2$, and let z_1, r_0, r_1, σ_0 and B_0 be as in Lemma 6.10. For $\beta \in \mathcal{K}$ and $|z| < z_1$ write $\zeta' = \zeta'(\beta,z)$ and $\rho_j = |\zeta_j|$, and put

$$\Psi(\beta,z,\varphi) = \Phi_z(\zeta' e^{\mathrm{i}\varphi}), \qquad \chi(\beta,z,\varphi) = b_z(\zeta' e^{\mathrm{i}\varphi}) \prod_{j\ge2} \zeta_j e^{\mathrm{i}\varphi_j}.$$

Step 1. The constants. Since θ_j is real, $|\rho_j - \theta_j| \le |\zeta_j - \theta_j| \le r_1$. So the points $\zeta' e^{\mathrm{i}\varphi}$ lie in U_β, and by Lemma 6.10(b), (c) the φ-derivatives of Ψ and χ are jointly continuous on the compact set $P_1 \times \mathbb{T}^m$, where $P_1 = \mathcal{K} \times \{|z| \le z_1/2\}$. Let B_Ψ and B_a bound those of Ψ up to order 4 and those of χ up to order 2. Since $\partial/\partial\varphi_j = \mathrm{i}w_j\,\partial/\partial w_j$ at $w = \zeta' e^{\mathrm{i}\varphi}$ and $\nabla\Phi_z(\zeta') = 0$, we have $\nabla_\varphi\Psi(\beta,z,0) = 0$ and $-\nabla_\varphi^2\Psi(\beta,z,0) = \mathcal{J}(\beta,z)$, as in the proof of Theorem 5.4(a). Fix $\varepsilon \in (0,1)$ with $m^3 B_\Psi \varepsilon \le \frac34\sigma_0$, and put $2\eta = \beta_{\min}\theta_{\min}(1-\cos\varepsilon)$, where $\beta_{\min}$ and $\theta_{\min}$ are the least entries of β and $\theta(\beta)$ over $\mathcal{K}$. By (P3), $\operatorname{Re}\Psi(\beta,0,\varphi) - \Psi(\beta,0,0) \le -2\eta$ for $\|\varphi\| \ge \varepsilon$. By uniform continuity on $P_1 \times \mathbb{T}^m$ we can choose $z_0 \in (0, z_1/4]$ such that for $\beta \in \mathcal{K}$ and $|z| \le 2z_0$

(i) $\operatorname{Re}\mathcal{J}(\beta,z) \ge \frac12\sigma_0 I$, since $\|\mathcal{J}(\beta,z) - \mathcal{J}(\beta,0)\| \le \frac12\sigma_0$ in the operator norm and $\mathcal{J}(\beta,0)$ is real,

(ii) $|\chi(\beta,z,0)|\ge c_\chi$, where $2c_\chi>0$ is the minimum over $\mathcal{K}$ of $\chi(\beta,0,0)=b_0(\theta')\prod_{j\ge2}\theta_j$, which is positive by (P4),

(iii) $\operatorname{Re}\Psi(\beta,z,\varphi)-\operatorname{Re}\Psi(\beta,z,0)\le-\eta$ for $\|\varphi\|\ge\varepsilon$.

The numbers z_0, ε, η and c_χ depend only on Q, μ, i and $\mathcal{K}$.

Step 2. The torus through the critical point. Let $\beta\in\mathcal{K}$, $|z|\le z_0$ and $S\ge1$. Since $\|\rho'-\theta'\|\le r_0$, Lemma 6.10(a) shows that the radii ρ' are admissible in Lemma A.3 with the dominating weights at $|z|$. With $t=S\beta$ it gives

$$e^{S\langle d(z),\beta\rangle}f_G^{(z)}(S\beta)=\frac{1}{(2\pi)^m}\int_{\mathbb{T}^m}\exp\bigl(S\Phi_z(w)\bigr)\,b_z(w)\prod_{j\ge2}w_j\,d\varphi,\qquad w_j=\rho_je^{\mathrm{i}\varphi_j}.$$

The translation of φ by the arguments of the ζ_j preserves the measure and turns w into $\zeta'e^{\mathrm{i}\varphi}$. So the integral is $\int_{\mathbb{T}^m}e^{S\Psi}\chi\,d\varphi$. No contour is moved. The torus representation holds at every admissible radius, and we use it at the radii ρ' of the complex critical point.

Step 3. The Laplace lemma. By Step 1, Lemma B.3 applies on $\mathcal{P}=\mathcal{K}\times\{|z|\le z_0\}$ with $\sigma_{\min}=\frac12\sigma_0$, ε and η. Since $\Psi(\beta,z,0)=\Phi_z(\zeta')$, it gives (19) with

$$J_G(\beta,z)=\langle d(z),\beta\rangle-\Phi_z(\zeta'),\qquad C_G(\beta,z)=(2\pi)^{-m/2}\det\mathcal{J}(\beta,z)^{-1/2}\,b_z(\zeta')\prod_{j\ge2}\zeta_j,$$

and with $|\varepsilon|\le C/(c_\chi S)$, where C is the constant of Lemma B.3.

Step 4. The functions. We define J_G and C_G on $\mathcal{K}\times\{|z|<2z_0\}$ by these formulas. Since ζ' is holomorphic in z and $g_z(z')$ and $b_z(z')$ are holomorphic in (z,z'), the numbers $\Phi_z(\zeta')$, $b_z(\zeta')$, $\prod_j\zeta_j$ and the entries of $\mathcal{J}(\beta,z)$ are holomorphic in z and jointly continuous. By (i) the matrix $\mathcal{J}(\beta,z)$ lies in the set $\mathcal{S}_m$ of Appendix B, so $\det\mathcal{J}(\beta,z)^{-1/2}$ is continuous, and as a continuous square root of a holomorphic function without zeros it is holomorphic in z. For real z, ζ' is real by Lemma 6.10(c), so J_G and C_G are real, and $\mathcal{J}(\beta,z)$ is real and positive definite, so the branch is the positive root. For real $0\le z<2z_0$, Lemma 6.10(e) gives $J_G(\beta,z)=\langle d(z)-\theta(\beta,z),\beta\rangle$, which is the rate of $Q_G(z)$ by the identity $J=\langle q-\theta,\alpha\rangle$ after Definition 5.1, since the diagonal of $Q_G(z)$ is $-d(z)$. It also gives $C_G(\beta,z)=(\mu^{(z)}\cdot r_\beta(z))(l_\beta(z)\cdot e^{(z)})/((2\pi)^{m/2}\sqrt{\det\Sigma_\beta(z)})$, the constant of Section 6.1. At $z=0$ it is $C_G(\beta)>0$, by (H). □

The proof does not use the irreducibility of F, which only makes c^- and c^+ nonzero.

6.8 Proofs of the main results

Proof of (a) Theorem 6.8 gives (19), with $\varepsilon=0$ for $k=2$. Take $\delta\le\frac12$, so $S\ge T/2$. Write $\Lambda(z)=S^{-(k-2)/2}C_G(\beta,z)e^{-SJ_G(\beta,z)}$ and $\varepsilon(z)=f_G^{(z)}(s)/\Lambda(z)-1$. By Step 1(i), (ii) of the proof of Theorem 6.8, $C_G(\beta,z)\ne0$ for $|z|\le2z_0$ if $|G|\ge2$, so ε is holomorphic there (for $|G|=1$, $\varepsilon=0$), and $|\varepsilon|\le C/S$ for $|z|\le z_0$. The Schwarz lemma for $(\varepsilon(z)-\varepsilon(0))S/(2C)$ on $|z|<z_0$ gives $|\varepsilon(z)-\varepsilon(0)|\le2C|z|/(z_0S)$ for $|z|\le z_0$. Since $f_G(s)=\Lambda(0)(1+\varepsilon(0))$, (16) gives for every $\rho\le z_0$

$$e^{q_ix}f_F(s,x)=(1+\varepsilon(0))\mathcal{A}+\mathcal{D},\qquad\mathcal{A}=\frac{1}{2\pi\mathrm{i}}\oint_{|z|=\rho}e^{x/z}\Lambda(z)\frac{dz}{z^2},\qquad|\mathcal{D}|\le\frac{2C\rho}{z_0S}\,\mathcal{A}^{\mathrm{abs}},$$

where $\mathcal{A}^{\mathrm{abs}}$ is the integral of $|e^{x/z}\Lambda(z)||z|^{-2}$ against $|dz|/(2\pi)$. By definition $\mathbf{K}_\beta(x,S)=e^{-q_ix}\mathcal{A}/\Lambda(0)$, and $|\varepsilon(0)|\le\frac12$ for T_0 large, so $f_F=f_G\mathbf{K}_\beta(1+E)$ with $|E|\le2|\mathcal{D}|/\mathcal{A}$ once $\mathcal{A}>0$. So it suffices to show that $\mathcal{A}>0$, $\mathcal{A}^{\mathrm{abs}}\le C'\mathcal{A}$ and $\rho/S\le C'(T^{-2}+x^{1/2}T^{-3/2})$. Put

$$\Xi(z)=\frac{C_G(\beta,z)}{C_G(\beta,0)}\,e^{S(J_G(\beta)-J_G(\beta,z)-\omega_\beta z)},$$

so that $\Lambda(z)/\Lambda(0) = e^{S\omega_\beta z}\Xi(z)$. By Proposition 6.6(ii), the holomorphy of C_G, and Cauchy estimates on the compact set $\mathcal{K} \times \{|z| \le \frac{3}{2}z_0\}$, the function $R = \log \Xi$ is holomorphic on $|z| \le z_0$ after shrinking z_0, and

$$R(z) = c_1 z - S\omega_\beta(\gamma_\beta - q_i)z^2 + O(|z|^2 + S|z|^3), \qquad |R''(z)| \le C(1+S).$$

We take δ so small that $z_* \le \sqrt{2\delta/\omega_{\min}} \le z_0/2$, where $\omega_{\min} = \min_{\mathcal{K}} \omega_\beta > 0$.

Few visits, $\zeta \le Z_0$ for a fixed large Z_0. Take $\rho = \max(z_*, 1/S)$, which is at most C/S. On the circle $|x/z| \le Z_0/2$, $|S\omega_\beta z| \le C$ and $|\Xi(z) - 1| \le C/S$. Let $\mathcal{A}_0$ be $\mathcal{A}$ with $\Xi = 1$. By (17) it equals $\Lambda(0)\sqrt{\omega_\beta S/x}\, I_1(\zeta)$, which is at least $\Lambda(0)\omega_\beta S$, because $\sqrt{\omega S/x}\, I_1(\zeta) = \omega S \sum_{n\ge 0} (\zeta/2)^{2n}/(n!(n+1)!)$. Also $|\mathcal{A} - \mathcal{A}_0| \le C\Lambda(0)\rho^{-1}/S \le C\Lambda(0)$ and $\mathcal{A}^{\mathrm{abs}} \le C\Lambda(0)\rho^{-1} \le C\Lambda(0)S$. So $\mathcal{A} \ge \Lambda(0)(\omega_\beta S - C) > 0$ and $\mathcal{A}^{\mathrm{abs}} \le C'\mathcal{A}$ for S large, and $\rho/S \le C/S^2 \le 4C/T^2$.

Many visits, $\zeta \ge Z_0$. Let $\lambda(z) = x/z + S\omega_\beta z + R(z)$, so that $e^{x/z}\Lambda(z) = \Lambda(0)e^{\lambda(z)}$. It is real on the real axis, and $\lambda'(z) = -x/z^2 + S\omega_\beta + O(1+S|z|)$. By the implicit function theorem it has a real critical point $\rho_s = z_*(1 + O(\sqrt{x/S} + 1/\zeta))$, and we take $\rho = \rho_s$. For $z = \rho e^{\mathrm{i}\phi}$,

$$\operatorname{Re}\lambda(z) - \lambda(\rho) = (x/\rho + S\omega_\beta\rho)(\cos\phi - 1) + \operatorname{Re}[R(z) - R(\rho)].$$

Since R is real on the real axis, $\operatorname{Re}[R'(\rho)(z-\rho)] = R'(\rho)\rho(\cos\phi - 1)$, and $|R''| \le C(1+S)$ bounds the rest by $2C(1+S)\rho^2(1-\cos\phi)$. Also $|R'(\rho)|\rho \le C(\rho + S\rho^2)$. Now $S\rho^2 \le Cx$, $x/\rho + S\omega_\beta\rho \ge \zeta$, and $x/\zeta \le C\sqrt{\delta}$. So for δ small $\operatorname{Re}\lambda(z) - \lambda(\rho) \le -\frac{\zeta}{2}(1-\cos\phi)$, and

$$\mathcal{A}^{\mathrm{abs}} \le \Lambda(0)\rho^{-1}e^{\lambda(\rho)}\frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-\frac{\zeta}{2}(1-\cos\phi)}d\phi \le C\Lambda(0)\rho^{-1}e^{\lambda(\rho)}\zeta^{-1/2}.$$

For the lower bound we write $\mathcal{A} = \Lambda(0)\rho^{-1}\frac{1}{2\pi}\int e^{\zeta\ell(\phi)}e^{-\mathrm{i}\phi}d\phi$ with $\ell(\phi) = \lambda(\rho e^{\mathrm{i}\phi})/\zeta$. Here $\ell(0)$ is real, $\ell'(0) = 0$, $-\ell''(0) = \rho^2\lambda''(\rho)/\zeta$ is real and lies in $[\frac{1}{2}, 2]$, the derivatives of ℓ up to order four are bounded uniformly in all parameters, and the gap above holds. The parameters (β, x, S) do not range over a compact set, so we use the uniform form of Lemma B.1 in Remark B.2, with $\sigma_{\min} = \frac{1}{2}$ and the gap $\frac{1}{2}(1-\cos\varepsilon)$. With $m = 1$, $\Psi = \ell$, $a = e^{-\mathrm{i}\phi}$ and ζ in place of T it gives $\mathcal{A} = \Lambda(0)\rho^{-1}e^{\lambda(\rho)}(2\pi\zeta(-\ell''(0)))^{-1/2}(1 + O(1/\zeta))$, which is positive and at least $c\,\mathcal{A}^{\mathrm{abs}}$. Finally $\rho_s \le 2z_*$ for δ small and Z_0 large, so $\rho/S \le 2x^{1/2}(\omega_{\min}S^3)^{-1/2} \le Cx^{1/2}T^{-3/2}$. □

Proof of (b) *and* (c) We use two facts on Bessel functions besides (17). For integers $n \ge 1$, $I_n(\zeta) = \frac{1}{\pi}\int_0^\pi e^{\zeta\cos t}\cos(nt)\,dt$ [36, Eq. 10.32.3]. So $I_n \le I_1$ for all $\zeta > 0$, and $|1 - \cos nt - (1-\cos t)| \le n^2t^2/2$ and a Laplace estimate give $|I_1(\zeta) - I_n(\zeta)| \le Cn^2 I_1(\zeta)/\zeta$ for $\zeta \ge 1$. Also $I_0/I_1 \le 1 + 2/\zeta$, as in the proof of Proposition 6.5.

The case $\pi_\beta^{\mathrm{in}} = \pi_\beta^{\mathrm{out}}$. By Proposition 6.6(v), $J_G(\beta, z) = J_G(\beta) - z\omega_\beta$ for real $0 \le z < 2z_0$, and by holomorphy on the whole disc, so $\Xi(z) = C_G(\beta, z)/C_G(\beta, 0) = 1 + c_1 z + \sum_{n\ge 2} c_n z^n$ with $|c_n| \le Cz_0^{-n}$. By (17) the term $c_n z^n$ contributes $c_n z_*^n I_{n-1}(\zeta)/I_1(\zeta)$ relative to the main term $z_*^{-1}I_1(\zeta)$. For $n \ge 2$ this is at most $|c_n|z_*^n$, and the sum over $n \ge 2$ is $O(z_*^2) = O(x/T)$, since $z_* \le z_0/2$. The term $n = 1$ is $c_1 z_* I_0(\zeta)/I_1(\zeta) = O(z_* + 1/(\omega_\beta S)) = O(\sqrt{x/T} + 1/T)$. With (a) this proves both claims of (c), for all $0 < x \le \delta T$.

The general case, $x \le 1$. Write $\Xi(z) = e^{-S\omega_\beta(\gamma_\beta - q_i)z^2}(1 + c_1 z + R_2(z))$ with $|R_2(z)| \le C(|z|^2 + S|z|^3)$. On $|z| = z_*$ the Gaussian factor has modulus at most $e^{(\gamma_\beta - q_i)x} \le C$. We expand it in powers of z^2 and use (17). With $y = (\gamma_\beta - q_i)x$ the term 1 gives the relative factor $\sum_{m\ge 0}(-y)^m I_{2m-1}(\zeta)/(m!\, I_1(\zeta))$. For $\zeta \ge 1$ it differs from e^{-y} by at most $C\sum_{m\ge 1} y^m m^2/(m!\,\zeta) \le Cx/\zeta = O(\sqrt{x/S})$. For $\zeta < 1$ we have $x < 1/(4\omega_\beta S)$, and

$|I_{2m-1}/I_1-1|\le 1$ bounds the difference by $e^y-1=O(1/S)$. The term $c_1 z$ times the Gaussian factor gives, by the same expansion, the relative factor $c_1 z_*\sum_{m\ge0}(-y)^m I_{2m}(\zeta)/(m!\,I_1(\zeta))$, since $S\omega_\beta z_*^2=x$. As $I_{2m}\le I_0$, its modulus is at most $|c_1|z_* e^y I_0(\zeta)/I_1(\zeta)\le Cz_*(1+2/\zeta)=O(\sqrt{x/S}+1/S)$. The term R_2 is bounded through the modulus of the integrand, and $\frac{1}{2\pi}\int e^{\zeta\cos\phi}d\phi = I_0(\zeta)$ gives $O\big((z_*^2+Sz_*^3)I_0(\zeta)/I_1(\zeta)\big)=O(1/S+\sqrt{x/S})$. So $\mathbf{K}_\beta=\mathbf{B}_\beta(1+O(1/T+\sqrt{x/T}))$ for $x\le 1$.

The general case, $1\le x\le \delta T$. Here $\zeta\ge 2\sqrt{\omega_{\min}S}\ge Z_0$. We use the evaluation of $\mathcal{A}$ in the proof of (a), and the same evaluation with $\Xi=1$, whose integral is exactly $\Lambda(0)z_*^{-1}I_1(\zeta)$. For $\Xi=1$ the critical point is z_*, the critical value is ζ, and $-\ell''(0)=1$. The ratio of the two evaluations is $e^{\lambda(\rho_s)-\zeta}(1+O(\sqrt{x/S}+1/\zeta))$, since ρ_s/z_*-1 and the relative change of $-\ell''(0)$ are $O(\sqrt{x/S}+1/\zeta)$. Now $x/z+S\omega_\beta z$ takes the value ζ at its critical point z_*, and the shift from z_* to ρ_s changes λ only by $O(\lambda''(\rho_s)(\rho_s-z_*)^2)=O(\zeta x/S+1/\zeta)$. So

$$\lambda(\rho_s)-\zeta=R(z_*)+O(\zeta x/S+1/\zeta)=-(\gamma_\beta-q_i)x+O\big(\sqrt{x/S}\,(1+x)\big),$$

because $c_1z_*=O(\sqrt{x/S})$, $S\omega_\beta z_*^2=x$, $Sz_*^3=O(x^{3/2}S^{-1/2})$, $\zeta x/S=O(x^{3/2}S^{-1/2})$ and $1/\zeta\le C\sqrt{x/S}$ for $x\ge1$. Hence $\mathbf{K}_\beta=\mathbf{B}_\beta\exp(O(\sqrt{x/T}(1+x)))$, and with (a) this proves (b). □

The proof of Proposition 6.2, which compares the critical value in the proof of Theorem 6.1(a) with Proposition 6.6(i), is in Section 6.8 of the arXiv version of this paper [40].

6.9 The counts near a facet

Now let n have support F. Put $N'=N-n_i$, let $\hat n$ be n restricted to G, and put $\beta=\hat n/N'$, $S=hN'$, $x=hn_i$ and $p_i=1-hq_i$. So $T=S+x$. Let $\widetilde\Lambda_\beta(z)=C_G(\beta,z)C_G(\beta,0)^{-1}e^{S(J_G(\beta)-J_G(\beta,z))}$ be the integrand of $\mathbf{K}_\beta$, and define the lattice kernel

$$\mathbf{K}^h_\beta(n_i,S)=\frac{h}{2\pi\mathrm{i}}\oint_{|z|=\rho}\Big(p_i+\frac{h}{z}\Big)^{n_i-1}\widetilde\Lambda_\beta(z)\,\frac{dz}{z^2}=\sum_{K=1}^{n_i}\binom{n_i-1}{K-1}p_i^{n_i-K}h^K\,[z^K]\widetilde\Lambda_\beta(z),$$

with $0<\rho<2z_0$. It is $h\mathbf{K}_\beta$ with $e^{x/z-q_ix}$ replaced by $(p_i+h/z)^{n_i-1}$, as Lemma 6.4 replaces Lemma 6.3. If $\widetilde\Lambda_\beta(z)=e^{S\omega_\beta z}\sum_{r\ge0}\xi_r z^r$, then $\mathbf{K}^h_\beta(n_i,S)=p_i^{n_i}\sum_r\xi_r(h/p_i)^r B^{(r)}_{n_i}(u)$ with $u=hS\omega_\beta/p_i$ and B_n of (22), so the Laguerre polynomials of Section 8 replace the Bessel kernel. The ratio of the two laws below carries one power of h for every k. Indeed $P_{N'}(\hat n)$ is close to $h^{k-2}f_G(s)$, by Lemma 5.5 for the face G if $k\ge3$ and by the path $j^{N'}$ if $G=\{j\}$, and Theorems 6.1(a) and 6.12(b) show that $P_N(n)$ is then within a factor $1+O(1/T+T^2/N)$ of $h^{k-1}f_F(s,x)$, also when n_i is small.

Theorem 6.11 (the counts on a pair face) *Let $F=\{j,i\}$ with $q_{ji},q_{ij}>0$ and $\mu_j>0$. Put $\omega=q_{ji}q_{ij}$, $c_0=1$, $c_1=q_{ji}+\frac{\mu_i}{\mu_j}q_{ij}$, $c_2=\frac{\mu_i}{\mu_j}\omega$ and $p_l=1-hq_l$, and let $hq_{\max}\le\frac12$ and $N',n_i\ge1$.*

(a) (The discrete Pedler formula.) *Exactly,* $P_{N'}(\hat{n}) = \mu_j p_j^{N'-1}$ *and*

$$\frac{P_N(n)}{P_{N'}(\hat{n})} = R := \sum_{K=1}^{n_i} \binom{n_i - 1}{K-1} p_i^{n_i - K} \sum_{e=0}^{2} c_e\, h^e \binom{N'-1}{K-e} \Big(\frac{h^2\omega}{p_j}\Big)^{K-e}.$$

(b) $\mathbf{K}^h(n_i, S) = p_i^{n_i}\big[B_{n_i}(u) + c_1 \frac{h}{p_i} B'_{n_i}(u) + c_2 \frac{h^2}{p_i^2} B''_{n_i}(u)\big]$ *with* $u = hS\omega/p_i$.

(c) With $U = n_i u$,

$$1 - \frac{U + 2\sqrt{U} + 2}{2N'} \le \frac{R}{\mathbf{K}^h(n_i, S)} \le e^{2hq_j(1+2\sqrt{U})}.$$

So if $n_i \le N'$ *and* $1 \le T \le N^{1/2}$, *then* $R = \mathbf{K}^h(n_i, S)(1 + O(h + h^2\sqrt{N' n_i}))$.

(d) If $n_i \le N'$ *and* $hS \le 1$, *then* $\mathbf{K}^h(n_i, S) = h\,\mathbf{K}(x, S)(1 + O(h + hS))$, *where* $\mathbf{K}(x, S) = f_F(S, x)/f_G(S)$ *is given by Proposition 6.5.*

Part (c) has no term $1/T$ and holds at every T, also when T^2/N is of order one, where (d) does not apply and Section 8 treats the complete graph.

Theorem 6.12 (the counts near a facet) *Let* F, i, G *and* $\mathcal{K}$ *be as in Theorem 6.1. There are* $\delta > 0$, $\varepsilon_1 > 0$, T_0 *and* C *such that the following hold uniformly for* $\operatorname{supp} n = F$ *with* $\beta \in \mathcal{K}$, $1 \le n_i \le \delta N$ *and* $T_0 \le T \le \varepsilon_1 N^{1/2}$.

(a) $P_N(n) = P_{N'}(\hat{n})\,\mathbf{K}^h_\beta(n_i, S)\,(1 + E)$ *with* $|E| \le C(T^{-2} + x^{1/2}T^{-3/2} + T^2/N) \le \frac{1}{2}$.

(b) $\mathbf{K}^h_\beta(n_i, S) = h\,\mathbf{K}_\beta(x', S)(1 + O(T^2/N))$ *for* $x' = x$ *and for* $x' = x - h$, *where* $\mathbf{K}_\beta(0, S) = S\omega_\beta + c_1(\beta)$ *by the same formula. Hence* $P_N(n) = hP_{N'}(\hat{n})\,\mathbf{K}_\beta(x, S)(1 + O(1/T + T^2/N))$.

(c) Parts (b) *and* (c) *of Theorem 6.1 hold with* $P_N(n)/(hP_{N'}(\hat{n}))$ *in place of* $f_F(s, x)/f_G(s)$ *and with* T^2/N *added to each error bound.*

For $k = 2$, Theorem 6.11 gives the ratio in (a) exactly. By (b) the lattice kernel matters only when T^2/N is not small. For $k \ge 3$ that regime is open, since (a) then has an error of order one. The proofs of both theorems are in Appendix E. The second one discretizes the states of G in the proof of Lemma 5.5, keeps the state i exact, and evaluates the contour integral in z as in the proof of Theorem 6.1(a).

7 The complete graph

On the complete graph every face ties at first order, and a few exact facts resolve the tie. The results of this section and the next are those of our preprint [39], and Table 1 says which of them follow from the general theory and which use the exact arguments of the preprint. On a complete face the tree sum of Proposition 5.2(b) has a closed form. It is classical.

Lemma 7.1 (weighted Cayley formula) *For* $x_1, \dots, x_k > 0$ *and the conductances* $c_{ij} = x_i x_j$,

$$\operatorname{tree}(c) = x_1 x_2 \cdots x_k\,(x_1 + \cdots + x_k)^{k-2}.$$

With every $x_i = 1$ *this is Cayley's count* k^{k-2} *of the spanning trees of the complete graph* K_k.

Its proof, by the matrix-tree theorem [7] and the matrix determinant lemma, is in Section 7 of the arXiv version of this paper [40].

Example 7.2 (the complete face) Let F be a face of $k \geq 2$ states with $q_{ij} = 1$ for all $i \neq j$ in F, and any exit rates, for example a face of $Q = J - dI$, where $q_i = d - 1$. Then F is reversible with $\nu = \mathbf{1}$, so by Proposition 5.2(b) we may take $r_\alpha = l_\alpha = \sqrt{\alpha}$ entrywise, and the conductances $c_{ij}(\alpha) = \sqrt{\alpha_i \alpha_j}$ are products. Put $A = \sum_{i \in F} \sqrt{\alpha_i}$. Lemma 7.1 gives $\operatorname{tree}(c(\alpha)) = (\prod_i \alpha_i)^{1/2} A^{k-2}$, so for every $\alpha \in \Delta_F^\circ$

$$C_F(\alpha) = \frac{(\mu \cdot \sqrt{\alpha})\, A^{k/2}}{(4\pi)^{(k-1)/2}\, (\prod_i \alpha_i)^{3/4}} = \frac{(\mu \cdot \sqrt{\alpha})\, A}{\sqrt{\det(2\pi \Sigma_\alpha)}}, \qquad J_F(\alpha) = \sum_{i \in F} q_i \alpha_i - (A^2 - 1).$$

The second form of $C_F(\alpha)$ uses $\det \Sigma_\alpha = 2^{k-1} (\prod_i \alpha_i)^2 / \operatorname{tree}(c(\alpha))$. For $Q = J - dI$ the rate is $J_F(\alpha) = d - A^2$, the rate of Dupuis and Liu [21, Eq. (3.2)]. At the center $\alpha = \mathbf{1}/k$ we have $A^2 = k$ and $C_F = \mu(F) k^{k-1/2} (4\pi)^{-(k-1)/2}$, which is Corollary 5.3 with $\operatorname{tree}(q_F) = k^{k-2}$.

From now on in this section $Q = J - dI$. The chain switches to each other state with probability h and keeps its state with probability $p = 1 - (d-1)h$. The following symbols of [39] are local to this section and to Appendix C. Put

$$\sigma = p - h, \qquad u = \frac{h}{p}, \qquad v = \frac{h}{\sigma} = \frac{u}{1-u}, \qquad V_N = \frac{p^{N-1}}{d}.$$

So u is the odds of a switch to one given state against no switch, and V_N is the probability of a vertex Ne_i for the uniform start, that is, $M_{\{i\}}$. For a composition n let $W_n(j)$ be the number of words with the counts n and j switches, and $S_n(u) = \sum_j W_n(j) u^j$. For a composition with some zero coordinates, S_n is the polynomial of its positive coordinates, which does not depend on d. By (2), the uniform start has

$$P_N(n) = V_N S_n(u) \qquad (0 < p \leq 1), \tag{20}$$

and so does every start that is constant on $\operatorname{supp} n$, with V_N replaced by $\mu_a p^{N-1}$ for $a \in \operatorname{supp} n$. A composition is *balanced* if its positive coordinates differ by at most 1. For the uniform start all faces of k states have the same face maximum M_F, and all balanced compositions with k positive coordinates have the same probability. For $p \geq 1/d$ this probability is M_F, by Theorem 7.5 below. Put

$$a_k = \frac{1}{2} \log(4\pi) - \frac{k + \frac{1}{2}}{k-1} \log k, \qquad D_k = e^{-(k-1)a_k} = \frac{k^{k+1/2}}{(4\pi)^{(k-1)/2}} \qquad (k \geq 2),$$

and $D_1 = 1$. Numerically $a_2 = -0.4674$, $a_3 = -0.6571$ and $a_4 = -0.8139$.

Corollary 7.3 (the complete graph) *A face F of k states has $\lambda_F = d - k$ and α_F^* uniform, and if $\mu(F) > 0$ then $K_F = \frac{\mu(F)}{k} D_k$. So $\Phi_1(F) = d - 1$ for every face, and all faces tie at $\tau = 1$. With the uniform start, the crossing of a face of $k \geq 2$ states with a vertex has $\Delta = \delta = k - 1$ and $b_{FG} = a_k$.*

Its proof, from Corollary 5.3 and Lemma 7.1, is in Section 7 of the arXiv version.

By Example 7.2 at the center, $D_k = dK_F = k/\sqrt{\det(2\pi\Sigma_k)}$ with $\Sigma_k = \Sigma_{\mathbf{1}/k} = (2/k^2)(I - J/k)$ in the coordinates α'. So D_k is the mass k of a face, in units of a vertex, over the Gaussian volume of the fluctuations of its fractions.

Theorem 7.4 (balanced crossings) *Let μ be uniform and let F be a face of $2 \le k \le d$ states.*

(i) For every $N \ge k$ there is exactly one $p = p_{N,k} \in (0,1)$ with $M_F = V_N$. The face maximum M_F is larger than V_N below $p_{N,k}$ and smaller above it, and so is the probability of a balanced composition of F. The crossing $p_{N,k}$ is strictly increasing in N, $p_{k,k} = 1/(1+(d-1)(k!)^{-1/(k-1)})$, and every crossing has $u \le (k!)^{-1/(k-1)} < 1$, so $p_{N,k} > 1/d$.

(ii) For large N, uniformly for $1 \le T \le N^{1/3}$,

$$\log \frac{M_F}{V_N} = (k-1)\Big[T - L + \frac{1}{2}\log T - a_k\Big] + O\Big(\frac{1}{T} + \frac{T^2}{N}\Big).$$

With $T = L - \frac{1}{2}\log L + t$, the ratio M_F/V_N tends to $e^{(k-1)(t-a_k)}$, uniformly for t in compact sets.

(iii) The value $T_{N,k} = N(1-p_{N,k})/(d-1)$ of T at the crossing satisfies

$$T_{N,k} + \frac{1}{2}\log T_{N,k} = L + a_k + O(1/L),$$

and

$$T_{N,k} = L - \frac{1}{2}\log L + a_k + \frac{\log L}{4L} + O\Big(\frac{1}{L}\Big). \tag{21}$$

(iv) $a_2 > a_3 > a_4 > \cdots$.

For $d = 2$ the chain gives the directions of the persistent walk, and $T_{N,2}$ is the crossing of its middle compositions and its endpoints, with $a_2 = \frac{1}{2}\log(\pi/8)$. The proof is in Appendix C of the arXiv version of this paper [40].

Theorem 7.5 (exact balancing) *Let $1/d \le p < 1$, and let μ be constant and positive on a face F. Moving one unit from a coordinate b of a composition with support F to another coordinate a, where $n_b \ge n_a + 2$ and $n_a \ge 1$, strictly increases the probability of the composition. So the most likely compositions with support F are exactly the balanced ones, and they all have the same probability.*

For an integer $n \ge 1$ put

$$B_n(w) = \sum_{m=1}^{n} \binom{n-1}{m-1} \frac{w^m}{m!} = \frac{w}{n} L_{n-1}^{(1)}(-w), \tag{22}$$

where $L_{n-1}^{(1)}$ is the Laguerre polynomial [36, Eq. 18.5.12]. The proof in Appendix C of the arXiv version of this paper [40] sums over the numbers of runs first. This writes the law as an integral of a product of one polynomial B_{n_i} per state, and $B_n(w)$ is log-concave in n for $w > 0$. The condition $p \ge 1/d$ holds for $T \le N/d$, and so at every crossing by Theorem 7.4(i).

Corollary 7.6 (exactly one mode change) *Let μ be uniform, and fix $d \geq 2$ and $C > 1$. There is N_0 such that for $N \geq N_0$ and every p with $0 < T \leq CL$ the following hold.*

(i) If $p > p_{N,d}$, the global modes are exactly the d vertices.

(ii) If $p < p_{N,d}$, they are exactly the balanced compositions with d positive coordinates.

(iii) If $p = p_{N,d}$, they are these two classes and nothing else.

Moreover $p_{N,2} < p_{N,3} < \cdots < p_{N,d}$ and $p_{N,d} - p_{N,k} = (d-1)(a_k - a_d + o(1))/N$. So all the crossings of balanced compositions with the vertices lie in a p-interval of length $(d-1)(a_2 - a_d + o(1))/N$, and as p decreases the balanced compositions with d positive coordinates cross first.

On the triangle the middle composition $(\frac{N}{3}, \frac{N}{3}, \frac{N}{3})$ overtakes the vertices while the middles of the edges are still below them by the factor $e^{-(a_2-a_3)} = 0.827$. The proof is in Appendix C of the arXiv version of this paper [40]. In the tie window at $\tau^* = 1$ this is Theorem 5.9, whose envelope E has a single kink, at $t = a_d$.

Remark 7.7 (an intermediate face can be a mode at finite N) Corollary 7.6 does not hold for every N. Take $d = 3$, $N = 4$ and $p = 10/17$, so $u = 7/20$ and $0 < T = 14/17 < \log 4$. Counting words gives $P_4(2,2,0)/V_4 = 2u + 2u^2 + 2u^3 = 4123/4000 > 1$ and $P_4(2,1,1)/V_4 = 6u^2 + 6u^3 = 3969/4000 < 1$. Since $p > 1/3$, Theorem 7.5 shows that every global mode is a vertex or a balanced composition, so the three edge centers are the only global modes. Hence $N_0 > 4$ when $d = 3$, for every $C > 1$.

8 Near a vertex

This section treats the boundary layer of Section 6 for the counts themselves, on the complete graph near a vertex, in the regime where T is of order $\sqrt{N}$. There the layer is discrete, and its Laguerre kernel cannot be replaced by the Bessel kernel.

We keep the notation of Section 7, namely $Q = J - dI$, the uniform start, p, u, v, and $S_n(u) = P_N(n)/V_N$ by (20). Let one count b be large and let the other positive counts $a_1, \ldots, a_\ell$ be small compared with b, with $\ell \geq 1$ fixed. We put $|a| = a_1 + \cdots + a_\ell$ and write $S_{(b,a)}$ for S_n with $n = (b, a_1, \ldots, a_\ell)$. These rare counts are not the constants a_k of Section 7. Since S_n does not depend on the start, $P_N(n) \leq (\max_i \mu_i) p^{N-1} S_n(u)$, so upper bounds on S_n hold for every start.

A rare run away from the ends needs two switches, which gives a factor u^2, and it can go in about b places. So the rare counts enter through bu^2, and for fixed rare counts the crossing moves to odds u of order $b^{-1/2}$, that is, to T of order $\sqrt{N}$. Use B_a from (22) and put

$$\mathcal{I}(x) = \sum_{j \geq 0} \frac{x^j}{j!\,(j+1)!} = \frac{I_1(2\sqrt{x})}{\sqrt{x}},$$

where I_1 is the modified Bessel function [36, Eq. 10.25.2], and $\mathcal{I}(0) = 1$. By the Laguerre form in (22), the next lemma is a quantitative form of the limit of Laguerre polynomials to Bessel functions.

Lemma 8.1 (a uniform Laguerre–Bessel inequality) *For every integer $a \geq 1$ and real $w > 0$,*

$$0 \leq 1 - \frac{B_a(w)}{w\,\mathcal{I}(aw)} \leq \frac{w}{2}. \tag{23}$$

The proof is in Appendix D of the arXiv version of this paper [40]. Put

$$\mathcal{B}(y) = \prod_{i=1}^{\ell} B_{a_i}(y), \qquad \kappa(y) = \frac{y\,\mathcal{B}'(y)}{\mathcal{B}(y)}.$$

The polynomial $\mathcal{B}$ has positive coefficients, $\mathcal{B}(0) = 0$ and $\mathcal{B}(1) \geq 1$, so $\mathcal{B}(y_a) = 1$ has exactly one positive root y_a, and $y_a \leq 1$. The polynomial $S_{(b,a)}$ has nonnegative coefficients, zero constant term and a positive coefficient at u^ℓ, so $S_{(b,a)}(u) = 1$ has exactly one positive root $u_{b,a}$, the crossing odds of the composition $(b, a_1, \dots, a_\ell)$ with a vertex.

Theorem 8.2 (the vertex law) *Fix $\ell \geq 1$ and put*

$$c_\ell = \ell + \sqrt{\ell} + 1, \qquad \varepsilon = \sqrt{\frac{|a|+1}{b}}, \qquad \delta = 16 c_\ell \varepsilon.$$

For all positive counts $b, a_1, \dots, a_\ell$ with $\delta \leq 1/2$,

$$(1-\delta)\sqrt{y_a/b} \;\leq u_{b,a} \leq (1+\delta)\sqrt{y_a/b}, \qquad \mathcal{B}(y_a) = 1. \tag{24}$$

In particular

$$\left| \frac{u_{b,a}}{\sqrt{y_a/b}} - 1 \right| \leq 16 c_\ell \sqrt{\frac{|a|+1}{b}} \qquad \text{whenever } \sqrt{\frac{|a|+1}{b}} \leq \frac{1}{32 c_\ell}, \tag{25}$$

so $u_{b,a} = \sqrt{y_a/b}\,(1 + o(1))$ as $(|a|+1)/b \to 0$. The constants are not optimized.

For fixed rare counts Lemma D.2 gives $S_{(b,a)}(\sqrt{y/b}) \to \mathcal{B}(y)$ as $b \to \infty$. The theorem shows that the crossing stays near $\sqrt{y_a/b}$ uniformly as long as $|a| = o(b)$. For $d = 2$ and $\ell = 1$ its leading term is that of the fixed-edge expansion of [38, Theorem 4.10]. The proof is in Appendix D of the arXiv version of this paper [40] and rests on two facts. First, $\log \mathcal{B}$ is concave, and for $0 < y \leq 1$ its elasticity $\kappa(y)$ lies between $\max\{\ell, \tfrac{1}{2}\sqrt{|a|y}\}$ and $\ell + \sqrt{\ell |a| y}$ (Lemma D.1). Second, $S_{(b,a)}(u) = (1-u)^{|a|}\mathbb{E}[\mathcal{B}(v\mathcal{X})]$ exactly, where $\mathcal{X}$ is a gamma variable plus a compound binomial sum with mean $2+(b-1)u$ (Lemma D.2). Jensen's inequality then gives a lower bound that exceeds 1 at $u = (1+\delta)\sqrt{y_a/b}$, and the tangent line of $\log \mathcal{B}$ at bu^2 with the moment generating function of $\mathcal{X}$ gives an upper bound that is below 1 at $u = (1-\delta)\sqrt{y_a/b}$.

With one rare count the vertex law is the discrete pair-face case of Section 6, with $F = \{1,2\}$, $G = \{1\}$, $i = 2$ and $n = (b, a)$. By Theorem 6.11(a) the main part of $P_N(b,a)$ is $\mu_1 p^{N-1} \sum_K \binom{a-1}{K-1}\binom{b-1}{K} u^{2K}$, which becomes $B_a(bu^2)$ when $\binom{b-1}{K}$ is replaced by $b^K/K!$. This is the layer with $\omega = 1$, $S = hb$ and $x = ha$, and $w = bu^2$ is ωSh up to the factor $p^{-2} = 1 + O(h)$. In Theorem 8.2 w is of order one, Lemma 8.1 pins $B_a(w)/(w\mathcal{I}(aw))$ between $1 - w/2$ and 1, and the continuum layer does not apply. When $T^2/N \to 0$, w tends to 0 and the lattice adds no separate regime.

9 Verification and open questions

Every theorem, proposition, lemma and corollary of this paper is proved in the text or in the appendices, here or in the arXiv version of this paper [40], apart from the results of [38] and [39] that we quote where we say so. No proof rests on a computation. The formal proofs and the scripts below check parts of the paper independently. All of them are in the repository named under Code availability.

Formal proofs

The folders lean/PaperC and lean/PaperB prove many finite and algebraic steps in Lean 4 (v4.33.1, with Mathlib), with no `sorry` or `admit`. The file lean/PaperC/MAP.md matches each statement to its Lean theorems. A positive Perron vector is a hypothesis where one is needed, since Mathlib has no Perron and Frobenius theorem for irreducible Metzler matrices, and the asymptotic results are not in Lean.

Computations

The command `py -I verify_all.py` in the folder paperC runs the 9 scripts that check this paper, in about four minutes, and writes each output, headed by the sha256 of its script, to paperC/data. Each check is an assertion, and the checks in floating point use a tolerance without interval certification. Section 9 of the arXiv version of this paper [40] says what each script checks and quotes its main numbers. The figures illustrate the theorems, and no statement is deduced from them or from the computations.

Open questions

A local law up to the lower-dimensional corners of a face is open. The results on reducible faces, Theorems 4.2(ii) and 4.5 and Corollary 4.6, and Theorem 7.4(ii) are proved for $T \le N^{1/3}$, and we have not extended them to $T \le h_* N$. We do not know whether the zero in Corollary 4.6 is unique, or where the most likely counts of a reducible face lie. For a facet of three or more states, the law of the counts near the facet when T^2/N is of order one is open. Theorem 6.12 has an error of order one there, and a statement would need a discrete form of Theorem 6.8. We do not know whether $r_\beta = 0$ forces $\pi_\beta^{\mathrm{in}} = \pi_\beta^{\mathrm{out}}$ in Proposition 6.2.

Acknowledgments. The question grew out of Problem 131 of Peter Kagey's Open Problem Collection [28], and I thank him for posing it. I thank Chase Hameetman and Aidan Deshong for conversations about some of the proofs of the preprint [39].

Statements and Declarations

- **Funding.** No funding was received for this work.
- **Competing interests.** The author has no competing interests to declare.
- **Ethics approval and consent to participate.** Not applicable.
- **Consent for publication.** Not applicable.

- **Data availability.** The outputs of the computations described in Section 9 are in the folder kagey-problems/problem131/paperC/data of the repository named under Code availability, each headed by the sha256 of the script that wrote it.
- **Materials availability.** Not applicable.
- **Code availability.** All code, data and Lean files supporting this paper are in the folder kagey-problems/problem131 of https://github.com/apemm/Kagey-Problems, with the scripts in the subfolder paperC and the Lean files in lean. Section 9 says how to run them. The scripts need Python 3 with `numpy`, `scipy`, `sympy` and `mpmath`, and the option `--full` adds the largest cases of four of them.
- **Author contribution.** I am the sole author and am responsible for the whole paper.

Appendix A Proofs of the local law and of the first-order bounds

This appendix proves Proposition 5.2(a), Lemma 4.1, Theorem 5.4, Lemma 5.5, Lemma A.5 and Lemma A.6. F is a face with $k \geq 2$ states, numbered $1, \dots, k$. For $x \in \mathbb{C}^F$ we write $x' = (x_2, \dots, x_k)$. We write i for the imaginary unit, since i denotes a state, and $\mathbb{T}^m = (\mathbb{R}/2\pi\mathbb{Z})^m$. We use two standard facts [3]. First, let $B \geq 0$ be square and S diagonal with positive diagonal. Then $\varrho(B - S) < 0$ if and only if $\varrho(S^{-1}B) < 1$, and also if and only if $(S - B)v > 0$ for some $v > 0$. In that case $(S - B)^{-1} = \sum_{j \geq 0} (S^{-1}B)^j S^{-1} \geq 0$. Second, if M is an irreducible Metzler matrix, then every nonnegative eigenvector of M is positive and belongs to $\varrho(M)$, and $\varrho(M)$ is larger than the Perron root of each proper principal submatrix of M (Lemma 3.2).

A.1 The Perron root under a diagonal perturbation

An irreducible generator on F is a Metzler matrix $\widetilde{Q}$ with $\widetilde{Q}\mathbf{1} = 0$ whose graph is strongly connected. It has a unique stationary law $\pi > 0$. The argument after Definition 5.1, with $(\widetilde{Q}, \pi)$ in place of $(\widehat{Q}_\alpha, \alpha)$, shows that $Z = (\mathbf{1}\pi^\top - \widetilde{Q})^{-1} - \mathbf{1}\pi^\top$ exists and satisfies (8). Put $D = \operatorname{diag} \pi$ and $\Gamma = DZ + Z^\top D$.

Lemma A.1 (the Perron root) *Let $\widetilde{Q}$, π, Z and Γ be as above. Then $\kappa(\eta) = \varrho(\widetilde{Q} + \operatorname{diag} \eta)$ is real-analytic near 0, with $\kappa(0) = 0$, $\nabla\kappa(0) = \pi$ and $\nabla^2\kappa(0) = \Gamma$. For $\eta \in \mathbb{R}^F$ and $u = Z\eta$,*

$$\eta^\top \Gamma \eta = \sum_{i \neq j} \pi_i \widetilde{Q}_{ij} (u_i - u_j)^2. \tag{A1}$$

So Γ is positive semidefinite with kernel $\mathbb{R}\mathbf{1}$, and Σ, which is Γ without the row and column of state 1, is positive definite.

The proof, which differentiates the eigenvalue equation of the simple Perron root twice at $\eta = 0$, is in Appendix A of the arXiv version of this paper [40].

Proof of Proposition 5.2(a) Step 1. The derivatives of the Perron root. For $\vartheta \in \mathbb{R}^F$ let $\Lambda(\vartheta) = \varrho(Q_F + \operatorname{diag} \vartheta)$, so that $\Lambda(\vartheta + c\mathbf{1}) = \Lambda(\vartheta) + c$, which we call the shift property. It is a

simple eigenvalue with positive right and left eigenvectors $\hat r, \hat l$, normalized by $\sum_i \hat r_i = 1$ and $\hat l \cdot \hat r = 1$, and by the implicit function theorem Λ, $\hat r$ and $\hat l$ are real-analytic in ϑ. Put $\pi = \hat l \circ \hat r$ and $\widetilde Q = \hat R^{-1}(Q_F + \operatorname{diag}\vartheta - \Lambda(\vartheta)I)\hat R$ with $\hat R = \operatorname{diag}\hat r$. Then $\widetilde Q \mathbf{1} = 0$ and $\pi^\top \widetilde Q = 0$ by the two eigenvector equations, so $\widetilde Q$ is an irreducible generator with stationary law π. Since $\hat R$ commutes with diagonal matrices, $\Lambda(\vartheta + \eta) = \Lambda(\vartheta) + \varrho(\widetilde Q + \operatorname{diag}\eta)$. So by Lemma A.1, $\nabla\Lambda(\vartheta) = \pi$ and $\nabla^2\Lambda(\vartheta) = \Gamma$, with Γ built from $(\widetilde Q, \pi)$.

Step 2. The gradient can be inverted. Let $\Lambda_0(\beta) = \Lambda(0, \beta)$ for $\beta \in \mathbb{R}^{k-1}$. Its Hessian is Γ without the row and column of state 1, which is positive definite by Lemma A.1. So Λ_0 is strictly convex, $\nabla\Lambda_0$ is injective, and by the inverse function theorem it has a real-analytic inverse on its open image.

Step 3. The tilt that belongs to α. Fix $\alpha \in \Delta_F^\circ$ and put $\vartheta(\alpha) = q - \theta(\alpha)$. Then $Q_F + \operatorname{diag}\vartheta(\alpha) = A_F - \operatorname{diag}\theta(\alpha)$ has the positive right and left eigenvectors r_α and l_α for the eigenvalue 0, by Definition 5.1 and (7). So its Perron root is 0, and at $\vartheta(\alpha)$ we have $\pi = \alpha$, $\widetilde Q = \widehat Q_\alpha$ and $\Gamma = \Gamma_\alpha$. Let $\beta(\alpha)$ be the last $k-1$ coordinates of $\vartheta(\alpha) - \vartheta_1(\alpha)\mathbf{1}$. A shift along $\mathbf{1}$ does not change the derivatives of Λ, so $\nabla\Lambda_0(\beta(\alpha)) = \alpha'$ and $\nabla^2\Lambda_0(\beta(\alpha)) = \Sigma_\alpha$. Hence Σ_α is positive definite, and $\beta(\alpha) = (\nabla\Lambda_0)^{-1}(\alpha')$ is real-analytic. The shift property and $\Lambda(\vartheta(\alpha)) = 0$ give $\vartheta_1(\alpha) = -\Lambda_0(\beta(\alpha))$. So $\theta(\alpha)$ is real-analytic, and then so are its Perron vectors and Σ_α.

Step 4. The Hessian of J_F. Since $\vartheta(\alpha) = \vartheta_1(\alpha)\mathbf{1} + (0, \beta(\alpha))$ and $\sum_i \alpha_i = 1$, we have $J_F(\alpha) = \langle \vartheta(\alpha), \alpha\rangle = \langle \beta(\alpha), \alpha'\rangle - \Lambda_0(\beta(\alpha))$. We differentiate in α'. The terms with the derivative of β cancel because $\nabla\Lambda_0(\beta(\alpha)) = \alpha'$, and $\nabla J_F = \beta$ remains. So $\nabla^2 J_F$ is the Jacobian of $(\nabla\Lambda_0)^{-1}$ at α', which is Σ_α^{-1}. The formula for C_F follows from Definition 5.1. □

A.2 The torus representation

The generating function of the words in H is Good's resolvent formula [23, (2)], and the torus integral below is his constant-term formula, here for the killed matrix A_F and with the circle of one state integrated out. We split

$$A_F = \begin{pmatrix} 0 & a_1^\top \\ b_1 & A' \end{pmatrix}, \qquad \mu_F = (\mu_1, \mu'),$$

so $a_1 = (q_{1j})_{j\ge 2}$, $b_1 = (q_{i1})_{i\ge 2}$ and $A' = (q_{ij})_{i,j\ge 2}$. For $z' \in \mathbb{C}^{k-1}$ let $\mathcal{Z}' = \operatorname{diag}(z')$. Where $\mathcal{Z}' - A'$ is invertible we put $X(z') = (\mathcal{Z}' - A')^{-1}$, $g(z') = a_1^\top X(z') b_1$ and

$$b(z') = \big(\mu_1 + \mu'^\top X(z') b_1\big)\big(1 + a_1^\top X(z')\mathbf{1}\big).$$

Lemma A.2 (torus representation) *Let $s' \in (0,\infty)^{k-1}$ satisfy $\varrho(A' - \operatorname{diag} s') < 0$. On the set of $z' \in \mathbb{C}^{k-1}$ with $|z_i| \ge s_i$ for every $i \ge 2$, the matrix $\mathcal{Z}' - A'$ is invertible, and X, g and b are given by absolutely convergent power series in $1/z_2, \dots, 1/z_k$ with nonnegative coefficients. Hence $|g(z')| \le g(s')$ and $|b(z')| \le b(s')$ there. For every $t \in (0,\infty)^F$,*

$$H(t) = \frac{1}{(2\pi)^{k-1}} \int_{\mathbb{T}^{k-1}} \exp\Big(t_1 g(z') + \sum_{i\ge 2} t_i z_i\Big)\, b(z') \prod_{i\ge 2} z_i \, d\varphi, \qquad z_i = s_i e^{\mathrm{i}\varphi_i}.$$

Let moreover $hq_{\max} \le 1$, *put* $p_i = 1 - hq_i$, *and let* $n \in \mathbb{Z}^F_{\ge 1}$. *Then the probability that* $X_1, \dots, X_{|n|}$ *have composition* n *is*

$$P_{|n|}(n) = \frac{h^{k-1}}{(2\pi)^{k-1}} \int_{\mathbb{T}^{k-1}} \bigl(p_1 + hg(z')\bigr)^{n_1 - 1} \prod_{i \ge 2} (p_i + hz_i)^{n_i - 1}\, b(z') \prod_{i \ge 2} z_i \, d\varphi.$$

Proof Step 1. The series. By the first fact, $\varrho(\operatorname{diag}(s')^{-1} A') < 1$. So the series $X(z') = \sum_{j \ge 0} (\mathcal{Z}'^{-1} A')^j \mathcal{Z}'^{-1}$ converges absolutely for $|z_i| \ge s_i$, since its terms are bounded entrywise in modulus by those at $z' = s'$. As a_1, b_1 and μ are nonnegative, this gives the claims about X, g and b.

Step 2. A radius for state 1. Choose $s_1 > g(s')$ and put $S = \operatorname{diag}(s_1, s')$. We show that $\varrho(S^{-1} A_F) < 1$. For $\varepsilon > 0$ let $v = (1, v')$ with $v' = X(s')(b_1 + \varepsilon \mathbf{1}) > 0$. Then $((S - A_F)v)_i = \varepsilon$ for $i \ge 2$, and $((S - A_F)v)_1 = s_1 - g(s') - \varepsilon a_1^{\top} X(s') \mathbf{1}$. For small ε both are positive, and the first fact gives the claim.

Step 3. The k *circles.* Let $\mathcal{Z} = \operatorname{diag}(z)$ with $|z_i| = s_i$ for every $i \in F$. The matrices $A_F S^{-1}$ and $S^{-1} A_F$ are similar, so $\varrho(A_F S^{-1}) = \varrho(S^{-1} A_F) < 1$, and expanding $(\mathcal{Z} - A_F)^{-1} = \mathcal{Z}^{-1}(I - A_F \mathcal{Z}^{-1})^{-1}$ gives

$$\mu_F^{\top} (\mathcal{Z} - A_F)^{-1} \mathbf{1} = \sum_c \mu_{c_1} \prod_{j < M} q_{c_j c_{j+1}} \prod_{i \in F} z_i^{-m_i(c)},$$

absolutely and uniformly on the torus. Here $c = c_1 \cdots c_M$ runs over the words in F with $M \ge 1$ and $c_j \ne c_{j+1}$, and $m_i(c)$ is the number of letters i in c. For an integer m and $t_i > 0$, Cauchy's formula gives $\frac{1}{2\pi} \int_0^{2\pi} e^{t_i z_i} z_i^{1-m} \, d\varphi_i = t_i^{m-1}/(m-1)!$ if $m \ge 1$ and 0 if $m \le 0$. We multiply the series by $e^{\langle t, z \rangle} \prod_i z_i$ and integrate term by term. The words that miss a state of F drop out, and grouping the others by $m(c)$ gives

$$H(t) = \frac{1}{(2\pi)^k} \int_{\mathbb{T}^k} e^{\langle t, z \rangle}\, \mu_F^{\top} (\mathcal{Z} - A_F)^{-1} \mathbf{1} \prod_{i \in F} z_i \, d\varphi. \tag{A2}$$

Step 4. The first circle. The Schur complement $z_1 - g(z')$ of $\mathcal{Z}' - A'$ in $\mathcal{Z} - A_F$ is not 0, since $|g(z')| \le g(s') < s_1$. Block inversion gives

$$\mu_F^{\top} (\mathcal{Z} - A_F)^{-1} \mathbf{1} = \frac{b(z')}{z_1 - g(z')} + \mu'^{\top} X(z') \mathbf{1}.$$

We integrate over φ_1 in (A2) and divide by 2π. The second term does not depend on z_1, so it contributes $\frac{1}{2\pi} \int_0^{2\pi} e^{t_1 z_1} z_1 \, d\varphi_1 = 0$. In the first term $g(z')$ lies inside the circle $|z_1| = s_1$, so Cauchy's formula gives $e^{t_1 g(z')} b(z')$. The factors without z_1 are unchanged, and this proves the formula for H.

The discrete formula. We repeat Steps 3 and 4 with $(p_i + hz_i)^{n_i - 1}$ in place of $e^{t_i z_i}$. The coefficient of z^{m-1} in the polynomial $(p + hz)^{n-1}$ is $h^{m-1} \binom{n-1}{m-1} p^{n-m}$, which is 0 for $m > n$. So $\frac{1}{2\pi} \int_0^{2\pi} (p_i + hz_i)^{n_i - 1} z_i^{1-m} \, d\varphi_i$ is this coefficient if $m \ge 1$ and 0 if $m \le 0$. The polynomial is bounded on the circle, so the term-by-term integration of Step 3 applies, and it gives

$$\frac{1}{(2\pi)^k} \int_{\mathbb{T}^k} \mu_F^{\top} (\mathcal{Z} - A_F)^{-1} \mathbf{1} \prod_{i \in F} (p_i + hz_i)^{n_i - 1} z_i \, d\varphi = \sum_m W(m) \prod_{i \in F} h^{m_i - 1} \binom{n_i - 1}{m_i - 1} p_i^{n_i - m_i}.$$

Since $h^{M-1} = h^{k-1} \prod_i h^{m_i - 1}$, this is $h^{-(k-1)} P_{|n|}(n)$ by Lemma 2.3 with $|n|$ in place of N. In Step 4 the second term gives $\frac{1}{2\pi} \int_0^{2\pi} (p_1 + hz_1)^{n_1 - 1} z_1 \, d\varphi_1 = 0$, and Cauchy's formula evaluates the polynomial $(p_1 + hz_1)^{n_1 - 1}$ at $z_1 = g(z')$. □

Section 6 needs the same formula for complex weights and an end vector other than $\mathbf{1}$. Let A be a complex matrix on F with zero diagonal and let $\nu, e \in \mathbb{C}^F$. Let $W_{A,\nu,e}(m)$ be the sum over the words c of $W(m)$ of $\nu_{c_1} \prod_{j<M} A_{c_j c_{j+1}} e_{c_M}$, and $H_{A,\nu,e}(t)$ the sum $H(t)$ with $W_{A,\nu,e}$ in place of W. So $H = H_{A_F,\mu,\mathbf{1}}$. We split A and ν as above and define X and g by the same formulas, and

$$b(z') = \big(\nu_1 + \nu'^{\top} X(z') b_1\big)\big(e_1 + a_1^{\top} X(z') e'\big).$$

Lemma A.3 (complex weights) *Let $\bar{A} \geq 0$ be a matrix on F with zero diagonal and $\bar{\nu}, \bar{e} \in [0,\infty)^F$ with $|A_{ij}| \leq \bar{A}_{ij}$, $|\nu_i| \leq \bar{\nu}_i$ and $|e_i| \leq \bar{e}_i$ for all i, j, and let $\bar{X}$, $\bar{g}$ and $\bar{b}$ be the functions above for $(\bar{A}, \bar{\nu}, \bar{e})$. Let $s' \in (0,\infty)^{k-1}$ satisfy $\varrho(\bar{A}' - \operatorname{diag} s') < 0$. Then the formula for H in Lemma A.2 holds for $H_{A,\nu,e}$ with these X, g and b, except that their series are dominated term by term in modulus by those of $\bar{X}$, $\bar{g}$ and $\bar{b}$ at s'. In particular $|g(z')| \leq \bar{g}(s')$ and $|b(z')| \leq \bar{b}(s')$ when $|z_i| \geq s_i$ for every $i \geq 2$. Where $\mathcal{Z}' - A'$ is invertible, X, g and b are rational functions of z' and of the entries of A, ν and e.*

Proof We follow the four steps of the proof of Lemma A.2. In Step 1 the first fact gives $\varrho(\operatorname{diag}(s')^{-1}\bar{A}') < 1$, and for $|z_i| \geq s_i$ the entries of $\mathcal{Z}'^{-1}A'$ are bounded in modulus by those of $\operatorname{diag}(s')^{-1}\bar{A}'$. So the series of X converges absolutely, and its terms are dominated by those of $\bar{X}(s') \geq 0$. Since a_1, b_1, ν and e are dominated by their barred versions, the same holds for g and for the two factors of b. Step 2 applies to $\bar{A}$ as it stands and gives $s_1 > \bar{g}(s')$ with $\varrho(S^{-1}\bar{A}) < 1$. In Step 3 the entries of $A\mathcal{Z}^{-1}$ are bounded in modulus by those of $\bar{A}S^{-1}$, so the word expansion of $\nu^{\top}(\mathcal{Z} - A)^{-1}e$ converges absolutely and uniformly on the torus, and the same integration gives (A2) with A, ν and e in place of A_F, μ_F and $\mathbf{1}$. In Step 4 the Schur complement $z_1 - g(z')$ is not 0, since $|g(z')| \leq \bar{g}(s') < s_1$, and block inversion gives $\nu^{\top}(\mathcal{Z} - A)^{-1}e = b(z')/(z_1 - g(z')) + \nu'^{\top} X(z') e'$. The rest of Step 4 is unchanged. The last claim is Cramer's rule. □

A.3 Proof of Lemma 4.1

We bound the integrand of the discrete formula of Lemma A.2 by its modulus at the real point. Step 2 bounds the continuum sums in the same way, for later use.

Step 1. The radii. Let F be irreducible and $\theta \in V_F$. Then $A_F - \operatorname{diag}\theta$ is an irreducible Metzler matrix with Perron root 0. So it has a right eigenvector $u > 0$ for 0, and $\theta_i = (A_F u)_i / u_i$. Each row of A_F has a positive entry, since G_F is strongly connected and $k \geq 2$, so $\theta > 0$. By the second fact, $\varrho(A' - \operatorname{diag}\theta') < 0$. The rows $i \geq 2$ of $(A_F - \operatorname{diag}\theta)u = 0$ give $u' = u_1 X(\theta') b_1$, and then row 1, which reads $a_1^{\top} u' = \theta_1 u_1$, gives $\theta_1 = g(\theta')$. Hence

$$\theta > 0, \qquad \varrho(A' - \operatorname{diag}\theta') < 0, \qquad g(\theta') = \theta_1 \qquad \text{for every } \theta \in V_F. \tag{A3}$$

Step 2. Continuum sums. For $\sigma > 0$ let H_σ be H for σA_F in place of A_F. We use Lemma A.2 for σA_F with the radii $\sigma\theta'$, which is allowed since $\varrho(\sigma A' - \operatorname{diag}\sigma\theta') = \sigma\varrho(A' - \operatorname{diag}\theta') < 0$. The functions of Lemma A.2 for σA_F satisfy $X_\sigma(\sigma z') = \sigma^{-1} X(z')$, so $g_\sigma(\sigma z') = \sigma g(z')$ and $b_\sigma(\sigma z') = b(z')$. So on the torus with radii $\sigma\theta'$ we

have $|g_\sigma| \le \sigma g(\theta') = \sigma\theta_1$ by (A3), $|b_\sigma| \le b(\theta')$ and $|z_i| = \sigma\theta_i$. Hence the integrand has modulus at most $e^{\sigma\langle t,\theta\rangle} b(\theta')\sigma^{k-1}\prod_{i\ge 2}\theta_i$, and for every $\sigma > 0$ and $t \in (0,\infty)^F$

$$H_\sigma(t) \le \sigma^{k-1} b(\theta') \prod_{i\ge 2} \theta_i \, e^{\sigma\langle t,\theta\rangle}. \tag{A4}$$

Step 3. The bound. Let n have support F, let $hq_{\max} \le 1$ and $p_i = 1 - hq_i$. We use the discrete formula of Lemma A.2 with the radii θ'. On that torus $|p_i + hz_i| \le p_i + h\theta_i$ for $i \ge 2$, $|p_1 + hg(z')| \le p_1 + h\theta_1$ since $|g(z')| \le g(\theta') = \theta_1$, and $|b(z')| \le b(\theta')$. So

$$P_{|n|}(n) \le h^{k-1} b(\theta') \prod_{i\ge 2} \theta_i \prod_{i\in F} (p_i + h\theta_i)^{n_i - 1} \qquad (\theta \in V_F,\ hq_{\max} \le 1). \tag{A5}$$

Now let $hq_{\max} \le \frac{1}{2}$. Since $\log(1+x) \le x$, we have $(p_i + h\theta_i)^{n_i} = (1 - h(q_i - \theta_i))^{n_i} \le e^{-hn_i(q_i - \theta_i)}$, and $p_i + h\theta_i \ge p_i \ge \frac{1}{2}$. So the lemma holds with $C_\theta = 2^k b(\theta')\prod_{i\ge 2}\theta_i$. By Lemma A.2, b is continuous on the open set of s' with $\varrho(A' - \operatorname{diag} s') < 0$, which contains θ' for every $\theta \in V_F$ by (A3). So C_θ is bounded on compact subsets of V_F. □

A.4 Proof of Theorem 5.4(a)

The field cancels from the two sides of Theorem 5.4, so we take $v = 0$. For $k = 1$ the formula is part of Lemma 2.4. Let $k \ge 2$ and $\alpha \in \mathcal{K}$, and write $\theta = \theta(\alpha)$, $r = r_\alpha$, $l = l_\alpha$ and $\Theta' = \operatorname{diag}(\theta')$. By (A3) we may use Lemma A.2 with the radii θ' and $t = T\alpha$. With Lemma 2.4 this gives

$$f_T(\alpha) = \frac{T^{k-1} e^{-T\langle q,\alpha\rangle}}{(2\pi)^{k-1}} \int_{\mathbb{T}^{k-1}} e^{T\Psi(\alpha,\varphi)} \chi(\alpha,\varphi)\, d\varphi, \tag{A6}$$

where $z_i = \theta_i e^{\mathrm{i}\varphi_i}$, $\Psi(\alpha,\varphi) = \alpha_1 g(z') + \sum_{i\ge 2}\alpha_i z_i$ and $\chi(\alpha,\varphi) = b(z')\prod_{i\ge 2} z_i$. We check the hypotheses of Lemma B.1 with $m = k - 1$.

Hypothesis (iii). By (A3), $\Psi(\alpha, 0) = \langle \alpha, \theta\rangle$ is real. By Lemma A.2, $|g(z')| \le g(\theta') = \theta_1$ on the torus, so

$$\operatorname{Re}\Psi(\alpha,\varphi) \le \Psi(\alpha,0) - \sum_{i\ge 2} \alpha_i\theta_i(1 - \cos\varphi_i).$$

All α_i and θ_i are positive, so the sum is positive unless $\varphi = 0$.

Hypotheses (i) *and* (ii). We need the first two derivatives of g at θ'. We get them by differentiating a relation $\lambda(g(z'), z') = 0$ for a Perron root λ, whose derivatives are known from Lemma A.1. For $z \in \mathbb{R}^F$ near θ let $\lambda(z) = \varrho(A_F - \operatorname{diag} z)$. With $R = \operatorname{diag} r$ we have $A_F - \operatorname{diag}(\theta - \eta) = R(\widehat{Q}_\alpha + \operatorname{diag}\eta)R^{-1}$, and $\widehat{Q}_\alpha$ is an irreducible generator with stationary law α. So $\lambda(\theta - \eta) = \varrho(\widehat{Q}_\alpha + \operatorname{diag}\eta)$, and Lemma A.1 gives $\nabla\lambda(\theta) = -\alpha$ and $\nabla^2\lambda(\theta) = \Gamma_\alpha$. For real z' near θ' we have $\varrho(A' - \operatorname{diag} z') < 0$, so $(1, X(z')b_1) \ge 0$ by the first fact. This vector lies in the kernel of $A_F - \operatorname{diag}(g(z'), z')$, in the rows $i \ge 2$ by the definition of X and in row 1 by the definition of g. By the second fact a nonnegative eigenvector belongs to the Perron root, so $\lambda(g(z'), z') = 0$.

Write g_j, g_{ij} for the derivatives of g at θ' and λ_j, λ_{ij} for those of λ at θ. Differentiating $\lambda(g(z'),z')=0$ in z_j gives $\lambda_1 g_j+\lambda_j=0$, and differentiating again in z_i gives $\lambda_1 g_{ij}+\lambda_{11}g_ig_j+\lambda_{1j}g_i+\lambda_{i1}g_j+\lambda_{ij}=0$. Put $w_j=e_j-(\alpha_j/\alpha_1)e_1\in\mathbb{R}^F$ for $j\ge 2$. Then the first equation is $g_j=-\alpha_j/\alpha_1$, so $w_j=e_j+g_je_1$, and the last four terms of the second equation are $w_i^\top\Gamma_\alpha w_j$. So

$$g_j=-\frac{\alpha_j}{\alpha_1},\qquad \alpha_1 g_{ij}=w_i^\top\Gamma_\alpha w_j.$$

Hence $\partial\Psi/\partial z_j=\alpha_1 g_j+\alpha_j=0$ at θ', and the Hessian $\mathcal{H}$ of Ψ in z' at θ' is $V^\top\Gamma_\alpha V$, where V has columns $w_2,\dots,w_k$. Since $\partial/\partial\varphi_j=\mathrm{i}z_j\,\partial/\partial z_j$ and the first derivatives vanish, $\nabla_\varphi\Psi(\alpha,0)=0$. We saw above that $\Psi(\alpha,0)$ is real, so this is hypothesis (i). Also $-\nabla^2_\varphi\Psi(\alpha,0)=\Theta'\mathcal{H}\Theta'$, which is real and symmetric, as (ii) requires.

For the rest of (ii), namely that this matrix is positive definite, let $B=I+\mathbf{1}\alpha'^\top/\alpha_1$. For $y\in\mathbb{R}^{k-1}$ the vector Vy has the entry $-\alpha'^\top y/\alpha_1$ at state 1 and the entries y elsewhere, so $Vy+(\alpha'^\top y/\alpha_1)\mathbf{1}=(0,By)$. Since $\Gamma_\alpha\mathbf{1}=0$, we may replace Vy by $(0,By)$ in $y^\top V^\top\Gamma_\alpha Vy$, which gives $\mathcal{H}=B^\top\Sigma_\alpha B$. The matrix determinant lemma gives $\det B=1+\alpha'^\top\mathbf{1}/\alpha_1=1/\alpha_1$. By Lemma A.1, Σ_α is positive definite, and hence so are $\mathcal{H}$ and $\Theta'\mathcal{H}\Theta'$. Also $\det\mathcal{H}=\det\Sigma_\alpha/\alpha_1^2$.

Regularity. By Proposition 5.2(a), $\theta(\alpha)$ is real-analytic. Fix $\alpha_0\in\mathcal{K}$. By (A3) and continuity there is $\delta\in(0,\min_i\theta_i(\alpha_0))$ with $\varrho(A'-\operatorname{diag}(\theta'(\alpha_0)-\delta\mathbf{1}))<0$. By Lemma A.2, X, g and b are holomorphic where $|z_i|>\theta_i(\alpha_0)-\delta$ for every $i\ge 2$, and this holds on the torus of radii $\theta'(\alpha)$ for α near α_0. So Ψ and χ are real-analytic near $\{\alpha_0\}\times\mathbb{T}^{k-1}$. Covering $\mathcal{K}$ by finitely many such sets shows that their φ-derivatives are jointly continuous on $\mathcal{K}\times\mathbb{T}^{k-1}$.

The constant. The matrix $\mathcal{J}=\Theta'\mathcal{H}\Theta'$ of Lemma B.1 has $\det\mathcal{J}=\prod_{i\ge2}\theta_i^2\,\det\Sigma_\alpha/\alpha_1^2$. So the lemma gives, uniformly for $\alpha\in\mathcal{K}$ and $T\ge1$,

$$\int_{\mathbb{T}^{k-1}} e^{T\Psi}\chi\,d\varphi=e^{T\langle\alpha,\theta\rangle}\Big(\frac{2\pi}{T}\Big)^{\frac{k-1}{2}}\frac{\alpha_1}{\prod_{i\ge2}\theta_i\sqrt{\det\Sigma_\alpha}}\Big(\chi(\alpha,0)+O\Big(\frac1T\Big)\Big).$$

Here $\chi(\alpha,0)=b(\theta')\prod_{i\ge2}\theta_i$. As in the derivation of (A3), $r'=r_1X(\theta')b_1$. In the same way the columns $j\ge2$ of $l^\top(A_F-\operatorname{diag}\theta)=0$, which is (7), give $l'=l_1X(\theta')^\top a_1$. Hence $\mu\cdot r=r_1(\mu_1+\mu'^\top X(\theta')b_1)$ and $l\cdot\mathbf{1}=l_1(1+a_1^\top X(\theta')\mathbf{1})$. Since $l_1r_1=\alpha_1$,

$$b(\theta')=\frac{(\mu\cdot r_\alpha)(l_\alpha\cdot\mathbf{1})}{\alpha_1}.$$

As $r_\alpha,l_\alpha>0$ and $\mu(F)>0$ by (H), $\chi(\alpha,0)$ is positive and continuous, hence bounded below on $\mathcal{K}$. So the last bracket is $\chi(\alpha,0)(1+O(1/T))$. Putting this into (A6) and using $J_F(\alpha)=\langle q-\theta,\alpha\rangle$ gives $f_T(\alpha)=T^{(k-1)/2}C_F(\alpha)e^{-TJ_F(\alpha)}(1+O(1/T))$, uniformly on $\mathcal{K}$. □

The proof uses A_F, μ and the end vector $\mathbf{1}$ only through their signs and the graph of A_F. Section 6 needs it for other weights.

Lemma A.4 (general weights) *Let $A \geq 0$ be a matrix on F with zero diagonal and a strongly connected graph, let $\nu, e \in [0,\infty)^F$ be nonzero, and let $\theta(\alpha)$, r_α, l_α and Σ_α be the objects of Definition 5.1 for A in place of A_F. Let g and b be the functions of Lemma A.3 for (A,ν,e). Then the identities (A3), $g_j = -\alpha_j/\alpha_1$, $\mathcal{H} = B^\top \Sigma_\alpha B$, $\det \mathcal{H} = \det \Sigma_\alpha / \alpha_1^2$ and $\alpha_1 b(\theta') = (\nu \cdot r_\alpha)(l_\alpha \cdot e)$ of the proof above hold, and uniformly for α in a compact $\mathcal{K} \subset \Delta_F^\circ$ and $T \geq 1$*

$$H_{A,\nu,e}(T\alpha) = (2\pi T)^{-(k-1)/2} e^{T\langle \alpha, \theta(\alpha)\rangle} \, \frac{(\nu \cdot r_\alpha)(l_\alpha \cdot e)}{\sqrt{\det \Sigma_\alpha}} \, \big(1 + O(1/T)\big).$$

Proof Definition 5.1, Proposition 5.2(a) and Lemma A.1 involve A_F only as a nonnegative matrix with zero diagonal and a strongly connected graph, so they hold for A. We repeat the proof above with Lemma A.3, for $(\bar A, \bar\nu, \bar e) = (A, \nu, e)$, in place of Lemma A.2. The paragraphs on the hypotheses and on the regularity do not change. In the paragraph on the constant the identities $r' = r_1 X(\theta') b_1$ and $l' = l_1 X(\theta')^\top a_1$ give $\nu \cdot r_\alpha = r_1(\nu_1 + \nu'^\top X(\theta') b_1)$ and $l_\alpha \cdot e = l_1(e_1 + a_1^\top X(\theta') e')$, and $l_1 r_1 = \alpha_1$ gives the formula for $b(\theta')$. It is positive since $r_\alpha, l_\alpha > 0$ and $\nu, e \geq 0$ are nonzero. Then (A6) without the factor $T^{k-1} e^{-T\langle q,\alpha\rangle}$ and the Laplace lemma give the display. □

A.5 Proof of Lemma 5.5

The discrete law of Lemma 2.3 and the continuum density of Lemma 2.4 are sums over the same vectors m of run counts, and we compare them term by term.

Step 1. The ratio of the terms. Let $t = hn$ and $\alpha = n/N \in \mathcal{K}$. Lemma 2.4 gives the identity $N^{-(k-1)} f_T(n/N) = h^{k-1} e^{-\langle q,t\rangle} H(t) = \sum_m v_m$ with

$$v_m = W(m)\, h^{k-1} \prod_{i \in F} e^{-q_i t_i} \frac{t_i^{m_i - 1}}{(m_i - 1)!}, \qquad m \in \mathbb{Z}_{\geq 1}^F.$$

Since $h^{m_i - 1}\binom{n_i - 1}{m_i - 1} = \frac{t_i^{m_i-1}}{(m_i-1)!} \prod_{1 \leq j < m_i}(1 - j/n_i)$, the m-th term of Lemma 2.3 is $\kappa_m v_m$ with

$$\kappa_m = \prod_{i \in F} \Big[\prod_{1 \leq j < m_i} \Big(1 - \frac{j}{n_i}\Big) \Big] (1 - hq_i)^{n_i - m_i} e^{hq_i n_i}.$$

Let $w_m = v_m / \sum_{m'} v_{m'}$, a probability on the vectors m since $H(t) > 0$ under (H). We write $\mathbb{E}_w$ and $\mathbb{P}_w$ for it and regard $M = \sum_i m_i$ as a random variable. By Lemma 2.3 the ratio in the lemma is $\sum_m \kappa_m v_m / \sum_m v_m = \mathbb{E}_w \kappa_m$.

Step 2. Bounds for κ_m. Let $n_{\min} = \min_i n_i$. We claim that for every m

$$\Big(1 - \frac{M^2}{2n_{\min}}\Big) e^{-q_{\max}^2 T^2/N} \leq \kappa_m \leq e^{hq_{\max} M}. \tag{A7}$$

First let $m_i \leq n_i$ for every i. The factors $1 - j/n_i$ are at most 1, and $(1 - hq_i)^{n_i - m_i} \leq e^{-hq_i(n_i - m_i)}$, so $\kappa_m \leq e^{hq_{\max} M}$. For the lower bound, $\log(1 - x) \geq -x - x^2$ for $0 \leq x \leq \frac{1}{2}$ gives $(1 - hq_i)^{n_i - m_i} e^{hq_i n_i} \geq e^{-n_i h^2 q_i^2}$, and the product of these over i is at least $e^{-q_{\max}^2 T^2/N}$. The factors $1 - j/n_i$ lie in $[0,1]$, so their product is at least 1 minus

the sum of the j/n_i, and hence at least $1-\sum_i m_i^2/(2n_i) \ge 1 - M^2/(2n_{\min})$. Now let $m_i > n_i$ for some i. Then $\kappa_m = 0$, while $M \ge n_{\min}+1$ gives $M^2/(2n_{\min}) \ge 2$, so the left side of (A7) is negative.

By (A7) it remains to bound $\mathbb{E}_w e^{hq_{\max}M}$ and $\mathbb{E}_w M^2$ from above.

Step 3. The number of runs under w. Since W for σA_F is $\sigma^{M-1}W(m)$, we have $\mathbb{E}_w\sigma^{M-1} = H_\sigma(t)/H(t)$. So the function $\Upsilon(c) = \log \mathbb{E}_w e^{c(M-1)} = \log H_{e^c}(t) - \log H(t)$ is finite. It is convex with $\Upsilon(0)=0$, as the logarithm of a moment generating function, and $\Upsilon(1) \ge 0$ since $M \ge 1$. So $\Upsilon(c) \le c\,\Upsilon(1)$ for $0 \le c \le 1$. We show that $\Upsilon(1) \le C_{\mathcal{K}}T$, with an upper bound for H_e and a lower bound for H at $t = T\alpha$. By (A4) with $\sigma = e$ and $\theta = \theta(\alpha)$, $H_e(T\alpha) \le e^{k-1}b(\theta')\prod_{i\ge 2}\theta_i\, e^{eT\langle\alpha,\theta\rangle}$. For the lower bound, $H(T\alpha)e^{-T\langle\alpha,\theta\rangle} = T^{-(k-1)}e^{TJ_F(\alpha)}f_T(\alpha)$, by the definition of f_T and $J_F(\alpha) = \langle q-\theta,\alpha\rangle$. By Theorem 5.4(a) this is at least $\frac12 T^{-(k-1)/2}C_F(\alpha)$ for $T \ge T_1$, and for $1 \le T \le T_1$ it is continuous and positive on $[1,T_1]\times\mathcal{K}$. So in both cases $H(T\alpha) \ge c_{\mathcal{K}}T^{-(k-1)/2}e^{T\langle\alpha,\theta\rangle}$. Since $\theta(\alpha)$ and $b(\theta'(\alpha))$ are bounded on $\mathcal{K}$, dividing the two bounds gives $\Upsilon(1) \le C_{\mathcal{K}}T$.

Step 4. The two averages. Let $x = hq_{\max} \le \frac12$. Then $\mathbb{E}_w e^{xM} = e^{x+\Upsilon(x)} \le e^{x(1+C_{\mathcal{K}}T)} = 1 + O(T^2/N)$, because $x \le q_{\max}T/N$ and $T^2/N \le 1$. For the second moment we use the Chernoff bound $\mathbb{P}_w(M-1 \ge y) \le e^{\Upsilon(1)-y}$ for $y \ge \Upsilon(1)$ and the bound 1 for $y < \Upsilon(1)$. This gives

$$\begin{aligned}\mathbb{E}_w(M-1)^2 &= \int_0^\infty 2y\,\mathbb{P}_w(M-1\ge y)\,dy \\ &\le \Upsilon(1)^2 + \int_{\Upsilon(1)}^\infty 2ye^{\Upsilon(1)-y}\,dy = \Upsilon(1)^2 + 2\Upsilon(1) + 2,\end{aligned}$$

so $\mathbb{E}_w M^2 = O(T^2)$. Finally $n_{\min} \ge \varepsilon_{\mathcal{K}}N$ with $\varepsilon_{\mathcal{K}} = \min_{\alpha\in\mathcal{K}}\min_i \alpha_i > 0$. So by (A7), $\mathbb{E}_w\kappa_m$ lies between $e^{-q_{\max}^2T^2/N}(1 - \mathbb{E}_w M^2/(2\varepsilon_{\mathcal{K}}N))$ and $\mathbb{E}_w e^{hq_{\max}M}$, and both are $1+O(T^2/N)$, uniformly for $\alpha\in\mathcal{K}$ and $1 \le T \le N^{1/2}$. □

A.6 The derivative in T

Theorem 5.8(iv) needs the derivative of the local law in T. For fixed N and n, $P_N(n)$ is a polynomial in T, and we differentiate it at fixed N and n.

Lemma A.5 (the derivative in T) *Let F satisfy* (H) *with $k \ge 2$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact. There are T_1, N_0 and C such that for $N \ge N_0$, $\operatorname{supp} n = F$, $n/N \in \mathcal{K}$ and $T_1 \le T \le N^{1/3}$,*

$$\left|\partial_T \log P_N(n) + J_F(n/N)\right| \le \frac{C}{T}.$$

Proof Put $\alpha = n/N$ and $t = hn$. Let $\tilde w$ be the probability on the vectors m proportional to the terms of Lemma 2.3. Since $T\partial_T = h\partial_h$ and the m-th term has the logarithmic derivative $(M-1) - \sum_i (n_i - m_i)hq_i/(1-hq_i)$ in $\log h$,

$$T\partial_T \log P_N(n) = \mathbb{E}_{\tilde w}(M-1) - \sum_{i\in F}\frac{hq_i}{1-hq_i}\,\mathbb{E}_{\tilde w}(n_i - m_i).$$

For $c \in \mathbb{R}$ let $P^{(c)}$ be the sum of Lemma 2.3 with $e^c A_F$ in place of A_F, that is with $e^{c(M-1)}W(m)$ in place of $W(m)$ and the same factors $(1-hq_i)^{n_i-m_i}$. Then $\Upsilon(c) = \log P^{(c)} - \log P^{(0)} = \log \mathbb{E}_{\tilde{w}} e^{c(M-1)}$ is convex, and $\Upsilon'(0) = \mathbb{E}_{\tilde{w}}(M-1)$.

Step 1. The asymptotics of Υ. The proof of Lemma 5.5 uses A_F only through the non-negative weights $W(m)$ and through bounds on its continuum sum H. For $e^c A_F$ this sum is $H_{e^c}(t) = e^{c(k-1)}H(e^c t)$, since a word with M letters has $M-1$ switches and $M-k$ is the total degree of $\prod_i t_i^{m_i-1}$. Let $|c| \le 1$ and $T \ge e$. Step 3 of that proof needs the ratio $H_{e^{c+1}}(t)/H_{e^c}(t) = e^{k-1}H(e^{c+1}t)/H(e^c t)$ to be at most $e^{C_{\mathcal{K}}T}$. This follows from (A4) and from the lower bound for H at time $e^c T \ge 1$ in that step. So Lemma 5.5 holds for the weights $e^c A_F$, uniformly for $|c| \le 1$, and gives $P^{(c)} = h^{k-1}e^{-\langle q,t\rangle}e^{c(k-1)}H(e^c t)(1+O(T^2/N))$. By Lemma 2.4 and $J_F(\alpha) = \langle q - \theta(\alpha), \alpha\rangle$, Theorem 5.4(a) at time $e^c T$ reads $H(e^c T\alpha) = (e^c T)^{-(k-1)/2} C_F(\alpha) e^{e^c T\langle\theta(\alpha),\alpha\rangle}(1+O(1/T))$. Hence, uniformly for $|c| \le 1$, $\alpha \in \mathcal{K}$ and $e \le T \le N^{1/3}$,

$$\Upsilon(c) = a(c) + \varepsilon(c), \qquad a(c) = \frac{k-1}{2}\,c + (e^c - 1)\,T\langle\theta(\alpha),\alpha\rangle, \qquad |\varepsilon(c)| \le \frac{C_1}{T},$$

where we used $T^2/N \le 1/T$.

Step 2. Convexity. We have $a'(0) = \frac{k-1}{2} + T\langle\theta(\alpha),\alpha\rangle$ and $0 \le a'' \le C_2 T$ on $[-1,1]$, since θ is bounded on $\mathcal{K}$. Let $\eta = 1/T$. By convexity, $(\Upsilon(0)-\Upsilon(-\eta))/\eta \le \Upsilon'(0) \le (\Upsilon(\eta)-\Upsilon(0))/\eta$. By Taylor's theorem both quotients differ from $a'(0)$ by at most $C_2 T\eta/2 + 2C_1/(T\eta)$, which is $C_2/2 + 2C_1$. So $\mathbb{E}_{\tilde{w}}(M-1) = T\langle\theta(\alpha),\alpha\rangle + O(1)$, and in particular $\mathbb{E}_{\tilde{w}} M = O(T)$.

Step 3. The other term. Since $hn_i = T\alpha_i$ and $hq_{\max} \le \frac{1}{2}$, $\sum_i hq_i n_i/(1-hq_i) = T\langle q,\alpha\rangle + O(Th)$ and $0 \le \sum_i hq_i \mathbb{E}_{\tilde{w}} m_i/(1-hq_i) \le 2hq_{\max}\mathbb{E}_{\tilde{w}} M = O(T^2/N)$. Altogether $T\partial_T \log P_N(n) = T\langle\theta(\alpha),\alpha\rangle - T\langle q,\alpha\rangle + O(1) = -TJ_F(\alpha) + O(1)$, and we divide by T. □

A.7 Proof of Theorem 4.2(ii)

Here Theorem 4.2(ii) follows from Theorem 4.5, and a direct proof by explicit paths is in this section of the arXiv version of this paper [40].

A.8 The local law over the whole switching range

Here F is irreducible with $k \ge 2$ states, h_* is fixed with $h_* q_{\max} < 1$, $p_i = 1 - hq_i$ and $p_{\min} = 1 - h_* q_{\max} > 0$. Let Ω be the open set of $s' \in (0,\infty)^{k-1}$ with $\varrho(A' - \operatorname{diag} s') < 0$, and for $s' \in \Omega$ put $s_1 = g(s')$ and

$$\Psi_h(\alpha; s') = -\sum_{i \in F} \alpha_i L_h(q_i - s_i),$$

with L_h from Section 3.3. For $h > 0$ this is $\frac{1}{h}\sum_i \alpha_i \log(p_i + hs_i)$.

Lemma A.6 (the discrete Hessian) *Let $(h,\alpha) \in [0,h_*] \times \Delta_F^\circ$, and let $\theta = \theta^{(h)}(\alpha)$, r, l and $c_h = c_h(\alpha)$ be as in the definition of $C_F^{(h)}$ in Section 5.2. Put $\pi = l \circ r$.*

(a) The point θ' is the only minimizer of $\Psi_h(\alpha;\cdot)$ on Ω, $\Psi_h(\alpha;\theta') = -J_F^{(h)}(\alpha)$, and the Hessian H_h of $\Psi_h(\alpha;\cdot)$ at θ' is positive definite.

(b) There is a $(k-1)\times(k-1)$ matrix K with $\det K = 1/(c_h \pi_1 \prod_{i\in F}(p_i + h\theta_i))$ and $\nabla^2 J_F^{(h)}(\alpha) = K^\top H_h^{-1} K$. So $\nabla^2 J_F^{(h)}(\alpha)$ is positive definite.

(c) $C_F^{(h)}(\alpha) = b(\theta')/\big((2\pi)^{(k-1)/2}\prod_{i\in F}(p_i+h\theta_i)\sqrt{\det H_h}\big)$*, and* $C_F^{(h)}$ *is positive and real-analytic on* $[0,h_*]\times\Delta_F^\circ$ *up to* $h=0$*.*

Proof (a) The map $\psi\mapsto\theta(\psi)'$ is a real-analytic diffeomorphism from $\{\psi\in\mathbb{R}^F:\psi_1=0\}$ onto Ω. Indeed $\theta(\psi)=A_Fe^\psi/e^\psi$ lies in V_F, so $\theta(\psi)'\in\Omega$ and $\theta_1(\psi)=g(\theta(\psi)')$ by (A3). Conversely, for $s'\in\Omega$ the vector $u=(1,X(s')b_1)$ is nonnegative by the first fact, and it is a null vector of $A_F-\operatorname{diag}(g(s'),s')$, as in the proof of Theorem 5.4(a). By the second fact it is positive, and $\psi=\log u$ has $\psi_1=0$ and $\theta(\psi)=(g(s'),s')$. Both maps are real-analytic. Since $\Psi_h(\alpha;\theta(\psi)')=-G_h(\alpha,\psi)$, part (a) follows from Proposition 3.5(b). The first derivatives vanish at the critical point, so there the two Hessians are congruent through the Jacobian of the diffeomorphism, which is invertible.

(b) Write $y_i(s')=L_h(q_i-s_i)$ with $s_1=g(s')$, so that, in the coordinates α', $-\Psi_h=y_1+\sum_{j\ge2}\alpha_j(y_j-y_1)$. By (a), $J_F^{(h)}(\alpha)=-\Psi_h(\alpha;\theta^{(h)}(\alpha)')$, and $\theta^{(h)}(\alpha)'$ is real-analytic in α. So the envelope theorem gives $\partial J_F^{(h)}/\partial\alpha_j=y_j(\theta')-y_1(\theta')$ for $j\ge2$. Let $K_{ij}=-\partial(y_j-y_1)/\partial s_i$ at θ', for $i,j\ge2$. Differentiating $\nabla_{s'}\Psi_h(\alpha;\theta^{(h)}(\alpha)')=0$ in α' gives $H_h\,\partial\theta'/\partial\alpha'=-K$, and hence $\nabla^2J_F^{(h)}=-K^\top\partial\theta'/\partial\alpha'=K^\top H_h^{-1}K$.

To compute K put $e_i=L_h'(q_i-\theta_i)=1/(p_i+h\theta_i)$, and let g_i be the derivatives of g at θ'. The vector $\psi=\log r$ attains the supremum defining $J_F(\pi)$ by the criterion in the proof of Proposition 3.3, since π is the stationary law of $R^{-1}(A_F-\operatorname{diag}\theta)R$ with $R=\operatorname{diag}r$. So $\theta=\theta(\pi)$, and the proof of Theorem 5.4(a) at the point π gives $g_i=-\pi_i/\pi_1$. Hence $K_{ij}=\delta_{ij}e_j+e_1\pi_i/\pi_1$, that is $K=\operatorname{diag}(e')+(e_1/\pi_1)\pi'\mathbf{1}^\top$, and the matrix determinant lemma gives $\det K=\prod_{i\in F}e_i\cdot\sum_{i\in F}\pi_i(p_i+h\theta_i)/\pi_1$. By Proposition 3.5(c) with $\psi=\log r$, $\alpha_i=c\,\pi_i(p_i+h\theta_i)$ for a constant c. Summing $\alpha_i/(p_i+h\theta_i)$ over i gives $c=c_h$, and summing α_i gives $\sum_i\pi_i(p_i+h\theta_i)=1/c_h$. This is the value of $\det K$, which is not 0, so $\nabla^2J_F^{(h)}(\alpha)$ is positive definite by (a).

(c) The identity $\alpha_1b(\theta')=(\mu\cdot r_\alpha)(l_\alpha\cdot\mathbf{1})$ of the proof of Theorem 5.4(a), at the point π, gives $b(\theta')=(\mu\cdot r)(l\cdot\mathbf{1})/\pi_1$, since $l\cdot r=1$. By (b), $\sqrt{\det\nabla^2J_F^{(h)}}=|\det K|/\sqrt{\det H_h}$. Putting both into the definition of $C_F^{(h)}$ gives the formula, which is positive. By Proposition 3.5(b), $\theta^{(h)}(\alpha)$ and $J_F^{(h)}$ are real-analytic up to $h=0$, and so are c_h and the normalized null vectors r and l, since 0 is a simple eigenvalue of $A_F-\operatorname{diag}\theta$. As $\det\nabla^2J_F^{(h)}>0$, $C_F^{(h)}$ is real-analytic up to $h=0$. □

Proof of Theorem 5.4(b) The field cancels from the two sides, so we take $v=0$. Let $\mathcal{K}$ be as in the theorem and put $\mathcal{P}=[0,h_*]\times\mathcal{K}$. For $(h,\alpha)\in\mathcal{P}$ let $\theta=\theta^{(h)}(\alpha)$ and $\Theta'=\operatorname{diag}(\theta')$.

The integral. Let $1\le T\le h_*N$, $\operatorname{supp}n=F$ and $\alpha=n/N\in\mathcal{K}$, so that $h=T/N\in(0,h_*]$ and $n_i=T\alpha_i/h$. By (A3) we may use the discrete formula of Lemma A.2 with the radii θ'. It gives $P_N(n)=h^{k-1}(2\pi)^{-(k-1)}\int_{\mathbb{T}^{k-1}}F_n\,d\varphi$, where F_n is the integrand. With $z_i=\theta_ie^{\mathrm{i}\varphi_i}$ and $z_1=g(z')$ put

$$\Psi_h(\alpha,\varphi)=-\sum_{i\in F}\alpha_iL_h(q_i-z_i),\qquad a_h(\alpha,\varphi)=\frac{b(z')\prod_{i\ge2}z_i}{\prod_{i\in F}(p_i+hz_i)}$$

where $\operatorname{Re}(p_i+hz_i)>0$ for every i. There $F_n=e^{T\Psi_h}a_h$, since $(p_i+hz_i)^{n_i}=\exp(n_i\log(p_i+hz_i))$ with the principal logarithm and $n_i\log(p_i+hz_i)=-T\alpha_iL_h(q_i-z_i)$. The logarithm has no branch on the whole torus when $h\theta_i>p_i$, so we use these formulas only near $\varphi=0$.

Regularity on a cube. The map $(h,\alpha)\mapsto\theta^{(h)}(\alpha)$ is continuous on the compact set $\mathcal{P}$, so $\theta_i\in[\theta_{\min},\theta_{\max}]$ with $\theta_{\min}>0$, and $\varrho(A'-\operatorname{diag}\theta')$ is bounded above by a negative constant. As in the regularity paragraph of the proof of (a), X, g and b are holomorphic near the tori of radii θ', uniformly on $\mathcal{P}$, and $g(z')$ is uniformly continuous in (h,α,φ). So there is $\varepsilon_1\in(0,1)$ such that $|g(z')-\theta_1|\le\theta_{\min}$ when $\|\varphi\|\le\varepsilon_1$. Then $\operatorname{Re}g(z')\ge0$ and $\operatorname{Re}z_i>0$ for $i\ge2$, so $\operatorname{Re}(p_i+hz_i)\ge p_{\min}$ for every i. Since $1-shy=(1-s)+s(1-hy)$, the function $L_h(y)=\int_0^1 y\,ds/(1-shy)$ is holomorphic in y and real-analytic in h where $\operatorname{Re}(1-hy)>0$, including $h=0$. So Ψ_h and a_h are defined and real-analytic on $\mathcal{P}\times\{\|\varphi\|\le\varepsilon_1\}$, and their φ-derivatives up to orders 4 and 2 are bounded there by constants B_Ψ and B_a. At $h=0$ they are $\Psi-\langle q,\alpha\rangle$ and χ of (A6).

Hypotheses (i) *and* (ii). At $\varphi=0$, $\Psi_h(\alpha,0)=\Psi_h(\alpha;\theta')=-J_F^{(h)}(\alpha)$ by Lemma A.6(a), which is real. Since $\partial/\partial\varphi_j=\mathrm{i}z_j\,\partial/\partial z_j$ and Ψ_h is holomorphic in z' with the derivatives of the real function $\Psi_h(\alpha;\cdot)$ at θ', Lemma A.6(a) gives $\nabla_\varphi\Psi_h(\alpha,0)=0$ and $-\nabla_\varphi^2\Psi_h(\alpha,0)=\Theta'H_h\Theta'$. This matrix is real, symmetric and positive definite, and it is continuous on $\mathcal{P}$, so its least eigenvalue is at least some $\sigma_{\min}>0$ on $\mathcal{P}$.

The gap. Choose $\varepsilon\in(0,\varepsilon_1]$ with $(k-1)^3B_\Psi\varepsilon\le\frac32\sigma_{\min}$. For $i\ge2$,

$$|p_i+h\theta_ie^{\mathrm{i}\varphi_i}|^2=(p_i+h\theta_i)^2-2p_ih\theta_i(1-\cos\varphi_i).$$

With $\log(1-x)\le-x$, and with $n_i-1\ge n_i/2=T\alpha_i/(2h)$, which holds for $N\ge N_0$ since then $n_i\ge2$, this gives $|p_i+hz_i|^{n_i-1}\le(p_i+h\theta_i)^{n_i-1}e^{-c_*T(1-\cos\varphi_i)}$ with $c_*=\min_{\mathcal{K}}\min_i\alpha_i\,p_{\min}\theta_{\min}/(2(1+h_*\theta_{\max})^2)>0$. The h in n_i cancels the h in $p_ih\theta_i$, so c_* does not depend on h. Also $|p_1+hg(z')|\le p_1+h\theta_1$ and $|b(z')|\le b(\theta')$ by Lemma A.2. If $\|\varphi\|\ge\varepsilon$, then $1-\cos\varphi_i\ge1-\cos\varepsilon$ for some $i\ge2$, so

$$|F_n(\varphi)|\le e^{-\eta T}F_n(0)=e^{-\eta T}e^{T\Psi_h(\alpha,0)}a_h(\alpha,0),\qquad\eta=c_*(1-\cos\varepsilon),$$

and $a_h(\alpha,0)$ is bounded on $\mathcal{P}$.

The expansion. By Remark B.2 in its localized form, with $m=k-1$, uniformly in (h,α) and $T\ge1$,

$$\int_{\mathbb{T}^{k-1}}F_n\,d\varphi=e^{-TJ_F^{(h)}(\alpha)}\Big(\frac{2\pi}{T}\Big)^{\frac{k-1}{2}}\det(\Theta'H_h\Theta')^{-1/2}\Big(a_h(\alpha,0)+O\Big(\frac1T\Big)\Big).$$

Here $a_h(\alpha,0)=b(\theta')\prod_{i\ge2}\theta_i/\prod_i(p_i+h\theta_i)$ is positive and continuous on $\mathcal{P}$, so the bracket is $a_h(\alpha,0)(1+O(1/T))$. Since $\det(\Theta'H_h\Theta')=\prod_{i\ge2}\theta_i^2\det H_h$ and $h^{k-1}T^{-(k-1)/2}=N^{-(k-1)}T^{(k-1)/2}$, Lemma A.6(c) gives the display of (b) with a factor $1+E$, $|E|\le C/T$.

Bounded T. It remains to bound $1+E$ below for $1\le T\le T_1=2C$. Let $N\ge T_1^2$. Then $T\le N^{1/2}$, and Lemma 5.5 and part (a) give $P_N(n)=N^{-(k-1)}f_T(\alpha)(1+O(T_1^2/N))$. The ratio of $f_T(\alpha)$ to $T^{(k-1)/2}C_F(\alpha)e^{-TJ_F(\alpha)}$ is continuous and positive on $[1,T_1]\times\mathcal{K}$, and the ratio of $C_F^{(h)}(\alpha)e^{-TJ_F^{(h)}(\alpha)}$ to $C_F(\alpha)e^{-TJ_F(\alpha)}$ is $1+O(T_1h)$ by Proposition 3.5(b) and Lemma A.6(c). So for large N, $1+E$ lies between two positive constants for $1\le T\le T_1$, and for $T\ge T_1$ it lies in $[\frac12,\frac32]$. □

Appendix B A Laplace lemma with a parameter

This appendix proves the Laplace lemma that Appendix A uses for Theorem 5.4, and its form with a complex Hessian that Section 6 uses for Theorem 6.8. The compact parameter space $\mathcal{K}$ can include the rates and the start when constants are needed uniformly over a class of chains. On $\mathbb{T}^m=(\mathbb{R}/2\pi\mathbb{Z})^m$ we use $\|\varphi\|=\max_i\operatorname{dist}(\varphi_i,2\pi\mathbb{Z})$. For $x\in\mathbb{R}^m$ we write $\|x\|=\max_i|x_i|$ and $|x|$ for the Euclidean norm. Repeated indices

are summed from 1 to m. The proofs of Lemmas B.1 and B.3, by the classical Laplace method, are in Appendix B of the arXiv version of this paper [40].

Lemma B.1 (Laplace integrals with a parameter) *Let $\mathcal{K}$ be a compact metric space and $m \geq 1$. Let $\Psi, a : \mathcal{K} \times \mathbb{T}^m \to \mathbb{C}$ be such that the φ-derivatives of Ψ up to order 4 and of a up to order 2 exist and are jointly continuous on $\mathcal{K} \times \mathbb{T}^m$. Assume that for every $\alpha \in \mathcal{K}$*

(i) the value $\Psi(\alpha, 0)$ is real and $\nabla_\varphi \Psi(\alpha, 0) = 0$,

(ii) the matrix $\mathcal{J}(\alpha) = -\nabla_\varphi^2 \Psi(\alpha, 0)$ is real, symmetric and positive definite, and

(iii) the inequality $\operatorname{Re} \Psi(\alpha, \varphi) < \Psi(\alpha, 0)$ holds for every $\varphi \neq 0$ in $\mathbb{T}^m$.

Then, uniformly for $\alpha \in \mathcal{K}$ and $T \geq 1$,

$$\int_{\mathbb{T}^m} e^{T\Psi(\alpha,\varphi)} a(\alpha, \varphi)\, d\varphi = e^{T\Psi(\alpha,0)} \Big(\frac{2\pi}{T}\Big)^{m/2} \det \mathcal{J}(\alpha)^{-1/2} \Big(a(\alpha, 0) + O\Big(\frac{1}{T}\Big)\Big).$$

Remark B.2 (the uniform form) The proof of Lemma B.1 of the arXiv version of this paper [40] uses compactness and joint continuity only to obtain the bounds B_Ψ and B_a on the derivatives, the lower bound $\sigma_{\min}$ on the least eigenvalue of $\mathcal{J}$, and a gap $\operatorname{Re} \Psi(\alpha, \varphi) - \Psi(\alpha, 0) \leq -\eta$ for $\|\varphi\| \geq \varepsilon$, where $\varepsilon \in (0, 1)$ satisfies $m^3 B_\Psi \varepsilon \leq \frac{3}{2}\sigma_{\min}$. So the following holds with the same proof. Let $\mathcal{P}$ be any set, and let $\Psi, a : \mathcal{P} \times \mathbb{T}^m \to \mathbb{C}$ satisfy (i) and (ii) for every $\alpha \in \mathcal{P}$, with φ-derivatives of Ψ up to order 4 and of a up to order 2 bounded by B_Ψ and B_a on $\mathcal{P} \times \mathbb{T}^m$, with $\mathcal{J}(\alpha) \geq \sigma_{\min} I$ for all α, and with the gap above for all α. Then the expansion of Lemma B.1 holds uniformly for $\alpha \in \mathcal{P}$ and $T \geq 1$, with a constant that depends only on m, B_Ψ, B_a, $\sigma_{\min}$, ε and η.

The proof uses the integrand off the cube $\|\varphi\| \leq \varepsilon$ only in the localization, through a bound on its modulus, and it uses the bounds on the derivatives only on the cube. So the expansion also holds in the following localized form, with a constant that depends in addition on a number B. Let Ψ and a be defined only on $\mathcal{P} \times \{\|\varphi\| \leq \varepsilon\}$, satisfy (i) and (ii), have φ-derivatives bounded by B_Ψ and B_a there, with $\mathcal{J}(\alpha) \geq \sigma_{\min} I$ and $m^3 B_\Psi \varepsilon \leq \frac{3}{2}\sigma_{\min}$, and let the integrand be $e^{T\Psi} a$ on the cube and any measurable function of modulus at most $B e^{T(\Psi(\alpha,0)-\eta)}$ off it. Appendix A.8 needs this form, since there the logarithm in the exponent has no branch on the whole torus.

Section 6 needs the same expansion when $\Psi(\alpha, 0)$ and the Hessian are complex. Let $\mathcal{S}_m$ be the set of complex symmetric $m \times m$ matrices M whose real part is positive definite. For $M \in \mathcal{S}_m$ and $x \in \mathbb{C}^m$ with $x \neq 0$ we have $\operatorname{Re}(\bar{x}^\top M x) = \bar{x}^\top (\operatorname{Re} M) x > 0$, because $\bar{x}^\top P x$ is real for a real symmetric P. So $\det M \neq 0$. The set $\mathcal{S}_m$ is convex, so $1/\det M$ has exactly two continuous square roots on it. We write $\det M^{-1/2}$ for the one that is positive on the real matrices of $\mathcal{S}_m$. Then

$$\int_{\mathbb{R}^m} e^{-\frac{1}{2} x^\top M x}\, dx = (2\pi)^{m/2} \det M^{-1/2} \qquad (M \in \mathcal{S}_m).$$

Indeed both sides are holomorphic functions of the entries M_{jk}, $j \leq k$, on the open set $\mathcal{S}_m$. For the left side we may differentiate under the integral, since $|e^{-x^\top M x/2}| = e^{-x^\top (\operatorname{Re} M) x/2}$ is bounded by a fixed Gaussian for M near a point of $\mathcal{S}_m$. The right side is a continuous square root of a holomorphic function without zeros. The two sides

agree on the real positive definite matrices, which form a nonempty open subset of the real points of $\mathcal{S}_m$. A holomorphic function on a connected open set that vanishes on such a subset vanishes everywhere, since its Taylor coefficients at a real point are real derivatives. So the two sides agree on $\mathcal{S}_m$.

Lemma B.3 (a complex Hessian) *Let $\mathcal{P}$ be a set and $m \geq 1$. Let $\Psi, a : \mathcal{P} \times \mathbb{T}^m \to \mathbb{C}$ have φ-derivatives up to order 4 and 2 that are continuous in φ and bounded in modulus by B_Ψ and B_a on $\mathcal{P} \times \mathbb{T}^m$. Let $\sigma_{\min} > 0$, $\eta > 0$ and $\varepsilon \in (0,1)$ with $m^3 B_\Psi \varepsilon \leq \frac{3}{2}\sigma_{\min}$. Assume that for every $p \in \mathcal{P}$*

(i) $\nabla_\varphi \Psi(p, 0) = 0$,

(ii) the complex symmetric matrix $\mathcal{J}(p) = -\nabla_\varphi^2 \Psi(p, 0)$ *satisfies* $\operatorname{Re} \mathcal{J}(p) \geq \sigma_{\min} I$, *and*

(iii) $\operatorname{Re} \Psi(p, \varphi) - \operatorname{Re} \Psi(p, 0) \leq -\eta$ *for* $\|\varphi\| \geq \varepsilon$.

Then the expansion of Lemma B.1 holds uniformly for $p \in \mathcal{P}$ and $T \geq 1$, with the branch of $\det \mathcal{J}^{-1/2}$ defined above and a constant that depends only on m, B_Ψ, B_a, $\sigma_{\min}$, ε and η.

Its proof is that of Lemma B.1 in the form of Remark B.2, with the bound $|e^{-x^\top \mathcal{J} x/2}| \leq e^{-\sigma_{\min}|x|^2/2}$ for real x and the Gaussian integral above.

Appendix C Proofs for the complete graph

In the notation of Section 7, the next lemma writes the law with blocks between refreshes in place of runs.

Lemma C.1 (the refresh form) *Let $N \geq 1$ and let n be a composition with support F. Sums over m run over the integer vectors with $1 \leq m_i \leq n_i$ for $i \in F$, and $M = \sum_i m_i$.*

(a) With $0^0 = 1$,

$$P_N(n) = \sum_m \Big(\sum_{i \in F} \mu_i m_i \Big) \frac{(M-1)!}{\prod_i m_i!}\, h^{M-1} \sigma^{N-M} \prod_{i \in F} \binom{n_i - 1}{m_i - 1}.$$

(b) Let $1/d < p < 1$, so that $0 < u < 1$ and $v > 0$. Then

$$S_n(u) = (1-u)^{N-1} \sum_m M!\, v^{M-1} \prod_{i \in F} \frac{1}{m_i!} \binom{n_i - 1}{m_i - 1}, \tag{C1}$$

which is a sum of positive terms, and

$$S_n(u) = \frac{(1-u)^{N-1}}{v} \int_0^\infty e^{-t} \prod_{i \in F} B_{n_i}(vt)\, dt, \tag{C2}$$

where B_n is defined in (22).

The proofs of Lemma C.1, Theorems 7.4 and 7.5 and Corollary 7.6 are in Appendix C of the arXiv version of this paper [40].

Appendix D Proofs near a vertex

In the notation of Section 8 we state the two lemmas behind Theorem 8.2.

Lemma D.1 (elasticity and concavity of the rare-count polynomial) *The polynomial $\mathcal{B}$ is convex, $\log \mathcal{B}$ is concave on $(0,\infty)$, and κ is nondecreasing. Also*

$$\frac{1}{|a|+1} \le y_a \le 1, \qquad \kappa(y) \le \ell + \sqrt{\ell |a| y} \quad (y>0). \tag{D1}$$

For $0 < y \le 1$ we also have $\kappa(y) \ge \max\{\ell, \frac{1}{2}\sqrt{|a|y}\}$.

Lemma D.2 (an exact compound-binomial representation) *For $0<u<1$, put $v = u/(1-u)$. Let Γ_2 have the gamma distribution with shape 2 and rate 1, let $E_1,\dots,E_{b-1}$ be independent rate-one exponential variables, and let $Z_1,\dots,Z_{b-1}$ be independent Bernoulli variables with parameter u. Take all these variables independent and set*

$$\mathcal{X} = \Gamma_2 + \sum_{j=1}^{b-1} Z_j E_j.$$

Then

$$S_{(b,a)}(u) = (1-u)^{|a|}\mathbb{E}[\mathcal{B}(v\mathcal{X})], \qquad \mathbb{E}e^{s\mathcal{X}} = (1-s)^{-2}\left(1+\frac{us}{1-s}\right)^{b-1} \quad (s<1). \tag{D2}$$

In particular, set $y = bu^2$ and $\kappa = \kappa(y)$. For every $0<u<1$,

$$(1-u)^{|a|}\mathcal{B}(y) \le S_{(b,a)}(u). \tag{D3}$$

If also $s = \kappa/[bu(1-u)] < 1$, then

$$S_{(b,a)}(u) \le (1-u)^{|a|}\mathcal{B}(y)e^{-\kappa}(1-s)^{-2}\left(1+\frac{us}{1-s}\right)^{b-1}. \tag{D4}$$

The proofs of Lemmas 8.1, D.1 and D.2 and of Theorem 8.2 are in Appendix D of the arXiv version of this paper [40].

Appendix E Proofs for reducible faces and for the counts near a facet

E.1 Blocks of a reducible face

In this subsection and the next, F is admissible, the notation is that of Section 4.4, and $hq_{\max} \le \frac{1}{2}$. For an irreducible C, states $y, x \in C$ and $m \in \mathbb{Z}_{\ge 1}^{C}$ with $|m| = l$, let $G_C(m)_{yx}$ be the probability that the chain with $X_1 = y$ has composition m in $X_1,\dots,X_l$ and $X_l = x$. So $G_{\{i\}}(m) = (1-hq_i)^{m-1}$. For n with support F let $n^{(j)}$ be n restricted to C_j, $N_j = |n^{(j)}|$, $s_j = hN_j$ and $G_j = G_{C_j}(n^{(j)})$. The blocks of a path with composition n occupy fixed times, a path that starts outside C_1 never visits C_1, and the Markov property at the $r-1$ switches gives

$$P_N(n) = \mu_{C_1}^{\top} G_1\,(hR_1)\,G_2\,(hR_2)\cdots(hR_{r-1})\,G_r\,\mathbf{1}.$$

The bridge run expansion. Let $|C| = k \geq 2$ and write $W_{yx} = W_{A_C,e_y,e_x}$. The proof of Lemma 2.3, with the first and the last letter of the words fixed, gives $G_C(m)_{yx} = \sum_{m'} W_{yx}(m') h^{M'-1} \prod_i \binom{m_i-1}{m'_i-1}(1-hq_i)^{m_i-m'_i}$. As in Step 1 of the proof of Lemma 5.5, with $t = hm$ the m'-th term is $\kappa_{m'} v_{m'}$, where $v_{m'} = W_{yx}(m') h^{k-1} \prod_i e^{-q_i t_i} t_i^{m'_i-1}/(m'_i-1)!$, so that $\sum_{m'} v_{m'} = h^{k-1}\Phi_C(t)_{yx}$, and $\kappa_{m'}$ is the factor of that proof. Its argument for (A7) gives

$$1 - \sum_i \frac{m'_i(m'_i-1)}{2m_i} - q_{\max}^2 h|t| \leq \kappa_{m'} \leq e^{hq_{\max}M'}.$$

Since $W_{yx}(m') \leq k^{M'} q_{\max}^{M'-1}$, the series of Φ_C converges for every $t \in \mathbb{R}^C$, and so do the series with the extra factors $M' e^{M'/2}$ or $m'_i(m'_i-1)t_i^{-1}$ in their terms. So Φ_C is real-analytic on $\mathbb{R}^C$, and these series are bounded on bounded sets. Also $\Phi_C(t)_{yx} > 0$ for $t \in (0,\infty)^C$, since some word from y to x uses every state.

Lemma E.1 (one bound for every block) *There is C_0 such that for $N \geq N_0$, $T \leq N^{1/3}$, every j, all $y, x \in C_j$ and every m with $\operatorname{supp} m = C_j$ and $s = h|m| \leq T$, we have $G_{C_j}(m)_{yx} \leq C_0 h^{k_j-1} w_j(s)$, where $w_j(s) = (1 \vee s)^{-(k_j-1)/2} e^{-\lambda_F s}$ if $\lambda_j = \lambda_F$ and $k_j \geq 2$, and $w_j(s) = e^{-\lambda_j s}$ otherwise.*

Proof For $k_j = 1$, $(1-hq_i)^{|m|-1} \leq e^{hq_i} e^{-q_i s}$. Let $k_j \geq 2$. The bridge is at most the probability that the chain started at y has composition m in its first $|m|$ visits. Lemma 4.1 with the start e_y and $\theta = q - \lambda_j \mathbf{1} \in V_{C_j}$ bounds it by $C h^{k_j-1} e^{-\lambda_j s}$. If $\lambda_j = \lambda_F$ and $s \geq 1$, we apply Theorem 5.7(b) with $v = 0$ to the chain started at y on the face C_j, which satisfies (H), with $l = |m|$ steps and time s in place of N and T, and the same h. Its range holds, since $l = sN/T \geq N^{2/3}$ and $s^3 \leq l$ is $s^2 T \leq N$. It gives $G_{C_j}(m)_{yx} \leq C' l^{-(k_j-1)} s^{(k_j-1)/2} e^{-\lambda_F s} = C' h^{k_j-1} s^{-(k_j-1)/2} e^{-\lambda_F s}$ with a constant C' for the start e_y. There are finitely many starts y. □

Lemma E.2 (blocks of bounded time) *For every R there is C_R such that $|G_C(m)_{yx}/h^{|C|-1} - \Phi_C(hm)_{yx}| \leq C_R h$ for all $y, x \in C$ and $\operatorname{supp} m = C$ with $h|m| \leq R$. Hence, if $D = \nu^\top \Psi_1 R'_1 \Psi_2 \cdots R'_{p-1} \Psi_p w$ with fixed nonnegative ν, R'_c and w and $\Psi_c = G_{C_c}(m^{(c)})/h^{|C_c|-1}$, and D^{c} is the same product with $\Phi_{C_c}(hm^{(c)})$, then $|D - D^{\mathrm{c}}| \leq C'_R h$ when $\sum_c h|m^{(c)}| \leq R$.*

Proof For a singleton, $e^{hq} e^{-qt} \geq (1-hq)^{m-1} \geq e^{-qt-q^2ht}$ with $t = hm$. Let $|C| = k \geq 2$. By the bound on $\kappa_{m'}$, $\kappa_{m'} - 1 \leq hq_{\max} M' e^{M'/2}$, which gives $G/h^{k-1} - \Phi_C \leq C_R h$. Also $1 - \kappa_{m'} \leq \sum_i m'_i(m'_i-1)/(2m_i) + q_{\max}^2 hR$, and since $1/m_i = h/t_i$, the sum of $v_{m'} m'_i(m'_i-1)/(2m_i)$ over m' is h^k times a series with the extra factor $m'_i(m'_i-1)/(2t_i)$, which is bounded on $|t| \leq R$. This gives $\Phi_C - G/h^{k-1} \leq C_R h$. For the product we telescope, since every factor is bounded on $|t| \leq R$. □

Lemma E.3 (the bridge law) *Let $|C| = k \geq 2$ and let $\mathcal{K} \subset \Delta_C^\circ$ be compact. Uniformly for $y, x \in C$, m with $m/l \in \mathcal{K}$ and $1 \leq s = hl \leq l^{1/2}$, $G_C(m)_{yx} = h^{k-1}\Phi_C(hm)_{yx}(1 + O(s^2/l))$.*

Proof This is the proof of Lemma 5.5 in Appendix A, with the bridge run expansion in place of Lemma 2.3, and l and s in place of N and T. Its Step 3 uses the weights only through an upper bound on H for eA_C and a lower bound on H at $t = s\alpha$. Step 2 of the proof of Lemma 4.1, with Lemma A.3 for (A_C, e_y, e_x) in place of Lemma A.2, gives (A4) for H_{A_C,e_y,e_x} with b_{yx} in place of b, and $b_{yx}(\theta'(\alpha))$ is bounded on $\mathcal{K}$. Lemma A.4 with $(\nu, e) = (e_y, e_x)$ gives $H_{A_C,e_y,e_x}(s\alpha)e^{-s\langle\alpha,\theta(\alpha)\rangle} \geq cs^{-(k-1)/2}$ for $s \geq s_1$, and for $1 \leq s \leq s_1$ the left side is continuous and positive on $[1, s_1] \times \mathcal{K}$. □

Lemma E.4 (bridge maxima) *Let $|C| = k \geq 2$, $\lambda = \lambda_C$ and $\Sigma = \Sigma_C$, and put $Y_{yx} = h^{k-1}s^{-(k-1)/2}r_{C,y}l_{C,x}e^{-\lambda s}/((2\pi)^{(k-1)/2}\sqrt{\det\Sigma})$. There are s_2, l_2 and C_3 such that the following hold for $y, x \in C$, $l \geq l_2$ and $s = hl \in [s_2, l^{1/3}]$.* (i) *$G_C(m)_{yx} \leq Y_{yx}(1+C_3(1/s+s^2/l))$ for every m with $|m| = l$ and $\operatorname{supp} m = C$.* (ii) *$G_C(m^0)_{yx} \geq Y_{yx}(1 - C_3(1/s + s^2/l))$ if $|m_i^0 - l\alpha_{C,i}^*| < 1$ for all i.*

Proof Put $c_{yx}(\alpha) = r_{\alpha,y}l_{\alpha,x}/((2\pi)^{(k-1)/2}\sqrt{\det\Sigma_\alpha})$, which is real-analytic and positive on Δ_C° by Proposition 5.2(a), with $c_{yx}(\alpha_C^*) = r_{C,y}l_{C,x}/((2\pi)^{(k-1)/2}\sqrt{\det\Sigma})$. Let $J^0 = J_C - \lambda$, which is nonnegative and vanishes only at α_C^* (Proposition 3.3(a)), and let ξ be the displacement $\alpha' - \alpha_C^{*\prime}$. Since $\nabla^2 J_C = \Sigma_\alpha^{-1}$, Taylor's theorem gives $\rho_0, c_U, C_U, C_1 > 0$ such that $U = \{|\xi| \leq \rho_0\} \subset \Delta_C^\circ$ and on U, for all y, x, $c_U|\xi|^2 \leq J^0 \leq C_U|\xi|^2$, $|\log(c_{yx}(\alpha)/c_{yx}(\alpha_C^*)) - g_{yx}\cdot\xi| \leq C_1|\xi|^2$ with $|g_{yx}| \leq C_1$, and $\log c_{yx}$ is C_1-Lipschitz.

Outside U. The function J^0 is lower semicontinuous and positive on the compact set $E = \Delta_C \setminus \{|\xi| < \rho_0\}$, so $J^0 \geq 3\eta$ there for some $\eta > 0$. Since $J_C = \sup_{\theta\in V_C}\langle q - \theta, \cdot\rangle$ (Section 4), each point of E has a neighborhood on which one $\theta \in V_C$ gives $\langle q-\theta, \cdot\rangle > \lambda+2\eta$, and finitely many $\theta_1, \dots, \theta_p$ cover E. For $m/l \in E$, the bridge is at most the probability that the chain started at y has composition m in its first l visits, and Lemma 4.1 bounds it by $Ch^{k-1}e^{-s(\lambda+2\eta)} = Y_{yx}\,O(s^{(k-1)/2}e^{-2\eta s}) = Y_{yx}\,O(1/s)$.

Inside U. Lemma E.3 and Lemma A.4 with $(\nu, e) = (e_y, e_x)$ at $t = s\alpha$ give $G_C(m)_{yx} = Y_{yx}(c_{yx}(\alpha)/c_{yx}(\alpha_C^*))e^{-sJ^0(\alpha)}(1 + E)$ with $|E| \leq C_2(1/s + s^2/l)$, where $\alpha = m/l$. For $s \geq 2C_1/c_U$ the logarithm of the two middle factors is at most $g_{yx}\cdot\xi - \frac{1}{2}c_U s|\xi|^2 \leq |g_{yx}|^2/(2c_U s)$. With the bound outside U this proves (i). For (ii), $|\xi| \leq k/l$ at m^0, so $c_{yx}(m^0/l) \geq c_{yx}(\alpha_C^*)e^{-kC_1/l}$ and $sJ^0(m^0/l) \leq C_U k^2 s/l^2$. □

E.2 Proofs of Theorem 4.5 and Corollary 4.6

Proof of Theorem 4.5 Write $a = a_F$, $K = K_F^{\mathrm{red}}$ and $X = N^{-(k_F-1)}T^{p_F}e^{-T\lambda_F} = h^{k_F-1}T^{-a}e^{-T\lambda_F}$. For $j \in S^*$ let $F_j^<(t^<)$ and $F_j^>(t^>)$ be the functions inside the suprema that define A_j and B_j. Call a block flat if it is a singleton $\{i\}$ with $q_i = \lambda_F$. Flat blocks exist only if $k^* = 1$, and then they are the blocks in S^*. Put $\tilde{w}_m(u) = e^{\lambda_F u}w_m(u) \leq 1$, with w_m from Lemma E.1, let $\delta_0 > 0$ be the least positive $\lambda_m - \lambda_F$ ($\delta_0 = 1$ if there is none), and $\eta(u) = \max(e^{-\delta_0 u}, u^{-1/2})$ for $u \geq 1$. A block that is not flat and has time $u \geq 1$ has $\tilde{w}_m(u) \leq \eta(u)$.

The constant. Since Φ_C is real-analytic on $\mathbb{R}^C$, $F_j^<$ and $F_j^>$ are Lipschitz on bounded parts of the closed orthants, and the suprema over the open and the closed orthants agree. Fix $t \in (0,\infty)^C$, let $h \to 0$ and $m = \lceil t/h \rceil$. By Lemma E.2, $G_C(m)/h^{|C|-1} \to \Phi_C(t)$, and Lemma E.1 with $T = |t|+1$ and $N = T/h$ gives $e^{\lambda_F|t|}\Phi_{C_m}(t) \le C_0\tilde{w}_m(|t|)$ entrywise. So every factor in $F_j^<$ and $F_j^>$ is bounded, and $A_j, B_j < \infty$. They are positive, since $\mu_{C_1} \ge 0$ and every $R_m \ge 0$ are nonzero, $r, l > 0$ and every $\Phi_{C_m} > 0$ on the open orthant. If $k^* \ge 2$, no factor is flat, so $F_j^<$ and $F_j^>$ tend to 0 at infinity, and the suprema are attained at points $t_*^<$ and $t_*^>$ of the closed orthants.

Upper bound. Let $\operatorname{supp} n = F$, so $\sum_m s_m = T$. By the factorization and Lemma E.1, summed over the finitely many switch states, and since $\sum_m (k_m - 1) + r - 1 = k_F - 1$,

$$P_N(n) \le C_1 h^{k_F-1} e^{-\lambda_F T} \prod_m \tilde{w}_m(s_m).$$

Let $k^* \ge 2$ and take m_0 with $s_{m_0} \ge T/r$. (A) If $m_0 \notin S^*$, then $\lambda_{m_0} > \lambda_F$, or $\lambda_{m_0} = \lambda_F$ and $k_{m_0} \ge k^* + 1$, so $P_N(n) \le C_1 X T^a \max(e^{-\delta_0 T/r}, (T/r)^{-a-1/2}) \le KX$ for $T \ge T_1$. (B1) If $m_0 = j \in S^*$ and $T - s_j \ge R_0$, another block has time at least R_0/r, and $\tilde{w}_j(s_j) \le (T/r)^{-a}$, so $P_N(n) \le C_1 r^a \eta(R_0/r) X \le KX$ once R_0 is fixed large. (B2) If $m_0 = j \in S^*$ and $T - s_j < R_0$, write $P_N(n) = \mathbf{a}^\top G_j \mathbf{b}$ with $\mathbf{a}^\top = \mu_{C_1}^\top G_1 h R_1 \cdots G_{j-1} h R_{j-1}$ and $\mathbf{b} = h R_j G_{j+1} \cdots G_r \mathbf{1}$, both nonnegative. Lemma E.4(i) applies with $l = N_j \ge N/2$ and $s = s_j \ge T - R_0$, since $s_j^3 \le N_j$ means $s_j^2 T \le N$, and $s_j^2/N_j = s_j T/N$. It gives

$$P_N(n) \le h^{k^*-1} s_j^{-a} e^{-\lambda_F s_j} \frac{(\mathbf{a}\cdot r)(l'\cdot \mathbf{b})}{(2\pi)^a \sqrt{\det \Sigma_{C_j}}} \Big(1 + C\Big(\frac{1}{T} + \frac{T^2}{N}\Big)\Big),$$

where $r = r_{C_j}$ and $l' = l_{C_j}$. Let $\varkappa_< = \sum_{m<j}(k_m - 1) + j - 1$ be the number of factors h in $\mathbf{a}$. Then $\mathbf{a}\cdot r/h^{\varkappa_<}$ is a product as in Lemma E.2 with total time $\rho_< < R_0$, whose continuum version is at most $e^{-\lambda_F \rho_<} A_j$. So $e^{\lambda_F \rho_<} \mathbf{a}\cdot r \le h^{\varkappa_<}(A_j + Ch)$, and likewise $e^{\lambda_F \rho_>} l'\cdot \mathbf{b} \le h^{\varkappa_>}(B_j + Ch)$. Since $\varkappa_< + \varkappa_> + k^* - 1 = k_F - 1$, $\rho_< + \rho_> + s_j = T$, $s_j^{-a} \le T^{-a}(1 + CR_0/T)$ and $h \le T^2/N$, we get $P_N(n) \le X(K + C(1/T + T^2/N))$.

Let $k^* = 1$, and let ρ be the total time of the blocks that are not flat. If $\rho \ge R_0$, one of them has time at least R_0/r, and $P_N(n) \le C_1 \eta(R_0/r) X \le KX$. If $\rho < R_0$, each flat block has $(1 - h\lambda_F)^{N_m - 1} \le e^{h\lambda_F} e^{-\lambda_F s_m}$, and the other blocks form products as in Lemma E.2 with total time below R_0. So $P_N(n) \le h^{k_F-1} e^{-\lambda_F T} e^{rhq_{\max}}(\Psi^* + Ch)$, where Ψ^* is the supremum of $e^{\lambda_F|t|}\mu_{C_1}^\top \Phi_{C_1} R_1 \cdots \Phi_{C_r}\mathbf{1}$, in which every flat factor equals 1. At a flat $j \in S^*$ this product splits into the products of A_j and B_j, in disjoint variables, so $\Psi^* = A_j B_j = K$.

Lower bound. Let $k^* \ge 2$, and take $j \in S^*$ that attains the maximum in K and the points $t_*^<$ and $t_*^>$, with total time τ_*. For $m \ne j$ put $n^{(m)} = \max(1, \lceil t_*^{(m)}/h \rceil)$ entrywise. Every entry of $hn^{(m)}$ is within h of that of $t_*^{(m)}$, so $F_j^<(hn^<) \ge A_j - Ch$ and $F_j^>(hn^>) \ge B_j - Ch$. Give block j the length $l = N - \sum_{m\ne j} N_m$ and a composition m^0 with $|m_y^0 - l\alpha^*_{C_j,y}| < 1$, which has full support for large N, so that $s = hl \in [T - \tau_* - Ch, T]$. By Lemma E.2, $\mathbf{a} \ge h^{\varkappa_<}(\mathbf{a}^{\mathrm{c}} - Ch)$ entrywise, with $e^{\lambda_F|hn^<|}\mathbf{a}^{\mathrm{c}}\cdot r = F_j^<(hn^<)$, and likewise for $\mathbf{b}$. Lemma E.4(ii) and nonnegativity give $P_N(n) \ge h^{k_F-1} s^{-a} e^{-\lambda_F T}(A_j - Ch)(B_j - Ch)((2\pi)^a\sqrt{\det\Sigma_{C_j}})^{-1}(1 - C(1/T + T^2/N))$, and $s^{-a} \ge T^{-a}$. So $M_F \ge X(K - C(1/T + T^2/N))$. Let $k^* = 1$. The supremum Ψ^* is attained at a point of the closed orthants. Indeed, with the flat factors equal to 1, every other factor is at most $C_0\tilde{w}_m \le C_0$, and a factor whose time grows tends to 0, since it is not flat. So the function tends to 0 at infinity. We take a flat $j \in S^*$, treat the blocks that are not flat in the same way at such a point, give every other flat block the length 1, and use $(1 - h\lambda_F)^{l-1} \ge e^{-\lambda_F s}(1 - CT^2/N)$ for block j. □

Proof of Corollary 4.6 Theorem 4.5 for F and G gives, uniformly on $[T_1, N^{1/3}]$,

$$D_N(T) = -\Delta L + \Xi(T) + \kappa + O\Big(\frac{1}{T} + \frac{T^2}{N}\Big), \qquad \Xi(T) = \delta T + \pi \log T.$$

(i) For fixed N, M_F and M_G are positive maxima of finitely many polynomials in h, so D_N is continuous. At $T = \frac{\Delta}{2\delta} L$ the display gives $D_N = -\frac{\Delta}{2} L + O(\log L) < 0$, and at $T = \frac{2\Delta}{\delta} L$ it gives $D_N = \Delta L + O(\log L) > 0$.

(ii) A zero has $\delta T_N + \pi \log T_N = \Delta L + O(1)$. If $\pi \geq 0$, then $\delta T_N \leq \Delta L + O(1)$. If $\pi < 0$, then $\pi \log T_N \geq \pi L/3$, since $T_N \leq N^{1/3}$, and $\delta T_N \leq (\Delta + |\pi|/3)L + O(1)$. So $T_N = O(L)$, then $T_N = \frac{\Delta}{\delta} L + O(\log L)$, and the error term is $O(1/L)$.

(iii) Let $\Xi(T^\circ) = \Delta L - \kappa$. As in (ii), $T^\circ = \frac{\Delta}{\delta} L + O(\log L)$, and $\Xi' = \delta + \pi/T \geq \delta/2$ on $[\frac{\Delta}{2\delta} L, \frac{2\Delta}{\delta} L]$ for large N. So every zero is within $O(1/L)$ of T°. Putting $\log T^\circ = \log L + \log \frac{\Delta}{\delta} + O(\log L/L)$ into $\delta T^\circ = \Delta L - \pi \log T^\circ - \kappa$ gives the formula. □

E.3 Proof of Theorem 6.11

The proof, from Lemmas 6.4 and 6.9 and bounds on the moments of the laws proportional to the terms of $\mathbf{K}^h$ and $\mathbf{K}$, is in this section of the arXiv version of this paper [40].

E.4 Proof of Theorem 6.12

Constants depend only on Q, μ, i and $\mathcal{K}$. Write $n = n_i$, $p = p_i$, $s = h\hat{n} = S\beta$, $\tilde{x} = (n-1)h$ and

$$w_d(z) = \Big(p + \frac{h}{z}\Big)^{n-1} = \sum_{K=1}^{n} b_K z^{-(K-1)}, \qquad b_K = \binom{n-1}{K-1} p^{n-K} h^{K-1},$$

so that $\mathbf{K}^h_\beta(n, S) = h \sum_K b_K [z^K] \widetilde{\Lambda}_\beta$. Let Λ, ε, Ξ and $R = \log \Xi$ be as in the proof of Theorem 6.1(a). So $|\varepsilon(z) - \varepsilon(0)| \leq C|z|/S$ for $|z| \leq z_0$, and R is real on the real axis with $|R(z)| \leq C(|z| + S|z|^2)$, $|R'(z)| \leq C(1 + S|z|)$ and $|R''| \leq C(1+S)$ on $|z| \leq z_0$. Throughout, $hS \leq T^2/N \leq 1$, $\tilde{x} \leq x \leq \delta T$ and $S \geq (1-\delta)T$.

The coefficient identity. The proof of Lemma 6.3 uses the rates only as weights of words. So for every matrix q' of nonnegative rates on F in place of Q, with the same q_G in the exponent, the coefficient of $x^{K-1}/(K-1)!$ in (15) gives

$$e^{-\langle q_G, s\rangle} \sum_{m:\, m_i = K} W'(m) \prod_{j \in G} \frac{s_j^{m_j - 1}}{(m_j - 1)!} = [z^K] f_G'^{(z)}(s),$$

where W' and $f_G'^{(z)}$ are W and $f_G^{(z)}$ built from q', with $d'(z) = q_G - z c'^- \circ c'^+$.

The radius. Put $\tilde{z}_* = \sqrt{\tilde{x}/(\omega_\beta S)}$ and $\tilde{\zeta} = 2\sqrt{\omega_\beta S \tilde{x}}$. If $\tilde{\zeta} \leq Z_0$, let $\rho = \max(\tilde{z}_*, 1/S)$. If $\tilde{\zeta} > Z_0$, let ρ be the critical point on $I = [\tilde{z}_*/2, 2\tilde{z}_*]$ of $\lambda_d(z) = (n-1)\log(p + h/z) + S\omega_\beta z + R(z)$. Let $\mathcal{A}_d = \frac{1}{2\pi \mathrm{i}} \oint_{|z|=\rho} w_d \Lambda \, dz/z^2$, let $\mathcal{A}_d^{\mathrm{abs}}$ be the integral of $|w_d \Lambda||z|^{-2}$ against $|dz|/(2\pi)$, and $\zeta_d = \max(1, S\omega_\beta \rho)$.

Lemma E.5 (the discrete contour) *For Z_0 large, then δ small and T_0 large, ρ is well defined, $\rho \le z_0/2$, $\mathcal{A}_d > 0$, $\mathcal{A}_d^{\mathrm{abs}} \le C\mathcal{A}_d$ and $\mathcal{A}_d \ge c\rho^{-1} w_d(\rho)\Lambda(\rho)\zeta_d^{-1/2}$. With $\mathbf{D} = \sum_K b_K[z^K] f_G^{(z)}(s) = \frac{1}{2\pi \mathrm{i}} \oint_{|z|=\rho} w_d f_G^{(z)}(s)\, dz/z^2$,*

$$\mathbf{D} = (1+\varepsilon(0))\mathcal{A}_d(1+O(\rho/S)), \qquad \rho/S \le C(T^{-2} + x^{1/2}T^{-3/2}).$$

Lemma E.6 (the facet states discretized) *Uniformly as in Theorem 6.12, $P_N(n) = h^{k-1}\mathbf{D}(1+O(T^2/N))$.*

The proofs of Lemmas E.5 and E.6 and of Theorem 6.12, which follow those of Theorem 6.1(a) and Lemma 5.5 with the state i kept exact, are in this section of the arXiv version.

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